

ON MODULAR FORMS

JOE ORII

MSci Thesis Supervised by Salvatore Mercuri
CID: 02207177

ABSTRACT. This thesis studies modular forms from both classical and modern perspectives. We begin with classical modular forms on the upper half-plane, focusing on their structural properties, including dimension formulae and the theory of Hecke operators. We then introduce the adèle ring and use it to build the adelic framework. Using this, we reformulate and generalize several classical notions, leading to the definition of automorphic forms on $GL(2, \mathbb{A}_{\mathbb{Q}})$. This perspective provides a more conceptual understanding of the objects arising in the classical theory. Finally, we consider quaternionic modular forms over $\mathbb{Q}$ and investigate the Jacquet–Langlands correspondence. In particular, we illustrate instances of this correspondence through explicit examples involving quaternion algebras of discriminants 11 and 19 and level 1, tying together the preceding theory.

CONTENTS

1. Introduction	2
1.1. The History of Modular Forms	2
1.2. What This Thesis Does	4
2. Classical Modular Forms	5
2.1. Congruence Subgroups	5
2.2. The Definition of a Modular Form	6
2.3. Some Examples	8
2.4. Dirichlet Character Decomposition	10
3. Dimension Formulae	11
3.1. Modular Curves	11
3.2. Classical Automorphic Forms	14
3.3. Meromorphic Differentials	16
3.4. The Dimension Formulae	18
4. Hecke Operators	21
4.1. Double Coset Operators	21
4.2. Eigenforms	25
4.3. Oldforms and Newforms	28
5. The Adelic Framework	29
5.1. Absolute Values	29
5.2. The Topology from an Absolute Value	30
5.3. The Adeles	31
5.4. The Group $GL(2, \mathbb{A}_{\mathbb{Q}})$	34
6. Automorphic Forms for $GL(2, \mathbb{A}_{\mathbb{Q}})$	37
6.1. Some Lie Theory	37
6.2. The Definition	41
6.3. The Adelic Lift	43
7. Quaternionic Modular Forms and the Jacquet–Langlands Correspondence	49
7.1. Quaternion Algebras	49
7.2. Fujisaki’s Lemma	54
7.3. Quaternionic Modular Forms	56
7.4. The Hecke Operators	58

7.5. The Jacquet-Langlands Correspondence	60
8. Conclusion and Future Directions	65
Appendix A. Reference Documentation for Code	67
References	68

Plagiarism Statement: This is my own work unless otherwise stated. I acknowledge the use of artificial intelligence tools, specifically ChatGPT and Claude, for grammatical corrections, L^AT_EX formatting, and in Section 7 to support the coding and computational work.

1. INTRODUCTION

1.1. The History of Modular Forms. We begin by giving a historical account of modular forms, based on the various sources in the references. Modular forms play a very important role in modern-day mathematics, especially in areas such as number theory or physics. One of the main origins of modular forms is the study of elliptic integrals in the 19th century. These were integrals which looked like

$$\int \frac{dx}{\sqrt{p(x)}},$$

where $p(x)$ is a polynomial of degree 3 or 4 without repeated roots. If we invert such integrals, we obtain elliptic functions, which were extensively studied by mathematicians such as Abel, Gauss, or Jacobi. Elliptic functions are doubly periodic functions on the complex plane, meaning that they admit two independent periods. This marked a fundamental shift, because it led to the study of functions with symmetry properties. Perhaps most significantly, Gauss and Jacobi introduced and studied theta functions, an example of which is

$$\theta(\tau) = \sum_{n \in \mathbb{Z}} e^{2\pi i n^2 \tau}, \quad (1)$$

where τ is in the upper half plane $\mathbb{H}$. They discovered that these functions satisfy specific transformation laws under a change of variables. For instance, we can calculate that

$$\theta\left(\frac{-1}{4\tau}\right) = \sqrt{-2i\tau} \theta(\tau) \quad (2)$$

Such symmetries are a precursor to the symmetry seen in modular forms.

Modular forms essentially refer to a more general class of functions on the upper half plane $\mathbb{H}$ which satisfy some symmetry and analytic properties (Definition 2.10). While this analytic origin may seem far from number theory, modular forms are of great importance to number theory because they can possess arithmetic information. To illustrate an instance of how this may occur, I discuss here the four-squares problem, which is a problem that can be solved using elementary methods such as those which come from a first course in number theory, based on Chapter 1 of [DS05]. The problem is this: given a positive integer n , is it possible to write n as a sum of four squares? The surprising fact is that we can use the theta series in Equation (1) to solve this problem. What is even nicer is that the theta series even provides the number of ways one can formulate such a sum. We briefly sketch how this works.

Definition 1.1. The *representation number* of n by k squares is the following number $r(n, k)$ defined by

$$r(n, k) = \left| \left\{ (v_1, \dots, v_k) \in \mathbb{Z}^k : n = v_1^2 + \dots + v_k^2 \right\} \right|.$$

One can show that then $r(n, k)$ satisfies the identity

$$r(n, i + j) = \sum_{l+m=n} r(l, i) r(m, j).$$

This looks very similar to how one would attain the coefficient of the product of a series, so we define a generating function $\theta(\tau, k)$ as

$$\theta(\tau, k) = \sum_{n=0}^{\infty} r(n, k) q^n, \quad q = e^{2\pi i \tau}, \quad \tau \in \mathbb{H}. \quad (3)$$

This is absolutely convergent (as can be shown in Chapter 4.9 of [DS05]), and so

$$\theta(\tau, k_1)\theta(\tau, k_2) = \theta(\tau, k_1 + k_2), \quad \tau \in \mathbb{H}. \quad (4)$$

Furthermore, one can show that $\theta(\tau + 1, k) = \theta(\tau, k)$ and $\theta(\tau, k) = (\theta(\tau))^k$, where $\theta(\tau)$ is as in Equation (1). Now, let us specialize to $k = 4$ because this is the appropriate setting for the four squares problem. Since $\theta(\tau, 4)$ satisfies both the relation $\theta(\tau + 1, 4) = \theta(\tau, 4)$ and the transformation law in Equation (2), one can deduce that $\theta(\tau, 4) = (\theta(\tau))^4$ is a modular form of weight 2 with respect to the congruence subgroup $\Gamma_0(4)$, as in Definition 2.10. Moreover, results in Chapter 4.6 of [DS05] imply that the space of weight-2 modular forms with respect to $\Gamma_0(4)$ is spanned by the following series:

$$G_{2,2}(\tau) = -\frac{\pi^2}{3} \left(1 + 24 \sum_{n=1}^{\infty} \left(\sum_{\substack{0 < d|n \\ d \text{ odd}}} d \right) q^n \right), \quad (5)$$

$$G_{2,4}(\tau) = -\pi^2 \left(1 + 8 \sum_{n=1}^{\infty} \left(\sum_{\substack{0 < d|n \\ 4|d}} d \right) q^n \right), \quad (6)$$

where $q = e^{2\pi i \tau}$ for $\tau \in \mathbb{H}$. Finding the coefficients of the linear combination and then equating coefficients of the series, we can find that

$$r_4(n) = 8 \sum_{\substack{d|n \\ 4 \nmid d}} d.$$

In particular, this solves the four-squares problem as we have a concrete formula for $r_4(n)$. We note that Jacobi found this formula in 1834, and this was the method he took. This solution is absolutely remarkable because it took an elementary number theory problem and applied these analytic methods to solve it completely. As to why arithmetic functions even arise in Equation (5) and Equation (6) in the first place, this requires knowledge about Eisenstein series, which we refer to Chapter 4 of [DS05]. While Eisenstein series is not a topic studied deeply in this thesis, Example 2.20 contain simple examples of Eisenstein series, in which arithmetic functions appear again.

To return to the historical development of modular forms, a major advancement in the arithmetic aspects of modular forms came with the work of Ramanujan. Ramanujan extensively studied the Fourier coefficients of the discriminant function Δ , defined in Example 2.22. Denoting these Fourier coefficients as $\tau(n)$, we often refer to this as the *Ramanujan τ -function*. Ramanujan conjectured in 1916 that $|\tau(p)| \leq 2p^{\frac{11}{2}}$, τ is multiplicative, and that

$$\tau(p)\tau(p^n) = \tau(p^{n+1}) + p^{11}\tau(p^{n-1}), \quad \text{if } p \text{ is prime and } n \geq 1.$$

A year later, Mordell proved the latter two of these three properties. This paper contains one of the first examples of Hecke operators, which is another significant object that enriched the theory of modular forms. The significance of these operators is that, after defining an inner product on the space of modular forms, there exist modular forms which are common eigenfunctions of these operators. We refer to such functions as eigenforms. Hecke went a step further and even associated L -functions with such eigenforms, sparking a connection between analytic number theory and modular forms.

While Hecke’s theory revealed the algebraic structure underlying modular forms, the work of Shimura and others in the 1950s demonstrated that modular forms could be interpreted geometrically in terms of algebraic varieties, and more generally, Shimura varieties. This geometric viewpoint was crucial in the work of Deligne, who proved the Weil conjectures in 1974 and, as a consequence, proved the first property of the conjecture by Ramanujan above.

The developments described above led to one of the most renowned conjectures in number theory, which is the Taniyama–Shimura–Weil conjecture. This conjecture was formulated in the 1950s by Taniyama and Shimura, then refined in the 1960s by Weil. This conjecture asserts that elliptic curves over $\mathbb{Q}$ are modular; vaguely speaking, it asserts that every elliptic curve over $\mathbb{Q}$ arises from a modular form. More precisely, it states that if we consider an L -function arising from an elliptic curve over $\mathbb{Q}$ and an L -function arising from a modular form in the sense above, these two L -functions coincide. This conjecture guided much of the development of number theory in the latter half of the 20th century. With a correction from Taylor, Wiles proved the conjecture for a large class of elliptic curves in 1995, enough for him to complete the proof of Fermat’s Last Theorem. The remaining cases were proven by Breuil, Conrad, Diamond, and Taylor in 2001 and this conjecture is now referred to as the modularity theorem.

The framework underlying these developments was anticipated, in a letter sent by Langlands to Weil in 1967, which can be found in [Lan67]. In this letter, Langlands proposed a vast generalization of this notion of modularity, which contains ideas that motivate an area of number theory referred to as the Langlands program. Particularly, Langlands wrote to Weil about reciprocity, which is a phenomenon that vaguely states that automorphic forms (or automorphic representations) should correspond to Galois representations. The modularity theorem is only an instance of this. The name is motivated by earlier results in number theory, most notably quadratic reciprocity; as a natural extension from the name, class field theory is viewed as the abelian case of reciprocity. There is another major pillar to the Langlands program besides reciprocity, which is the notion of functoriality. Very broadly, the aim of this aspect is to relate the automorphic representations of different reductive groups. In [Lan70], Langlands himself goes into these ideas in more detail, subdividing them into seven major questions in Section 7.

1.2. What This Thesis Does. While what we have studied in this dissertation is extremely far from many of the topics just mentioned, it is a starting point for studying the central objects of the Langlands program. In Sections 2 to 4, we have focused on classical modular forms. These refer to modular forms defined on the upper half plane with respect to the group $\mathrm{SL}(2, \mathbb{Z})$. We will see that modular forms are instances of much more general functions called classical automorphic forms, and we will generalize this notion even further in Sections 5 and 6 to formulate an adelic definition. Finally, in Section 7, we talk about the Jacquet–Langlands correspondence and run through an example which illustrates the result. This brings together our previous theory because we define automorphic forms on quaternion algebras using our adelic framework and observe how they correspond to objects on the classical side. We note that this correspondence can be thought of as an instance of Langlands functoriality, as described in [Gre15].

I would like to finish this introduction with two remarks. Firstly, I would like to express my deepest gratitude towards my supervisor Dr. Salvatore Mercuri for his supervision for the past academic year. I truly appreciate how supportive, patient, and kind he has been in discussing mathematics and giving me advice. Secondly, I would like to end this introductory section with a quote attributed to Eichler which conveys the significance of modular forms:

“There are five fundamental operations of arithmetic: addition, subtraction, multiplication, division, and modular forms.”

2. CLASSICAL MODULAR FORMS

In this first section, we introduce the basic definitions and some examples of modular forms over the upper half plane $\mathbb{H}$. Note that $\mathbb{H}$ is the subset of $\mathbb{C}$ containing elements with strictly positive imaginary part. Often, modular forms in this setting are referred to as *classical modular forms*, and we will use this term frequently as well.

2.1. Congruence Subgroups. Modular forms come with some symmetrical properties, but to discuss this, we first introduce the action of the modular group $\mathrm{SL}(2, \mathbb{Z})$ on $\mathbb{H}$. We define this via

$$\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}, \quad (7)$$

where $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ and $\tau \in \mathbb{H}$. One can check that $\gamma \cdot \tau$ lies in $\mathbb{H}$ for any $\gamma \in \mathrm{SL}(2, \mathbb{Z})$, providing a well-defined action on $\mathbb{H}$. We note that this action is also associative. Moreover, the fundamental domain of $\mathbb{H}$ under this action by $\mathrm{SL}(2, \mathbb{Z})$ is given by the set

$$D = \left\{ \tau \in \mathbb{H} : |\tau| > 1, |\mathrm{Re}(\tau)| < \frac{1}{2} \right\}.$$

If we further consider the following union of D with some boundary identifications

$$D' := D \cup \left\{ \tau \in \mathbb{H} : |\tau| \geq 1, \mathrm{Re}(\tau) = -\frac{1}{2} \right\} \cup \left\{ \tau \in \mathbb{H} : |\tau| = 1, -\frac{1}{2} \leq \mathrm{Re}(\tau) \leq \frac{1}{2} \right\},$$

then there exists a unique representative in D' for every orbit of $\mathrm{SL}(2, \mathbb{Z})$. To gain some intuition about why this is true, we note that $\mathrm{SL}(2, \mathbb{Z})$ has the two generators $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, which we shall call α, β for now. Then, if $\tau \in \mathbb{H}$, α and β act by

$$\alpha \cdot \tau = \tau + 1, \quad \beta \cdot \tau = -\frac{1}{\tau}.$$

Geometrically, what this means is that α translates τ horizontally. As for β , if $|\tau| < 1$, β maps τ to a region in $\mathbb{H}$ with complex absolute value greater than 1. By composing these transformations, we can bring τ into D . As for the boundary identifications, these come from analyzing the points of D' which lie in the same $\mathrm{SL}(2, \mathbb{Z})$ -orbit. This is in Lemma 2.3.2 of [DS05].

Definition 2.1. The *principal congruence subgroup of level N* is defined as

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

Here, we take the congruence entry-wise, so $a \equiv 1 \pmod{N}$, $b \equiv 0 \pmod{N}$, and so forth.

Definition 2.2 (Definition 1.2.1 in [DS05]). A subgroup $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ is a *congruence subgroup* if $\Gamma(N) \subset \Gamma$, for some positive integer N . In this case, we say Γ is a *congruence subgroup of level N* .

Example 2.3. Two important examples of congruence subgroups are the following groups.

$$\Gamma_0(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}$$

$$\Gamma_1(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}$$

Proposition 2.4 (Exercise 1.2.3 in [DS05]). *If Γ is any of the congruence subgroups introduced above, then $[\mathrm{SL}(2, \mathbb{Z}) : \Gamma]$ is finite and we have the following chain of subgroups:*

$$\Gamma(N) \trianglelefteq \Gamma_1(N) \trianglelefteq \Gamma_0(N) \subseteq \mathrm{SL}(2, \mathbb{Z})$$

Furthermore, $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}(2, \mathbb{Z})$.

Proof. The main idea to see how one is a normal subgroup of another is to realize each normal subgroup as the kernel of some group homomorphism. First, $\Gamma(N)$ is the kernel of the surjection $\Gamma_1(N) \rightarrow \mathbb{Z}/N\mathbb{Z}$ defined by $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto b \pmod{N}$. Secondly, $\Gamma_1(N)$ is the kernel of the surjection $\Gamma_0(N) \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$ defined by $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N}$. It is then immediate from these that $[\Gamma_0(N) : \Gamma_1(N)]$ and $[\Gamma_1(N) : \Gamma(N)]$ are finite. It remains to check that $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}(2, \mathbb{Z})$ and that $[\mathrm{SL}(2, \mathbb{Z}) : \Gamma] < \infty$. This second claim will follow from the fact that $[\mathrm{SL}(2, \mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p|N} (1 - p^{-2}) < \infty$.

We first note that $\Gamma(N)$ is the kernel of the natural surjection $\mathrm{SL}(2, \mathbb{Z}) \rightarrow \mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z})$. What remains then is to show the claim about the cardinality. We do this by proving the claim for prime powers and then applying the Chinese remainder theorem. We proceed by induction. When $n = 1$, we use the first isomorphism theorem on $\det: \mathrm{GL}(2, \mathbb{Z}/p\mathbb{Z}) \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$, which we note is surjective. We can calculate that $|\mathrm{GL}(2, \mathbb{Z}/p\mathbb{Z})| = (p^2 - 1)(p^2 - p)$ by analyzing when a matrix in $M(2, \mathbb{Z}/p\mathbb{Z})$ has linearly independent rows. There are $p^2 - 1$ possibilities for the first row by excluding the case when both entries of the row are 0, and there are $p^2 - p$ possibilities for the second row by omitting scalar multiples of the first row. Since $\mathrm{SL}(2, \mathbb{Z}/p\mathbb{Z}) = \ker \det$, we then find that

$$|\mathrm{SL}(2, \mathbb{Z}/p\mathbb{Z})| = \frac{(p^2 - 1)(p^2 - p)}{p - 1} = p^3(1 - p^{-2}).$$

Now assume the result holds up to some n . Then, we consider the reduction modulo p^n map $\pi_n: \mathrm{SL}(2, \mathbb{Z}/p^{n+1}\mathbb{Z}) \rightarrow \mathrm{SL}(2, \mathbb{Z}/p^n\mathbb{Z})$, which is surjective. We now want to compute the order of the kernel. The kernel contains matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}/p^{n+1}\mathbb{Z})$ subject to the condition

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{p^n}.$$

We now parametrize a, b, c, d as

$$a = a_0 p^n + 1, \quad b = b_0 p^n, \quad c = c_0 p^n, \quad d = d_0 p^n + 1, \quad a_0, b_0, c_0, d_0 \in \mathbb{Z}/p\mathbb{Z},$$

and find the cardinality of numbers a_0, b_0, c_0, d_0 which work. Note that this matrix must have determinant 1, so these variables must satisfy

$$p^n(a_0 + d_0) \equiv 0 \pmod{p^{n+1}}.$$

In other words, the only constraint on a_0, b_0, c_0, d_0 is the equation $a_0 \equiv -d_0 \pmod{p}$, so $|\ker \pi_n| = p^3$. To complete the induction, we have that

$$|\mathrm{SL}(2, \mathbb{Z}/p^{n+1}\mathbb{Z})| = |\mathrm{SL}(2, \mathbb{Z}/p^n\mathbb{Z})| \cdot |\ker \pi_n| = p^{3n}(1 - p^{-2}) \cdot p^3 = p^{3(n+1)}(1 - p^{-2}).$$

We now know $|\mathrm{SL}(2, \mathbb{Z}/p^n\mathbb{Z})| = p^{3n}(1 - p^{-2})$ for any positive integer n , so the Chinese remainder theorem tells us that $|\mathrm{SL}(2, \mathbb{Z}/N\mathbb{Z})| = N^3 \prod_{p|N} (1 - p^{-2})$ as desired. $\square$

Lemma 2.5 (Exercise 1.5 in [DS05]). *Let $\alpha \in \mathrm{SL}(2, \mathbb{Z})$ and $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ be a congruence subgroup. Then, $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$.*

Proof. Γ is a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$, so $\Gamma(N) \subseteq \Gamma$, for some positive integer N . By Proposition 2.4, $\Gamma(N) \trianglelefteq \mathrm{SL}(2, \mathbb{Z})$, so $\alpha^{-1}\Gamma(N)\alpha = \Gamma(N)$. Therefore, $\Gamma(N) = \alpha^{-1}\Gamma(N)\alpha \subseteq \alpha^{-1}\Gamma\alpha$. $\square$

2.2. The Definition of a Modular Form.

Definition 2.6. For any $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$, we define the *factor of automorphy* $j(\gamma, \tau) \in \mathbb{C}$ for $\tau \in \mathbb{H}$ as

$$j(\gamma, \tau) := c\tau + d.$$

Furthermore, we define the *slash operator of weight k* on $f: \mathbb{H} \rightarrow \mathbb{C}$ as

$$(f|_k \gamma)(\tau) = j(\gamma, \tau)^{-k} f(\gamma \cdot \tau), \quad \tau \in \mathbb{H}.$$

Remark 2.7. Note that the slash operator of weight k is a linear operator. Let $f, g: \mathbb{H} \rightarrow \mathbb{C}$, $a, b \in \mathbb{C}$, and $\gamma \in \mathrm{SL}(2, \mathbb{Z})$. We have as follows:

$$\begin{aligned} ((a \cdot f + b \cdot g)|_k \gamma)(\tau) &= j(\gamma, \tau)^{-k} (a \cdot f(\gamma \cdot \tau) + b \cdot g(\gamma \cdot \tau)) \\ &= a \cdot (j(\gamma, \tau)^{-k} f(\gamma \cdot \tau)) + b \cdot (j(\gamma, \tau)^{-k} g(\gamma \cdot \tau)) = a \cdot (f|_k \gamma)(\tau) + b \cdot (g|_k \gamma)(\tau). \end{aligned}$$

Definition 2.8 (Definition 1.1.1 in [DS05]). Let k be an integer and $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ be a congruence subgroup. A meromorphic function $f: \mathbb{H} \rightarrow \mathbb{C}$ is said to be *weakly modular of weight k* if

$$f(\gamma \cdot \tau) = j(\gamma, \tau)^k f(\tau),$$

where $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and $\tau \in \mathbb{H}$.

Remark 2.9. It is clear that f being weakly modular of weight k is equivalent to saying that f is invariant under the slash operator of weight k ; that is, $(f|_k \gamma)(\tau) = f(\tau)$. We may also call this property *weight- k invariance* under Γ .

Definition 2.10 (Definition 1.1.2 in [DS05]). Let k be an integer and $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ be a congruence subgroup. A holomorphic function $f: \mathbb{H} \rightarrow \mathbb{C}$ is a *modular form of weight k with respect to Γ* if f is weakly modular of weight k and $(f|_k \alpha)$ is holomorphic at ∞ for all $\alpha \in \mathrm{SL}(2, \mathbb{Z})$. Furthermore, if $a_0 = 0$ in the Fourier expansion of $(f|_k \alpha)$ for all $\alpha \in \mathrm{SL}(2, \mathbb{Z})$, we say that f is a *cusp form*.

Notation 2.11. Given some congruence subgroup $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$, we shall denote the spaces of modular forms and cusp forms of weight k with respect to Γ by $\mathcal{M}_k(\Gamma)$ and $\mathcal{S}_k(\Gamma)$, respectively. Furthermore, we denote the spaces of modular forms and cusp forms of all weights with respect to Γ by $\mathcal{M}(\Gamma)$ and $\mathcal{S}(\Gamma)$, respectively.

Remark 2.12. We refer the condition of $f|_k \alpha$ being holomorphic at ∞ for all $\alpha \in \mathrm{SL}(2, \mathbb{Z})$ as *holomorphy at the cusps*. We note that this naming comes from the fact that any cusp (as in Definition 3.5) can be moved to the point at ∞ via some $\mathrm{SL}(2, \mathbb{Z})$ -action.

Remark 2.13. To check holomorphy at the cusps, it suffices to check the condition for a finite set of representatives of the quotient $\Gamma \backslash \mathrm{SL}(2, \mathbb{Z})$. This can be further refined by quotienting on the right by Γ_∞ , where $\Gamma_\infty = \{\pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z}\}$.

Remark 2.14. Given a modular form of weight k with respect to a congruence subgroup $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$, say f , we note that $f|_k \alpha$ does indeed admit a Fourier expansion, for $\alpha \in \mathrm{SL}(2, \mathbb{Z})$. We shall first show that f admits a Fourier expansion. As noted in page 16 of [DS05], every congruence subgroup contains a matrix of the form $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$. Therefore, $f(\tau + h) = f(\tau)$, so the function f descends through $q_h = e^{\frac{2\pi i \tau}{h}}$ to a holomorphic function g on the punctured disk $0 < |q_h| < 1$, satisfying $f(\tau) = g(q_h)$. Since g is holomorphic on the punctured disk, it has a Laurent expansion

$$g(q_h) = \sum_{n \in \mathbb{Z}} a_n q_h^n. \quad (8)$$

Since $|q_h| = e^{\frac{-2\pi \mathrm{Im}(\tau)}{h}}$, we have that $q_h \rightarrow 0$ if $\mathrm{Im}(\tau) \rightarrow \infty$. Viewing ∞ as some point far in the imaginary direction, we have that f is holomorphic at ∞ precisely when g extends holomorphically to $q_h = 0$; equivalently, when no negative powers occur. In other words, the Laurent series expansion in Equation (8) sums over nonnegative integers:

$$f(\tau) = \sum_{n=0}^{\infty} a_n q_h^n$$

Therefore, f admits a Fourier expansion. To deduce that $f|_k \alpha$ has a Fourier expansion, we note that $f|_k \alpha$ is holomorphic on $\mathbb{H}$ and weakly modular with respect to $\alpha^{-1} \Gamma \alpha$, which is a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$ by Lemma 2.5. Applying the preceding reasoning to $f|_k \alpha$, it admits a Laurent expansion and holomorphy at ∞ makes sense for $f|_k \alpha$ as well. Consequently, $f|_k \alpha$ also admits a Fourier expansion.

Remark 2.15. In practice, since $q \rightarrow 0$ precisely when $\text{Im}(\tau) \rightarrow \infty$, a weakly modular holomorphic function is holomorphic at ∞ whenever it remains bounded as $\text{Im}(\tau) \rightarrow \infty$.

The following result provides us with a sufficient analytic condition to check holomorphy at the cusps.

Proposition 2.16 (Proposition 1.2.4 in [DS05]). *Let k be an integer and $\Gamma \subseteq \text{SL}(2, \mathbb{Z})$ be a congruence subgroup. Let $f: \mathbb{H} \rightarrow \mathbb{C}$ be holomorphic and weakly modular of weight k . If the Fourier coefficients a_n satisfy $|a_n| \leq Cn^r$, for $n > 0$ and some positive constants C, r , then f is holomorphic at the cusps.*

Remark 2.17. In fact, the converse to this result holds as well. This is discussed on page 200 of [DS05].

Before going on to examples, we shall list some important properties. First, we note that $\mathcal{M}_k(\Gamma)$ forms a vector space over $\mathbb{C}$ with the $\mathbb{C}$ -action being defined as $(\lambda \cdot f)(\tau) := \lambda f(\tau)$, where $\lambda \in \mathbb{C}$, $f \in \mathcal{M}_k(\Gamma)$, and $\tau \in \mathbb{H}$.

Lemma 2.18. *Let $f \in \mathcal{M}_k(\Gamma)$, $g \in \mathcal{M}_\ell(\Gamma)$, where $\Gamma \subseteq \text{SL}(2, \mathbb{Z})$ is a congruence subgroup and k, ℓ are integers. Then, $fg \in \mathcal{M}_{k+\ell}(\Gamma)$.*

Proof. First note that since f, g are holomorphic on $\mathbb{H}$, their product is as well. Next, we prove that fg is weakly modular of weight $k + \ell$. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$. Then,

$$((fg)|_{k+\ell}\gamma)(\tau) = (c\tau + d)^{-(k+\ell)} f(\gamma\tau)g(\gamma\tau) = \left((c\tau + d)^{-k} f(\gamma\tau)\right) \left((c\tau + d)^{-\ell} g(\gamma\tau)\right) = (fg)(\tau).$$

The final condition to check is holomorphy at the cusps. For any $\alpha \in \text{SL}(2, \mathbb{Z})$, we have $(fg)|_{k+\ell}\alpha = (f|_k\alpha)(g|_\ell\alpha)$ by a similar calculation to the above. Now, since f and g are holomorphic at the cusps, both $f|_k\alpha$ and $g|_\ell\alpha$ are holomorphic at ∞ . Therefore, their product is holomorphic at ∞ . $\square$

By Lemma 2.18, it also follows that the space of modular forms $\mathcal{M}(\Gamma)$ has a graded ring structure indexed by the weights:

$$\mathcal{M}(\Gamma) = \bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\Gamma)$$

Since the space of cusp forms is a vector subspace of the space of modular forms, the analogous statement of Lemma 2.18 for cusp forms and the graded structure hold for the space of cusp forms as well.

2.3. Some Examples. We shall introduce some examples of modular forms on $\mathbb{H}$ now.

Proposition 2.19 (Exercise 1.1.4 in [DS05]). *Let k be an even integer greater than 2. The weight- k Eisenstein series is defined as*

$$G_k(\tau) = \sum_{d \neq 0} \frac{1}{d^k} + \sum_{c \neq 0} \sum_{d \in \mathbb{Z}} \frac{1}{(c\tau + d)^k}.$$

Then, $G_k(\tau)$ is a modular form of weight k with respect to $\text{SL}(2, \mathbb{Z})$.

Proof. We begin by showing holomorphy. We will show that $G_k(\tau)$ converges absolutely and uniformly on compact subsets of $\mathbb{H}$, which will imply holomorphy. We first note that the series

$$\sum_{(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}} (\sup\{|c|, |d|\})^{-k} \tag{9}$$

converges. This is because there are $8n$ lattice points which satisfy $\sup\{|c|, |d|\} = n$, so the sum above is equal to

$$8 \sum_{n=1}^{\infty} n^{1-k}.$$

This converges since $k > 2$. Next, consider the compact subset $\Omega \subseteq \mathbb{H}$ defined by

$$\Omega = \{\tau \in \mathbb{H}: |\text{Re}(\tau)| \leq A, \text{Im}(\tau) \geq B\}, \tag{10}$$

for some fixed positive reals A, B . We show that there is some constant $C > 0$ such that $|\tau + \delta| > C \sup\{1, |\delta|\}$ for all $\tau \in \Omega$ and $\delta \in \mathbb{R}$. Write $\tau = x + iy \in \Omega$. Then, for any $\delta \in \mathbb{R}$, it follows that $|\tau + \delta| \geq B$. We now case work. If $|\delta| \geq 2A$, then the triangle inequality implies $|x + \delta| \geq \frac{|\delta|}{2}$. Therefore,

$$|\tau + \delta| \geq \frac{|\delta|}{2}.$$

If $|\delta| < 2A$, then $\sup\{1, |\delta|\} \leq \sup\{1, 2A\}$, so

$$|\tau + \delta| \geq B \geq \frac{B}{\sup\{1, 2A\}} \sup\{1, |\delta|\}.$$

If we take $C = \min\left\{\frac{1}{2}, \frac{B}{\sup\{1, 2A\}}\right\} > 0$, we attain

$$|\tau + \delta| \geq C \sup\{1, |\delta|\} \quad (11)$$

for every $\tau \in \Omega$ and every $\delta \in \mathbb{R}$.

We now have the results needed to prove holomorphy. Consider the summands $(c\tau + d)^{-k}$ in the definition of $G_k(\tau)$. If $c \neq 0$, then Equation (11) with $\delta = \frac{d}{c}$ tells us that $|c\tau + d| \geq C \sup\{|c|, |d|\}$. If $c = 0$ and $d \neq 0$, then $|c\tau + d| = |d| = \sup\{|c|, |d|\}$. Replacing the constant C by $\min\{C, 1\}$ if necessary, we have that $|(c\tau + d)^{-k}| \leq C^{-k} (\sup\{|c|, |d|\})^{-k}$. We can bound $|G_k(\tau)|$ by the series in Equation (9), which converges, so the series defining G_k converges absolutely and uniformly on Ω . Noting that every compact subset of $\mathbb{H}$ is contained in a set of the form given in Ω , we can conclude that G_k converges uniformly on compact subsets of $\mathbb{H}$. It follows that G_k is holomorphic on $\mathbb{H}$.

Let $L' := \mathbb{Z}^2 \setminus \{(0, 0)\}$, so that $G_k(\tau) = \sum_{(c,d) \in L'} (c\tau + d)^{-k}$. To show that $G_k(\tau)$ is weakly modular of weight k , we first show that right multiplication by $\gamma \in \text{SL}(2, \mathbb{Z})$ is a bijection from L' to itself. Note that if $(c', d') \in L'$, then $(c', d')\gamma$ cannot equal $(0, 0)$ because γ is invertible, implying that $(c', d') = (0, 0)$. This is a contradiction. Furthermore, this is bijective because the inverse of right multiplication by γ is right multiplication by γ^{-1} . With the claim in hand, we calculate $G_k(\gamma \cdot \tau)$:

$$G_k(\gamma \cdot \tau) = \sum_{(m,n) \in L'} \frac{1}{(m(\gamma \cdot \tau) + n)^k} = (c\tau + d)^k \sum_{(m,n) \in L'} \frac{1}{(m(a\tau + b) + n(c\tau + d))^k}$$

Since $(m, n)\gamma = (ma + nc, mb + nd)$ and right multiplication by γ is a bijection, it follows that the summation at the very right is simply $G_k(\tau)$. Therefore, $G_k(\tau)$ is weakly modular of weight k .

The final check is holomorphy at infinity. Let Ω be the compact subset of $\mathbb{H}$ from Equation (10). In the above, we showed that there exists a constant $C > 0$ such that $|G_k(\tau)|$ is bounded above by the summation from Equation (9). Therefore, $G_k(\tau)$ is bounded on Ω . Now take $A = \frac{1}{2}$ for Ω . Then, for any $\tau = x + iy \in \mathbb{H}$, choose $n \in \mathbb{Z}$ such that $|x - n| \leq \frac{1}{2}$. Since $G_k(\tau)$ is weakly modular of weight k , we have $G_k(\tau) = G_k(\tau - n)$. If $y \geq B$, then $|\text{Re}(\tau - n)| \leq \frac{1}{2}$ and $\text{Im}(\tau - n) = y \geq B$, so $\tau - n \in \Omega$. It follows that there exists a constant $M > 0$ such that $|G_k(\tau - n)| \leq M$; hence, $|G_k(\tau)| \leq M$ if $\text{Im}(\tau) \geq B$. Thus, $G_k(\tau)$ is bounded as $\text{Im}(\tau) \rightarrow \infty$. $\square$

Example 2.20. We can normalize these Eisenstein series, in the sense that we make the constant part equal to 1. This is referred to as the *normalized Eisenstein series* $E_k(\tau) = \frac{G_k(\tau)}{2\zeta(k)}$. Note that this $2\zeta(k)$ comes from the Fourier expansion of G_k , which is calculated on Page 5 of [DS05]:

$$G_k(\tau) = 2\zeta(k) + 2 \frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n, \quad q = e^{2\pi i \tau}, \quad (12)$$

where $\sigma_{k-1}(n)$ is the arithmetic function

$$\sigma_{k-1}(n) = \sum_{\substack{m|n \\ m>0}} m^{k-1}.$$

Remark 2.21. These series $G_k(\tau)$ fit into a wider class of Eisenstein series, as discussed throughout Chapter 4 in [DS05].

Example 2.22. Using $g_2(\tau) = 60G_4(\tau)$, $g_3(\tau) = 140G_6(\tau)$, one can define the *discriminant function*

$$\Delta(\tau) = (g_2(\tau))^3 - 27(g_3(\tau))^2.$$

By the graded ring structure of the space of modular forms, $\Delta(\tau)$ is a cusp form of weight 12 over $\mathrm{SL}(2, \mathbb{Z})$. We note that this has the Fourier expansion

$$\Delta(\tau) = (2\pi)^{12} \sum_{n=0}^{\infty} \tau(n) q^n, \quad q = e^{2\pi i \tau},$$

as calculated in Exercise 1.1.7 of [DS05].

Example 2.23. The *Dedekind eta function* refers to the infinite product

$$\eta(\tau) = q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1 - q^n), \quad q = e^{2\pi i \tau}.$$

Then, we have that $\eta^{24} \in \mathcal{S}_{12}(\mathrm{SL}(2, \mathbb{Z}))$. Later, we shall see that $\mathcal{S}_{12}(\mathrm{SL}(2, \mathbb{Z}))$ has dimension 1, which would imply that $\Delta(\tau)$ and $(\eta(\tau))^{24}$ are scalar multiples of each other. In particular, the identity $\Delta = (2\pi)^{12} \eta^{24}$ holds. We leave the details of this example to page 20 in [DS05].

2.4. Dirichlet Character Decomposition. In this section, we shall discuss how the space of modular forms can decompose into subspaces indexed by a Dirichlet character. The reason we even consider Dirichlet characters is due to the group isomorphism $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$ shown in Proposition 2.4.

Definition 2.24. Let N be a positive integer. A *Dirichlet character modulo N* is a morphism of multiplicative groups $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$.

If we consider d to be a positive divisor of N , we can see that a Dirichlet character χ_d modulo d lifts to a Dirichlet character χ_N modulo N via

$$\chi_N(n \pmod{N}) := \chi_d(n \pmod{d}),$$

for all integers n coprime to N . The reverse process of descending a Dirichlet character modulo N to a Dirichlet character modulo d is not always possible. This motivates the following definition.

Definition 2.25. The *conductor* of a Dirichlet character χ_N modulo N is the smallest positive divisor d of N such that $\chi_N = \chi_d \circ \pi_{N,d}$, where $\pi_{N,d}: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow (\mathbb{Z}/d\mathbb{Z})^\times$ is the natural projection.

Example 2.26. The character $\mathbf{1}: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ sending every element of $(\mathbb{Z}/N\mathbb{Z})^\times$ to 1 is called the *trivial character* modulo N .

Recall that the isomorphism $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$ was constructed by taking a matrix in $\Gamma_0(N)$ and mapping it to its bottom right entry modulo N . This motivates the following definition in our context.

Definition 2.27. Let χ be a Dirichlet character modulo N . The χ -*eigenspace* of $\mathcal{M}_k(\Gamma_1(N))$ is defined as

$$\mathcal{M}_k(N, \chi) := \left\{ f \in \mathcal{M}_k(\Gamma_1(N)) : f|_k \gamma = \chi(d_\gamma) f, \text{ for all } \gamma = \begin{pmatrix} * & * \\ * & d_\gamma \end{pmatrix} \in \Gamma_0(N) \right\}$$

Remark 2.28 (Exercise 4.3.3 in [DS05]). Note that if we consider the trivial character modulo N in Definition 2.27, it is immediate by definition that $\mathcal{M}_k(N, \mathbf{1}) = \mathcal{M}_k(\Gamma_0(N))$. On the other hand, note that $\mathcal{M}_k(N, \chi) = 0$ unless $\chi(-1) = (-1)^k$. Let us prove this quickly here. Let $f \in \mathcal{M}_k(N, \chi)$ and consider $-I \in \Gamma_0(N)$. The condition on the Dirichlet character gives $(f|_k - I) = \chi(-1)f$, but the slash operator applied normally gives $f|_k(-I) = (-1)^{-k}f = (-1)^k f$. Therefore, if $\chi(-1) \neq (-1)^k$, then f is zero.

Proposition 2.29 (Exercise 4.3.3 in [DS05]). *We have the following decomposition:*

$$\mathcal{M}_k(\Gamma_1(N)) = \bigoplus_{\chi} \mathcal{M}_k(N, \chi)$$

Consequently, the analogous decomposition holds for cusp forms as well.

Proof. Through the isomorphism $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$, $(\mathbb{Z}/N\mathbb{Z})^\times$ acts on $\mathcal{M}_k(\Gamma_1(N))$ by the slash operator. The group $(\mathbb{Z}/N\mathbb{Z})^\times$ is finite and abelian, so every finite dimensional complex representation of $(\mathbb{Z}/N\mathbb{Z})^\times$ decomposes as a direct sum of its character eigenspaces.

For the statement of $\mathcal{S}_k(\Gamma_1(N))$, we note that $\mathcal{S}_k(\Gamma_1(N))$ is a vector subspace of $\mathcal{M}_k(\Gamma_1(N))$ and is stable under the action of $\Gamma_0(N)/\Gamma_1(N)$. Therefore, the same decomposition holds for the invariant subspace. $\square$

3. DIMENSION FORMULAE

The first major result which we shall work towards is the finite dimensionality of the space of modular forms. Not only do we show that it is finite dimensional, but we will find explicit formulae of the dimension using topological invariants. Note that we shall fix Γ to be a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$ throughout this section.

3.1. Modular Curves. We will first study the symmetry of the congruence subgroup on which the modular form is defined on. By interpreting the actions of congruence subgroups on the upper half plane as Riemann surfaces and compactifying them, this will allow us to use useful theorems, such as the Riemann-Roch Theorem.

3.1.1. Elliptic Points. We shall first determine the fixed points under the action of Γ .

Definition 3.1 (Definition 2.2.1 in [DS05]). For each $\tau \in \mathbb{H}$, let $\Gamma_\tau := \{\gamma \in \Gamma : \gamma \cdot \tau = \tau\}$ denote the stabilizer of τ under the action of Γ , which we refer to as the *isotropy subgroup* of τ . If $\{\pm I\} \subsetneq \Gamma_\tau \subset \Gamma_\tau\{\pm I\}$, we say that τ is an *elliptic point* of Γ . Here, $\Gamma_\tau\{\pm I\} := \Gamma_\tau \cup \{-\gamma : \gamma \in \Gamma_\tau\}$.

Let us now find the elliptic points of $\mathrm{SL}(2, \mathbb{Z})$.

Lemma 3.2 (Exercises 2.3.2 and 2.3.3 in [DS05]). *Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ be a non-scalar matrix and $\tau \in \mathbb{H}$. If $\gamma \cdot \tau = \tau$, then the order of γ as a matrix is 3, 4, or 6.*

Proof. $\gamma \cdot \tau = \tau$, so $a\tau + b = c\tau^2 + d\tau$. Noting that τ has positive imaginary part, we require the quadratic in τ to have negative discriminant. This provides the inequality $|a + d| < 2$. Since $\mathrm{tr}(\gamma) = a + d$, $\mathrm{tr}(\gamma) \in \{0, \pm 1\}$. Now, the characteristic polynomial of γ is $\gamma^2 - \mathrm{tr}(\gamma)\gamma + I = 0$, since $\det \gamma = 1$, so we now case work amongst possible traces of γ . If $\mathrm{tr}(\gamma) = -1$, the equation becomes $\gamma^2 + \gamma + I = 0$. Thus, $\gamma^3 = I$, after multiplying the equation by $(\gamma - I)$. Next, if $\mathrm{tr}(\gamma) = 0$, the characteristic polynomial is $\gamma^2 + I = 0$, so $\gamma^4 = I$. Finally, if $\mathrm{tr}(\gamma) = 1$, then the equation becomes $\gamma^2 - \gamma + I = 0$. In this case, multiplying by $(\gamma + I)$ gives $\gamma^3 = -I$, so $\gamma^6 = I$. This completes the proof. $\square$

Now that we know the possible orders of a matrix which fix a point, we can make them concrete, as listed in the following proposition.

Proposition 3.3 (Proposition 2.3.3 in [DS05]). *Let $\gamma \in \mathrm{SL}(2, \mathbb{Z})$.*

- *If γ has order 3, then γ is conjugate to $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}^{\pm 1}$ in $\mathrm{SL}(2, \mathbb{Z})$.*
- *If γ has order 4, then γ is conjugate to $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{\pm 1}$ in $\mathrm{SL}(2, \mathbb{Z})$.*
- *If γ has order 6, then γ is conjugate to $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}^{\pm 1}$ in $\mathrm{SL}(2, \mathbb{Z})$.*

Analyzing the fixed points of the matrices above in each case, we can deduce that the elliptic points for $\mathrm{SL}(2, \mathbb{Z})$ are precisely $i = \sqrt{-1}$ and $e^{\frac{2\pi i}{3}}$. In the case of congruence subgroups, the finiteness of $[\mathrm{SL}(2, \mathbb{Z}) : \Gamma]$ implies that Γ has finitely many elliptic points. In fact, Proposition 2.2.2 in [DS05] says more: the isotropy subgroup Γ_τ is finite cyclic, for every elliptic point τ .

3.1.2. *Modular Curves as Riemann Surfaces.* We shall now build some of the geometric intuition.

Definition 3.4. The *modular curve* $Y(\Gamma)$ is defined as the quotient space of orbits under Γ :

$$Y(\Gamma) := \Gamma \backslash \mathbb{H} = \{\Gamma\tau : \tau \in \mathbb{H}\}.$$

Recalling that Riemann surfaces are one-dimensional connected complex manifolds, we wish to view $Y(\Gamma)$ as a Riemann surface. For this, we first endow $Y(\Gamma)$ with the quotient topology under the natural surjection

$$\begin{aligned} \pi: \mathbb{H} &\longrightarrow Y(\Gamma) \\ \tau &\longmapsto \Gamma\tau, \end{aligned} \tag{13}$$

where $\mathbb{H}$ is endowed with the Euclidean topology. This makes π a continuous and open map, which shows that $Y(\Gamma)$ is second countable, Hausdorff, and connected. We must also show that $Y(\Gamma)$ admits a coordinate chart around every point of $Y(\Gamma)$; that is, an open neighborhood around every point that is homeomorphic to some open subset of $\mathbb{C}$.

The simple case is for the point $\pi(\tau)$, where τ is not an elliptic point. Here, we may take some small enough open neighborhood $U \subseteq \mathbb{H}$ containing τ which is homeomorphic to its image $\pi(U)$ in Y . Using a local inverse $\varphi: \pi(U) \rightarrow U$ to π , we can take $(\pi(U), \varphi)$ to be a chart in this case.

Next consider a point $\pi(\tau)$, where τ is an elliptic point. For this, we use Corollary 2.2.3 in [DS05], which states that there is some open neighborhood $U \subseteq \mathbb{H}$ containing τ such that U contains no other elliptic points besides τ . Note that this corollary follows from the more general result, which is Proposition 2.1.1 in [DS05]. It states that for any two not necessarily distinct points τ_1, τ_2 in $\mathbb{H}$, there exist open neighborhoods $U_1, U_2 \subseteq \mathbb{H}$ containing τ_1, τ_2 , respectively, such that for every $\gamma \in \text{SL}(2, \mathbb{Z})$ satisfying $\gamma(U_1) \cap U_2 \neq \emptyset$, $\gamma(\tau_1) = \tau_2$. Next, consider the straightening map δ defined on $\mathbb{C}$ by $\delta(z) := \begin{pmatrix} 1 & -\tau \\ 0 & 1 \end{pmatrix} \cdot z$ (as in Equation (7)). Note that this maps τ to 0. Moreover, if $z \in \mathbb{H}$, then $|\delta(z)| = \frac{|z-\tau|}{|z-\bar{\tau}|} < 1$, so δ maps $\mathbb{H}$ to the open unit disk around 0. We then compose this with the h -fold map $\rho(z) = z^h$, where

$$h = |\Gamma_\tau \{\pm I\} / \{\pm I\}| = \begin{cases} \frac{|\Gamma_\tau|}{2} & \text{if } -I \in \Gamma_\tau, \\ |\Gamma_\tau| & \text{if } -I \notin \Gamma_\tau. \end{cases} \tag{14}$$

Consider now the composition $\rho \circ \delta: U \rightarrow \mathbb{C}$. By the open mapping theorem, its image is open, which we call $V \subseteq \mathbb{C}$. In page 50 of [DS05], it is shown that restricting to U , we have that $\rho \circ \delta$ and π have the same fibers. In other words, for $\tau_1, \tau_2 \in U$, $\rho \circ \delta(\tau_1) = \rho \circ \delta(\tau_2)$ if and only if $\pi(\tau_1) = \pi(\tau_2)$. Consequently, it follows that there exists an injective map $\varphi: \pi(U) \rightarrow V$ defined by the relation $\varphi \circ \pi = \rho \circ \delta$. This is the chart map we wanted, so in the case of an elliptic point, we take our coordinate chart to be $(\pi(U), \varphi)$ as discussed above.

To conclude that $Y(\Gamma)$ is a Riemann surface, we need to show that the transition maps of the charts defined as above are holomorphic, but we leave this check to page 51 in [DS05]. Our next task is to compactify $Y(\Gamma)$ as a Riemann surface. This is done by adding in finitely many points, which we call cusps.

Definition 3.5. Let $\mathbb{H}^* := \mathbb{H} \cup \mathbb{Q} \cup \{\infty\}$. Then, we define

$$X(\Gamma) := \Gamma \backslash \mathbb{H}^* = Y(\Gamma) \cup \Gamma \backslash (\mathbb{Q} \cup \{\infty\}).$$

The orbits $\Gamma \backslash (\mathbb{Q} \cup \{\infty\})$ are called the *cusps* of $X(\Gamma)$.

Notation 3.6. We denote the compactified modular curves for $\Gamma_0(N)$, $\Gamma_1(N)$, and $\Gamma(N)$ as follows:

$$X_0(N) := \Gamma_0(N) \backslash \mathbb{H}, \quad X_1(N) := \Gamma_1(N) \backslash \mathbb{H}, \quad X(N) := \Gamma(N) \backslash \mathbb{H}$$

Remark 3.7. The fact that only finitely many points (or orbits) are added follows from the fact that the modular curve $X(1) := X(\text{SL}(2, \mathbb{Z}))$ has only one cusp, which is at ∞ . Then, since

any congruence subgroup has finite index in $\mathrm{SL}(2, \mathbb{Z})$, it follows that any modular curve $X(\Gamma)$ has finitely many cusp points.

We want this $X(\Gamma)$ to be our compactified Riemann surface, so we must put a topology on $X(\Gamma)$. For this, we first make $\mathbb{H}^*$ a topological space. In addition to the usual open sets in $\mathbb{H}$, we also declare sets of the form

$$\gamma(\{\tau \in \mathbb{H} : \mathrm{Im}(\tau) > M\} \cup \{\infty\}),$$

where $M > 0$ and $\gamma \in \mathrm{SL}(2, \mathbb{Z})$, to be open in $\mathbb{H}^*$. These new sets we have adjoined are tangent disks to the real axis for certain γ . Given this topology on $\mathbb{H}^*$, we endow $X(\Gamma)$ with the quotient topology via the extended map $\pi: \mathbb{H}^* \rightarrow X(\Gamma)$ from Equation (13). This makes $X(\Gamma)$ a compact Riemann surface, whose details are contained in Chapter 2.4 of [DS05].

3.1.3. The Genus. Since $X(\Gamma)$, as defined above, is a compact Riemann surface, it comes with a topological invariant—the genus, which we shall denote as g . Topologically speaking, the genus of a compact Riemann surface denotes the number of *holes* or *handles* on the object. This will become a key constant in the dimension formulae for the vector space of modular forms. Let us first recall the definition of the degree of a map in the context of Riemann surface theory.

Definition 3.8. Let $f: X \rightarrow Y$ be a nonconstant holomorphic map between compact Riemann surfaces. The *degree* of f is the positive integer d such that $|f^{-1}(y)| = d$ for all but finitely many $y \in Y$.

Remark 3.9. We note that the existence of such a value d is justified in pages 65–66 in [DS05].

As done in Section 3.1 of [DS05], one can apply the Riemann-Hurwitz relation to deduce the following result.

Theorem 3.10 (Theorem 3.1.1 in [DS05]). *Let Γ be a congruence subgroup, and $f: X(\Gamma) \rightarrow X(1)$ be the natural projection, with d the degree of f . Let ϵ_2 and ϵ_3 denote the number of elliptic points of period 2 and 3, respectively, and ϵ_∞ the number of cusps of $X(\Gamma)$. Then, the genus of $X(\Gamma)$ is*

$$g = 1 + \frac{d}{12} - \frac{\epsilon_2}{4} - \frac{\epsilon_3}{3} - \frac{\epsilon_\infty}{2}.$$

Example 3.11 (Exercise 3.1.4 in [DS05]; Genus of $X_0(p)$). Using Theorem 3.10, we compute the genus of the modular curve $X_0(p)$, where p is a prime number. First, we show that

$$\mathrm{SL}(2, \mathbb{Z}) = \bigsqcup_{j=0}^{p-1} \Gamma_0(p)\alpha_j \sqcup \Gamma_0(p)\alpha_\infty,$$

where $\alpha_\infty = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ and $\alpha_j = \begin{pmatrix} 1 & 0 \\ j & 1 \end{pmatrix}$, for $j = 0, \dots, p-1$. We do this by taking an arbitrary element $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ and focusing on whether p divides d or not. If $p \nmid d$, then we can calculate that $\gamma\alpha_j^{-1} = \begin{pmatrix} a-bj & b \\ c-dj & d \end{pmatrix}$. This is in $\Gamma_0(p)$ if $j \in \{0, \dots, p-1\}$ is the unique solution to $dj \equiv c \pmod{p}$. In this case, $\gamma \in \Gamma_0(p)\alpha_j$. If $p \mid d$, then $\gamma\alpha_\infty^{-1} = \begin{pmatrix} -b & a+b \\ -d & c+d \end{pmatrix}$, so $\gamma\alpha_\infty^{-1} \in \Gamma_0(p)$; hence, $\gamma \in \Gamma_0(p)\alpha_\infty$. This tells us that the degree (in the sense of maps between Riemann surfaces) of the natural projection $f: X_0(p) \rightarrow X(1)$ is $p+1$. Then, we note that $X_0(p)$ has precisely two cusps represented by the rationals $\frac{r_1}{s_1}$ and $\frac{r_2}{s_2}$, where $p \nmid s_1$ and $p \mid s_2$, respectively. Using Proposition 3.3, we can show that the number of elliptic points of period 2 and period 3 depend on the number of solutions to the congruences $x^2 + 1 \equiv 0 \pmod{p}$ and $x^2 - x + 1 \equiv 0 \pmod{p}$, respectively. Explicitly, for the equation $x^2 + 1 \equiv 0 \pmod{p}$, we can use the fact that for odd primes, -1 is a quadratic residue if and only if $p \equiv 1 \pmod{4}$, in which there are two solutions. Therefore, if $p \equiv 3 \pmod{4}$, then there are no solutions. If $p = 2$, then there is clearly only one solution given by $x = 1$. As for $x^2 - x + 1 \equiv 0 \pmod{p}$, we perform a similar analysis by considering when -3 is a quadratic residue modulo p . Then, in the notation of Theorem 3.10,

we have as follows:

$$\epsilon_2 = \begin{cases} 2 & \text{if } p \equiv 1 \pmod{4}, \\ 0 & \text{if } p \equiv 3 \pmod{4}, \\ 1 & \text{if } p = 2, \end{cases} \quad \epsilon_3 = \begin{cases} 2 & \text{if } p \equiv 1 \pmod{3}, \\ 0 & \text{if } p \equiv 2 \pmod{3}, \\ 1 & \text{if } p = 3. \end{cases}$$

Applying Theorem 3.10 in all possible cases, we can deduce that

$$g = \begin{cases} \lfloor \frac{p+1}{12} \rfloor - 1 & \text{if } p \equiv 1 \pmod{12}, \\ \lfloor \frac{p+1}{12} \rfloor & \text{otherwise.} \end{cases}$$

3.2. Classical Automorphic Forms. In this section, we introduce the notion of classical automorphic forms, which will be used to define meromorphic differentials in the coming section. We will also see that classical automorphic forms are far more general objects than modular forms. Moving forward, let us denote the Riemann sphere $\mathbb{C} \cup \{\infty\}$ by $\hat{\mathbb{C}}$.

Definition 3.12. Let $f: \mathbb{H} \rightarrow \hat{\mathbb{C}}$ be a meromorphic function. Assume also that f has no poles in the region $\{\tau \in \mathbb{H}: \text{Im}(\tau) > c\}$, where c is some positive real. Then, f admits a Laurent series on the corresponding punctured disk about 0,

$$f(\tau) = \sum_{n=-\infty}^{\infty} a_n q_h^n, \quad \text{if } \text{Im}(\tau) > c, \quad q_h = e^{2\pi i \tau / h},$$

where h is the smallest integer such that $(\frac{1}{0} \frac{h}{1}) \in \Gamma$. Then, we say that f is *meromorphic at ∞* if this series truncates from the left, starting at some integer m where $a_m \neq 0$, or if $f = 0$.

Definition 3.13 (Definition 3.2.1 in [DS05]). Let k be an integer. A function $f: \mathbb{H} \rightarrow \hat{\mathbb{C}}$ is a *classical automorphic form of weight k with respect to Γ* if f is meromorphic, weight- k invariant under Γ , and $f|_k \gamma$ is meromorphic at ∞ for all $\gamma \in \text{SL}(2, \mathbb{Z})$. Similar to Remark 2.12, we may refer to this last condition as *meromorphy at the cusps*.

Notation 3.14. We shall denote the space of classical automorphic forms of weight k with respect to Γ by $\mathcal{A}_k(\Gamma)$ and the space of classical automorphic forms with respect to Γ of all weights by $\mathcal{A}(\Gamma)$.

Analogous to the case for modular forms or cusp forms, we find that $\mathcal{A}(\Gamma)$ forms a graded ring $\mathcal{A}(\Gamma) = \bigoplus_{k \in \mathbb{Z}} \mathcal{A}_k(\Gamma)$. If the congruence subgroup is $\Gamma_1(N)$, then the analogous statement of Proposition 2.29 for automorphic forms holds as well. That is, if we define $\mathcal{A}_k(N, \chi)$ similarly to Definition 2.27, then we have the decomposition $\mathcal{A}_k(\Gamma_1(N)) = \bigoplus_{\chi} \mathcal{A}_k(N, \chi)$. Furthermore, $\mathcal{A}_k(\Gamma)$ is nonempty for all positive integers k , which is crucial in computing the dimension formulae, as we will need to work on a nonempty space in the first place. Example 3.16 shows this result is true for even k , but for odd k , this will require more Riemann surface theory, which we refer to pages 91-92 in [DS05] for.

Remark 3.15. $\mathcal{A}_0(\Gamma)$ can be naturally identified with $\mathbb{C}(X(\Gamma))$, the field of meromorphic functions on $X(\Gamma)$. This is because if $f \in \mathcal{A}_0(\Gamma)$, then f is Γ -invariant and meromorphic. If f is nonzero, then $\frac{1}{f} \in \mathcal{A}_0(\Gamma)$ as well. Particularly, $\frac{1}{f}$ is meromorphic on $\mathbb{H}$ and at the cusps, since the zeros of f become poles of $\frac{1}{f}$. Furthermore, for every $\gamma \in \Gamma$, Γ -invariance follows immediately from Γ -invariance for f , so $\frac{1}{f} \in \mathcal{A}_0(\Gamma)$. Conversely, every meromorphic function $g \in \mathbb{C}(X(\Gamma))$ can be identified with a weight-0 automorphic form with respect to Γ via the composition $g \circ \pi$, where $\pi: \mathbb{H}^* \rightarrow X(\Gamma)$ is the natural quotient map. We can restrict this statement further to say that a modular form of weight 0 corresponds to a holomorphic function on $X(\Gamma)$. If $f \in \mathcal{M}_0(\Gamma)$, then $f(\gamma\tau) = f(\tau)$, for all $\gamma \in \Gamma$, so it descends to a well-defined map on $Y(\Gamma)$. Holomorphy at the cusps then implies that f extends holomorphically to the compactification of $Y(\Gamma)$, which is exactly $X(\Gamma)$. Conversely, if g is a holomorphic function on $X(\Gamma)$, then $g \circ \pi$ is in $\mathcal{M}_0(\Gamma)$.

Example 3.16. We have that $\mathcal{A}_0(\text{SL}(2, \mathbb{Z})) = \mathcal{A}_0(\text{SL}_2(\mathbb{Z}))j$. Here, $j: \mathbb{H} \rightarrow \hat{\mathbb{C}}$ is the modular j -invariant defined by $j = 1728 \frac{g_2^3}{\Delta}$, where g_2 and Δ are from Example 2.22. j has only one pole

that is simple, which can be observed from its Laurent series

$$j(\tau) = \frac{1}{q} + \sum_{n=0}^{\infty} a_n q^n, \quad a_n \in \mathbb{Z}, \quad q = e^{2\pi i \tau}.$$

To attain the equality $\mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z})) = \mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z}))j$, the inclusion $\mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z})) \subseteq \mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z}))j$ is the one which requires some work. To show this, we must first view j as a meromorphic function on the compact Riemann surface $X(1)$. Since the one pole is simple, j has degree 1. Now, let f be a nonconstant meromorphic function on $X(1)$ with a finite number of zeroes and poles, which we denote by $z_1, \dots, z_m$ and $p_1, \dots, p_n$, respectively. We now construct a function $g(\tau) \in \mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z}))j$ which has the same zeroes and poles as f away from the cusp at infinity. As justified in page 73 of [DS05], the function

$$g(\tau) = \frac{\prod_{i=1}^m (j(\tau) - j(z_i))}{\prod_{i=1}^n (j(\tau) - j(p_i))}$$

works. By analyzing the vanishing order of f and g at the $\mathrm{SL}(2, \mathbb{Z})$ -orbit of ∞ , it follows that the rational function $\frac{f}{g}$ as a map of Riemann surfaces from $X(1) \rightarrow \hat{\mathbb{C}}$, is constant. Therefore, $f \in \mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z}))j$. The details can be found in page 73 of [DS05].

Remark 3.17. The significance of the modular invariant is that its derivative j' lies in $\mathcal{A}_2(\Gamma)$. This implies that $\mathcal{A}_{2k}(\Gamma)$ is nonzero, by taking integer powers of j' .

The above example told us that $\mathcal{A}_0(\mathrm{SL}(2, \mathbb{Z}))$ is generated by one classical automorphic form, j . More generally, we have the following result.

Proposition 3.18 (Exercise 3.2.4 in [DS05]). *If f is a nonzero element of $\mathcal{A}_k(\Gamma)$, then $\mathcal{A}_k(\Gamma) = \mathcal{A}_0(\Gamma)f$ (for k an integer).*

Proof. The inclusion $\mathcal{A}_k(\Gamma) \supseteq \mathcal{A}_0(\Gamma)f$ is immediate, so we will prove the reverse inclusion here. Let $g \in \mathcal{A}_k(\Gamma)$. Since f and g are meromorphic, the quotient $h = \frac{g}{f}$ is also meromorphic. h is also meromorphic at ∞ because the Laurent expansions for f and g both have finitely many negative powers, so h has finitely many negative powers as well. Finally, h is clearly weight-0 invariant under Γ , so $h \in \mathcal{A}_0(\Gamma)$. Since $g = hf$, we have shown that $\mathcal{A}_k(\Gamma) \subseteq \mathcal{A}_0(\Gamma)f$. $\square$

Since we want to apply Riemann-Roch, we need to introduce the order of a meromorphic function and its divisors.

Definition 3.19. Let $f: V \rightarrow \hat{\mathbb{C}}$ be meromorphic on an open subset $V \subseteq \mathbb{H}$. Then, its Laurent series expansion about $\tau \in \mathbb{H}$ is truncated from the left:

$$f(t) = \sum_{n=m}^{\infty} a_n (t - \tau)^n, \quad m \in \mathbb{Z}, \quad a_n \in \mathbb{C} \text{ with } a_m \neq 0,$$

for all t in some disk about τ . We define the *order of f at τ* , denoted $\nu_{\tau}(f)$, to be the integer m . Analogously, we define the *order of f at ∞* , denoted $\nu_{\infty}(f)$, to be the starting index m in Definition 3.12, with the exception that, when $f = 0$, we define $\nu_{\infty}(f) = \infty$.

We now want to determine the order of f at a point $\pi(\tau) \in X(\Gamma)$ using the order of f at τ . First, if $\pi(\tau) \in X(\Gamma)$ is not a cusp, then we attain that

$$\nu_{\pi(\tau)}(f) = \frac{\nu_{\tau}(f)}{h}, \tag{15}$$

where h is as in (14) when τ is viewed as an elliptic point. It follows that in this case the order is integral at non-elliptic points, but at elliptic points, $\nu_{\pi(\tau)}(f)$ can be half-integral or third-integral. We also wish to attain a similar relation for cusps, but we must first define the order of f at a cusp $s \in \mathbb{Q} \cup \{\infty\}$.

Definition 3.20. For any cusp $s \in \mathbb{Q} \cup \{\infty\}$, the *order of f at s* is defined as

$$\nu_s(f) := \nu_{\infty}(f|_k \gamma),$$

where $\gamma \cdot \infty = s$, for some $\gamma \in \mathrm{SL}(2, \mathbb{Z})$. Here, we have extended the action of γ on $\mathbb{H}$ as in Equation (7) to $\mathbb{H}^*$. In particular, if $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\gamma \cdot \infty = \begin{cases} \frac{a}{c} & \text{if } c \neq 0, \\ \infty & \text{if } c = 0, \end{cases}$$

Remark 3.21. We note that the order of f at a cusp s is independent of the choice of $\gamma \in \mathrm{SL}(2, \mathbb{Z})$ such that $\gamma(\infty) = s$. To see this, note that if $\gamma_1, \gamma_2 \in \mathrm{SL}(2, \mathbb{Z})$ such that $\gamma_1(\infty) = \gamma_2(\infty) = s$, then $\gamma_2^{-1}\gamma_1(\infty) = \infty$, so $\gamma_2^{-1}\gamma_1 = \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, for some $n \in \mathbb{Z}$. Now, $f|_k\gamma_1 = (f|_k\gamma_2)|_k(\gamma_2^{-1}\gamma_1)$, but the slash operator of $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ sends τ to $\tau + n$ and the slash operator of $\pm I$ just adds a sign of $(\pm 1)^k$. Neither of these affect the order of vanishing or pole at infinity, so this is well-defined.

Using this, we want to define the order of f at $\pi(s) \in X(\Gamma)$. As explained in pages 74-75 of [DS05], we can deduce that

$$\nu_{\pi(s)}(f) := \begin{cases} \frac{\nu_s(f)}{2} & \text{if } (\gamma^{-1}\Gamma\gamma)_\infty = \langle -(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}) \rangle \text{ and } k \text{ is odd,} \\ \nu_s(f) & \text{otherwise.} \end{cases} \quad (16)$$

Here, $(\gamma^{-1}\Gamma\gamma)_\infty$ is the isotropy subgroup of ∞ with respect to $\gamma^{-1}\Gamma\gamma$, as in Definition 3.1.

Definition 3.22. Let s be a cusp. We call $\pi(s)$ an *irregular cusp* if there exists some $f \in \mathcal{A}_k(\Gamma)$ such that $\nu_{\pi(s)}(f)$ is half-integral as in Equation (16). Otherwise, we call $\pi(s)$, or s , a *regular cusp*.

As for the details to derive (15) and (16), we shall refer to Chapter 3.2 in [DS05].

Remark 3.23. We note that Definition 3.22 is independent of the choice of the classical automorphic form f of weight k . To see this, let $s = \gamma(\infty)$ for some $\gamma \in \mathrm{SL}(2, \mathbb{Z})$ and set $g = f|_k\gamma$. Further, suppose that $(\gamma^{-1}\Gamma\gamma)_\infty = \langle -(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}) \rangle$ and that k is odd. Since f is automorphic, g is invariant under this generator. Hence $g|_k(-(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix})) = g$, so $g(\tau + h) = -g(\tau)$. Further, $g(\tau + 2h) = g(\tau)$ so g has a Fourier expansion

$$g(\tau) = \sum_n a_n e^{\frac{2\pi i n \tau}{2h}}.$$

Replacing τ by $\tau + h$ gives

$$g(\tau + h) = \sum_n (-1)^n a_n e^{\frac{2\pi i n \tau}{2h}}.$$

Since $g(\tau + h) = -g(\tau)$, we can compare the coefficients to deduce that $(-1)^n a_n = -a_n$. Therefore, $a_n = 0$ whenever n is even. Hence, every nonzero term in the Fourier expansion has odd exponent, so $\nu_s(f)$ is odd. In particular, we didn't use anything specific about f , so we have shown that Definition 3.22 is independent of f .

Example 3.24 (Exercise 3.2.5 in [DS05]). We will show that when k is odd, $\frac{1}{2}$ is an irregular cusp of $\Gamma_1(4)$. Firstly, $\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \cdot \infty = \frac{1}{2}$. Then, we claim that $(\gamma^{-1}\Gamma_1(4)\gamma)_\infty = \langle -(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}) \rangle$, where $\gamma := \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$. To show this, we calculate that $\pm\gamma\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\gamma^{-1} = \pm\begin{pmatrix} 1-2h & h \\ -4h & 1+2h \end{pmatrix}$. For this to be contained in $\Gamma_1(4)$, h must be odd. In particular, we may take $h = 1$ to get a group generator. We have shown what is desired.

Remark 3.25. In fact, when k is odd, it is shown in Chapter 3.8 of [DS05] that $\Gamma_1(4)$ is the only example of a congruence subgroup with an irregular cusp among the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, and $\Gamma(N)$.

3.3. Meromorphic Differentials. We will now introduce the notion of meromorphic differentials. This is the key object which relates our previous discussions of modular curves to modular forms. In particular, it will be a corollary of the Riemann-Roch theorem which will allow us to use the topological invariants previously defined.

Definition 3.26. Let $V \subseteq \mathbb{C}$ be open, and n be a nonnegative integer. The space of **meromorphic differentials on V of degree n** refers to the $\mathbb{C}$ -vector space $\Omega^{\otimes n}(V)$ defined as the set

$$\Omega^{\otimes n}(V) := \{f(q)(dq)^n : f \text{ is meromorphic on } V\}$$

with the natural definitions of addition and scalar multiplication as follows: let f, g be meromorphic on V and $c \in \mathbb{C}$.

$$\begin{aligned} f(q)(dq)^n + g(q)(dq)^n &:= (f + g)(q)(dq)^n \\ c(f(q)(dq)^n) &:= (cf)(q)(dq)^n \end{aligned}$$

Here, we note that q is the variable on V .

Remark 3.27. We note that any holomorphic map $\varphi: V_1 \rightarrow V_2$ between open sets of $\mathbb{C}$ induces a pullback map on meromorphic differentials on V_2 .

$$\begin{aligned} \varphi^*: \Omega^{\otimes n}(V_2) &\rightarrow \Omega^{\otimes n}(V_1) \\ f(q_2)(dq_2)^n &\mapsto f(\varphi(q_1))(\varphi'(q_1))^n(dq_1)^n \end{aligned}$$

Here, $\varphi'(q_1)$ denotes the derivative of φ with respect to the coordinate q_1 on V_1 .

Definition 3.28. Given a Riemann surface X , let $\{(U_j, \varphi_j: U_j \rightarrow V_j)\}$ be coordinate charts for X . A *meromorphic differential on X of degree n* is a collection of local meromorphic differentials of degree n ,

$$(\omega_j)_{j \in J} \in \prod_{j \in J} \Omega^{\otimes n}(V_j),$$

that is compatible in the sense that if we let $\varphi_{k,\ell}: \varphi_\ell(U_\ell \cap U_k) \rightarrow \varphi_k(U_\ell \cap U_k)$ be the transition map, then the following diagram commutes.

$$\begin{array}{ccc} & \prod_{j \in J} \Omega^{\otimes n}(V_j) & \\ \text{res}_{\varphi_k(U_\ell \cap U_k)} \circ p_k \swarrow & & \searrow \text{res}_{\varphi_\ell(U_\ell \cap U_k)} \circ p_\ell \\ \Omega^{\otimes n}(V_k) & \xrightarrow{\varphi_{k,\ell}^*} & \Omega^{\otimes n}(V_\ell) \end{array}$$

Here, $\text{res}_{\varphi_k(U_\ell \cap U_k)}$ denotes the restriction map to $\varphi_k(U_\ell \cap U_k)$, and p_k denotes the projection on the k -th component of $(\omega_j)_{j \in J}$. In other words, we wish for $\varphi_{k,\ell}^*(\omega_\ell|_{\varphi_\ell(U_\ell \cap U_k)}) = \omega_k|_{\varphi_k(U_\ell \cap U_k)}$ to hold. The meromorphic differentials of degree n on X form a vector space, denoted $\Omega^{\otimes n}(X)$.

Given this framework, we go back to our setting with the modular curves and the projection map $\pi: \mathbb{H} \rightarrow X(\Gamma)$. Let $\omega = (\omega_j)_{j \in J}$ be a meromorphic differential on $X(\Gamma)$. Then,

$$\pi^*(\omega)|_{U_j \cap \mathbb{H}} = \psi_j^*(\omega_j|_{\varphi_j(U_j \cap \mathbb{H})}).$$

Here, ψ_j^* is the pullback of the map $\psi_j: U_j \rightarrow V_j$ defined by $\psi_j = \varphi_j \circ \pi$, where φ_j is the local coordinate chart for $X(\Gamma)$ and U_j is a neighborhood of some point in $\mathbb{H}^*$. Analyzing the action of π^* on a meromorphic differential then leads to the following theorem, which is the key to relating classical automorphic forms to modular curves.

Theorem 3.29 (Theorem 3.3.1 in [DS05]). *Let $k \in \mathbb{Z}_{\geq 0}$ be even and Γ be a congruence subgroup. The map*

$$\begin{aligned} \omega: \mathcal{A}_k(\Gamma) &\rightarrow \Omega^{\otimes \frac{k}{2}}(X(\Gamma)) \\ f &\mapsto (\omega_j)_{j \in J} \text{ where } (\omega_j) \text{ pulls back under } \pi^* \text{ to } f(\tau)(d\tau)^{\frac{k}{2}} \in \Omega^{\otimes \frac{k}{2}}(\mathbb{H}) \end{aligned}$$

is an isomorphism of complex vector spaces.

Remark 3.30. The inverse map of the isomorphism of Theorem 3.29 is what one would expect. If $(\omega_j)_{j \in J} \in \Omega^{\otimes \frac{k}{2}}(X(\Gamma))$ and $\pi^*((\omega_j)) = f(\tau)(d\tau)^{\frac{k}{2}}$, then $(\omega_j) \mapsto f$.

Given the isomorphism in Theorem 3.29, in order to compute the dimensions of $\mathcal{M}_k(\Gamma)$ and $\mathcal{S}_k(\Gamma)$ (which are indeed subspaces of $\mathcal{A}_k(\Gamma)$), we can simply compute the dimension of the space of meromorphic differentials of degree n . This is an approachable computation, thanks to a corollary of the Riemann-Roch theorem. To state it, we need a few preliminary definitions.

Definition 3.31. Let X be a compact Riemann surface.

- A *divisor* D of X is a finite formal sum of integer multiples of points of X . That is, $D = \sum_{x \in X} n_x x$, $n_x \in \mathbb{Z}$ for all x and $n_x = 0$ for all but finitely many x . The *degree* of D , denoted $\deg(D)$, is the quantity $\deg(D) = \sum_{x \in X} n_x$.
- If $f: X \rightarrow \hat{\mathbb{C}}$ is meromorphic, we define $\operatorname{div}(f) := \sum_{x \in X} \nu_x(f)x$, where $\nu_x(f)$ denotes the order of f at x .

Definition 3.32. The *linear space* of any divisor D on a compact Riemann surface X is the following set:

$$L(D) = \{f \in \mathbb{C}(X) : f = 0 \text{ or } \operatorname{div}(f) + D \geq 0\}.$$

Remark 3.33. Note that in the definition of $L(D)$ above, the condition $\operatorname{div}(f) + D \geq 0$ is an inequality on the pointwise coefficients. If $D = \sum_{x \in X} n_x x$, then we are requiring $\nu_x(f) + n_x \geq 0$, for all $x \in X$.

Remark 3.34. Note that if we have nonzero $f, g \in L(D)$, then

$$\nu_x(f + g) \geq \min\{\nu_x(f), \nu_x(g)\} \geq -D$$

for all $x \in X$ because the order of a meromorphic function at x determines the smallest degree present in the Laurent expansion. This implies that $L(D)$ is a $\mathbb{C}$ -vector space because $f + g \in L(D)$. We denote the dimension of $L(D)$ by $\ell(D)$. Note that by Corollary 3.37, $\ell(D)$ is finite.

Definition 3.35. Let $\omega \in \Omega^{\otimes n}(X)$ be a nonzero meromorphic differential on X , where n is a positive integer. Then, ω has a local representation $\omega_x = f_x(q)(dq)^n$ about each point $x \in X$, where q is the local coordinate about x . We define $\operatorname{div}(\omega) := \sum_{x \in X} \nu_0(f_x)x$.

Note that because orders of meromorphic functions add under multiplication, we can deduce that $\operatorname{div}(\omega_1 \omega_2) = \operatorname{div}(\omega_1) + \operatorname{div}(\omega_2)$, where $\omega_1 \in \Omega^{\otimes n}(X)$ and $\omega_2 \in \Omega^{\otimes m}(X)$.

Definition 3.36. A *canonical divisor* on a compact Riemann surface X is a divisor of the form $\operatorname{div}(\lambda)$, where λ is a nonzero element of $\Omega^1(X)$.

Corollary 3.37 (Corollary 3.4.2 in [DS05]). *Let X be a compact Riemann surface of genus g , $\operatorname{div}(\lambda)$ be a canonical divisor on X , and D be a divisor. Then, the following hold:*

- (1) $\ell(\operatorname{div}(\lambda)) = g$,
- (2) $\deg(\operatorname{div}(\lambda)) = 2g - 2$,
- (3) If $\deg(D) < 0$, then $\ell(D) = 0$,
- (4) If $\deg(D) > 2g - 2$, then $\ell(D) = \deg(D) - g + 1$.

3.4. The Dimension Formulae. We are now ready to compute the dimension formulae. For this, we will need to split cases depending on the parity of the weight k , because when k is even, all cusps are regular and when k is odd, we may have irregular cusps. Before splitting cases, we will establish a framework which holds for both cases.

Let $f \in \mathcal{A}_k(\Gamma)$, which we shall assume to be nonzero for now. Then, we have that $\mathcal{A}_k(\Gamma) = \mathcal{A}_0(\Gamma)f$ by Proposition 3.18. Then, note that modular forms are automorphic forms which are holomorphic. Equivalently, modular forms have no poles, so if $f_0 \in \mathcal{A}_0(\Gamma)$, then $f_0 f \in \mathcal{M}_k(\Gamma)$ if and only if $\operatorname{div}(f_0 f) \geq 0$. Therefore,

$$\mathcal{M}_k(\Gamma) = \{f_0 f \in \mathcal{A}_k(\Gamma) : f_0 \in \mathbb{C}(X(\Gamma)) \text{ and } f_0 f = 0 \text{ or } \operatorname{div}(f_0 f) \geq 0\}.$$

In the language of the previous subsection, this space is isomorphic to $L(\operatorname{div}(f))$. Therefore,

$$\mathcal{M}_k(\Gamma) \cong \{f_0 \in \mathbb{C}(X(\Gamma)) : f_0 = 0 \text{ or } \operatorname{div}(f_0) + \operatorname{div}(f) \geq 0\} = L(\operatorname{div}(f)). \quad (17)$$

Now, if $\operatorname{div}(f)$ were integral, then we can apply Corollary 3.37, which solves our problems. However, $\operatorname{div}(f)$ is not necessarily integral. Since $f \in \mathcal{A}_k(\Gamma)$, there is a well-defined divisor $\sum_{x \in X(\Gamma)} \nu_x(f)x$ on $X(\Gamma)$, regardless of whether f is well-defined on the Riemann surface. However, Equation (15) and (16) tells us that $\nu_x(f)$ is not necessarily integral and may possibly be rational. Therefore, $\operatorname{div}(f)$ is possibly a $\mathbb{Q}$ -linear sum. On the other hand, $\operatorname{div}(f_0)$ is integral because f_0 is meromorphic on $X(\Gamma)$. Keeping this in mind, we notice the equivalence

$$\operatorname{div}(f_0) + \operatorname{div}(f) \geq 0 \iff \operatorname{div}(f_0) + \lfloor \operatorname{div}(f) \rfloor \geq 0,$$

where $\lfloor \operatorname{div}(f) \rfloor := \sum_{x \in X(\Gamma)} \lfloor \nu_x(f) \rfloor x$. This means that we may replace $\operatorname{div}(f)$ in Equation (17) with $\lfloor \operatorname{div}(f) \rfloor$, providing the isomorphism

$$\mathcal{M}_k(\Gamma) \cong L(\lfloor \operatorname{div}(f) \rfloor). \quad (18)$$

We can provide a similar isomorphism in the case of cusp forms; the correct analogue of Equation (17) for cusp forms is

$$\mathcal{S}_k(\Gamma) \cong \{f_0 \in \mathbb{C}(X(\Gamma)) : f_0 = 0 \text{ or } \operatorname{div}(f_0) + \operatorname{div}(f) - \sum_i x_i - \sum_i x'_i \geq 0\},$$

where x_i and x'_i denote the regular and irregular cusps, respectively. Note that the extra terms $-\sum_i x_i - \sum_i x'_i$ are present because any product $f_0 f$, for some $f_0 \in \mathbb{C}(X(\Gamma))$, must vanish at the cusps. Applying the same reasoning as above that $\operatorname{div}(f)$ is not necessarily a $\mathbb{Z}$ -linear sum, we deduce the following isomorphism:

$$\mathcal{S}_k(\Gamma) \cong L(\lfloor \operatorname{div}(f) \rfloor - \sum_i x_i - \sum_i x'_i) \quad (19)$$

In order to study $\lfloor \operatorname{div}(f) \rfloor$, we appeal to meromorphic differentials. Let $x_{2,i}, x_{3,i}, x_i, x'_i$ denote the period 2 elliptic points, period 3 elliptic points, regular cusps, and irregular cusps, respectively. We make the formal definition

$$\operatorname{div}(d\tau) := -\sum_i \frac{1}{2} x_{2,i} - \sum_i \frac{2}{3} x_{3,i} - \sum_i x_i - \sum_i x'_i, \quad (20)$$

as an element of the group of $\mathbb{Q}$ -linear divisors on $X(\Gamma)$. We are now ready to split between when k is odd or even.

3.4.1. *Even k .* Suppose k is even now. Then, there are no irregular cusps, so (19) reduces to

$$\mathcal{S}_k(\Gamma) \cong L(\lfloor \operatorname{div}(f) \rfloor - \sum_i x_i)$$

and (20) reduces to

$$\operatorname{div}(d\tau) = -\sum_i \frac{1}{2} x_{2,i} - \sum_i \frac{2}{3} x_{3,i} - \sum_i x_i.$$

Furthermore, we know that $\mathcal{A}_k(\Gamma)$ is nonempty due to Example 3.16. This leads to the following theorem.

Theorem 3.38 (Theorem 3.5.1 in [DS05]). *Let k be an even integer, Γ be a congruence subgroup of $\operatorname{SL}(2, \mathbb{Z})$, g the genus of $X(\Gamma)$, $\epsilon_2, \epsilon_3, \epsilon_\infty$ be the number of elliptic points of period 2, elliptic points of period 3, and cusps, respectively. Then,*

$$\dim(\mathcal{M}_k(\Gamma)) = \begin{cases} (k-1)(g-1) + \lfloor \frac{k}{4} \rfloor \epsilon_2 + \lfloor \frac{k}{3} \rfloor \epsilon_3 + \frac{k}{2} \epsilon_\infty & \text{if } k \geq 2, \\ 1 & \text{if } k = 0, \\ 0 & \text{if } k < 0, \end{cases}$$

and

$$\dim(\mathcal{S}_k(\Gamma)) = \begin{cases} \dim(\mathcal{M}_k(\Gamma)) - \epsilon_\infty & \text{if } k \geq 4, \\ g & \text{if } k = 2, \\ 0 & \text{if } k \leq 0. \end{cases}$$

Proof. First, let h be some weight-2 classical automorphic form. Then, $\omega(h) \in \Omega^1(X(\Gamma))$ by Theorem 3.29, where $\omega(h)$ pulls back under π^* to $h(\tau)(d\tau) \in \Omega^1(\mathbb{H})$. It follows by definition that $\text{div}(\omega(h))$ is a canonical divisor. Therefore, by Corollary 3.37, its degree is $2g - 2$. Furthermore, $(\omega(h))^{\frac{k}{2}} \in \Omega^{\otimes \frac{k}{2}}(X(\Gamma))$, so $\deg(\text{div}((\omega(h))^{\frac{k}{2}})) = k(g - 1)$. Now, note that any meromorphic function in $\mathbb{C}(X(\Gamma))$ has equally many zeroes and poles, counted with multiplicity; hence, the degree of the divisor is equal to 0. By Proposition 3.18, it follows that any nonzero differential in $\Omega^{\otimes \frac{k}{2}}(X(\Gamma))$ has a divisor of degree $k(g - 1)$.

Using this calculation, we compute $\deg(\lfloor \text{div}(f) \rfloor)$, where f is a weight- k classical automorphic form. First, note that we have the identity $\text{div}(\omega(f)) = \text{div}(f) + \frac{k}{2}\text{div}(d\tau)$. This is because Theorem 3.29 implies that $\pi^*(\omega(f)) = f(\tau)(d\tau)^{\frac{k}{2}}$ and divisors turn products into sums of divisors. If we then shift the term $\frac{k}{2}\text{div}(d\tau)$ to the left hand side and take the floor of the expression, we have that

$$\lfloor \text{div}(f) \rfloor = \text{div}(\omega(f)) + \sum_i \left\lfloor \frac{k}{4} \right\rfloor x_{2,i} + \sum_i \left\lfloor \frac{k}{3} \right\rfloor x_{3,i} + \sum_i \frac{k}{2} x_i. \quad (21)$$

Consequently,

$$\deg(\lfloor \text{div}(f) \rfloor) > 2g - 2, \quad k \geq 2. \quad (22)$$

Equation (22) satisfies the hypothesis of the fourth statement of Corollary 3.37, so we use Equation (21) and the isomorphism in Equation (18) to deduce that

$$\dim(\mathcal{M}_k(\Gamma)) = \ell(\lfloor \text{div}(f) \rfloor) = (k - 1)(g - 1) + \left\lfloor \frac{k}{4} \right\rfloor \epsilon_2 + \left\lfloor \frac{k}{3} \right\rfloor \epsilon_3 + \frac{k}{2} \epsilon_\infty,$$

for $k \geq 2$. For $k = 0$, we note that $f \in \mathcal{M}_0(\Gamma)$ corresponds to a meromorphic function on $X(\Gamma)$ with no poles, that is, a holomorphic map. By Remark 3.15, we may view f as a holomorphic map from $X(\Gamma)$ to $\hat{\mathbb{C}}$, that is, f is a holomorphic map between compact Riemann surfaces. Therefore, f is either surjective or constant, but the image of f is contained in $\mathbb{C}$, so f is not surjective. Hence, f must be constant, so $\mathcal{M}_0(\Gamma) = \mathbb{C}$, since the map is constant. For $k < 0$, we have that $\mathcal{M}_k(\Gamma) = 0$ because if f is such a modular form, then $f^{12}\Delta^{-k}$ is in $\mathcal{S}_0(\Gamma)$. However, $\mathcal{S}_0(\Gamma) = 0$ because cusp forms have constant term equal to zero.

The formula for $\dim(\mathcal{S}_k(\Gamma))$ then comes from (19) and seeing that $\ell(\lfloor \text{div}(f) - \sum_i x_i \rfloor) = \ell(\lfloor \text{div}(f) \rfloor) - \epsilon_\infty$ for $k \geq 4$. For $k = 2$, $\lfloor \text{div}(f) - \sum_i x_i \rfloor$ is a canonical divisor, so Corollary 3.37 says that its dimension is the genus. $\square$

3.4.2. Odd k . Now let k be odd. As mentioned, we do not yet know whether $\mathcal{A}_k(\Gamma)$ is nonempty, which is a major problem. However, these spaces are nonempty. Pages 91-92 of [DS05] construct a classical automorphic form of weight-1, after which we can take powers to deduce the existence of any odd-weight classical automorphic form. We also need to be careful about what cusps we have; particularly, we have both regular and irregular cusps, so we need (19) and (20) fully. However, before diving in, we shall make a quick reduction. If $-I \in \Gamma$, then the slash operator applied to $-I$ for any $f \in \mathcal{A}_k(\Gamma)$ tells us that f must be zero. Thus, if $-I \in \Gamma$, then $\dim(\mathcal{A}_k(\Gamma)) = 0$. To avoid this case, we can just assume that $-I \notin \Gamma$.

Now, let $\omega(f^2) \in \Omega^{\otimes k}(X(\Gamma))$ be the differential which pulls back to $f^2(d\tau)^k$ under the isomorphism in Theorem 3.29. Then, (20) allows us to conclude that

$$\text{div}(f) = \frac{1}{2}\text{div}(\omega(f^2)) + \sum_i \frac{k}{3} x_{3,i} + \sum_i \frac{k}{2} x_i + \sum_i \frac{k}{2} x'_i.$$

By methods similar to Theorem 3.38 and being careful about what occurs at the cusps, we reach the following theorem.

Theorem 3.39 (Theorem 3.6.1 in [DS05]). *Let k be an odd integer. Let $\Gamma \subseteq \text{SL}(2, \mathbb{Z})$ be a congruence subgroup. If $-I \in \Gamma$, then $\mathcal{M}_k(\Gamma) = \mathcal{S}_k(\Gamma) = 0$. If $-I \notin \Gamma$, let g be the genus of*

$X(\Gamma)$, ϵ_3 be the number of elliptic points with period 3, $\epsilon_\infty^{\text{reg}}$ be the number of regular cusps, and $\epsilon_\infty^{\text{irreg}}$ be the number of irregular cusps. Then, we attain

$$\dim(\mathcal{M}_k(\Gamma)) = \begin{cases} (k-1)(g-1) + \left\lfloor \frac{k}{3} \right\rfloor \epsilon_3 + \frac{k}{2} \epsilon_\infty^{\text{reg}} + \frac{k-1}{2} \epsilon_\infty^{\text{irreg}} & \text{if } k \geq 3, \\ 0 & \text{if } k < 0, \end{cases}$$

and

$$\dim(\mathcal{S}_k(\Gamma)) = \begin{cases} (k-1)(g-1) + \left\lfloor \frac{k}{3} \right\rfloor \epsilon_3 + \frac{k-2}{2} \epsilon_\infty^{\text{reg}} + \frac{k-1}{2} \epsilon_\infty^{\text{irreg}} & \text{if } k \geq 3, \\ 0 & \text{if } k < 0. \end{cases}$$

If $\epsilon_\infty^{\text{reg}} > 2g - 2$ then

$$\dim(\mathcal{M}_1(\Gamma)) = \frac{\epsilon_\infty^{\text{reg}}}{2}, \quad \dim(\mathcal{S}_1(\Gamma)) = 0.$$

If $\epsilon_\infty^{\text{reg}} \leq 2g - 2$ then

$$\dim(\mathcal{M}_1(\Gamma)) \geq \frac{\epsilon_\infty^{\text{reg}}}{2}, \quad \dim(\mathcal{S}_1(\Gamma)) = \dim(\mathcal{M}_1(\Gamma)) - \frac{\epsilon_\infty^{\text{reg}}}{2}.$$

Example 3.40. Consider the full level congruence subgroup $\text{SL}(2, \mathbb{Z})$ and let k be odd, then the dimension formulae imply that $\dim(\mathcal{M}_k(\text{SL}(2, \mathbb{Z}))) = \dim(\mathcal{S}_k(\text{SL}(2, \mathbb{Z}))) = 0$. This is because $-I \in \text{SL}(2, \mathbb{Z})$ and so $f = f|_k(-I) = (-1)^k f$, but k being odd implies that f is zero.

Example 3.41. In Section 7.5.1, we will be interested in the spaces $\mathcal{S}_2(\Gamma_0(11))$ and $\mathcal{S}_2(\Gamma_0(19))$. Using Exercise 3.11, the genus of $X_0(11)$ and $X_0(19)$ are both 1. Therefore, Theorem 3.38 implies that $\dim(\mathcal{S}_2(\Gamma_0(11))) = \dim(\mathcal{S}_2(\Gamma_0(19))) = 1$.

Remark 3.42 (Eisenstein Spaces). To conclude this section, we see that Theorems 3.38 and 3.39 imply that $\mathcal{M}_k(\Gamma)/\mathcal{S}_k(\Gamma)$ is generally nonzero. It is natural to study this space, and this quotient is naturally represented by the Eisenstein space. Formally, the Eisenstein space $\mathcal{E}_k(\Gamma)$ is defined as the orthogonal complement of $\mathcal{S}_k(\Gamma)$ in $\mathcal{M}_k(\Gamma)$ with respect to the Petersson inner product (Definition 4.17). In other words, we have the decomposition

$$\mathcal{M}_k(\Gamma) = \mathcal{S}_k(\Gamma) \oplus \mathcal{E}_k(\Gamma).$$

This is extensively studied in Chapters 4 and 5.11 of [DS05].

4. HECKE OPERATORS

In this section, we shall define and study Hecke operators. These operators are more general than the slash k operator, in the following sense. Let $\Gamma_1, \Gamma_2 \subseteq \text{SL}(2, \mathbb{Z})$ be distinct congruence subgroups. While the slash k operators send a form in $\mathcal{M}_k(\Gamma_1)$ to $\mathcal{M}_k(\Gamma_1)$, Hecke operators may possibly send a form in $\mathcal{M}_k(\Gamma_1)$ to $\mathcal{M}_k(\Gamma_2)$. We will touch upon one of the topics mentioned in the introduction, which is that there are modular forms that are eigenvectors for these operators. We will also study and provide results of the effect of the Hecke operators on the Fourier coefficients of a modular form.

4.1. Double Coset Operators. We begin by generalizing our notion of the slash operators to matrices in $\text{GL}(2, \mathbb{Q})^+$ (the subset of matrices of $\text{GL}(2, \mathbb{Q})$ which have positive determinant).

Definition 4.1. If $f: \mathbb{H} \rightarrow \mathbb{C}$, then for $\gamma \in \text{GL}(2, \mathbb{Q})^+$, we define

$$(f|_k \gamma)(\tau) = (\det \gamma)^{k-1} j(\gamma, \tau)^{-k} f(\gamma(\tau)), \quad \tau \in \mathbb{H}.$$

Remark 4.2. We note that the way we extend the slash operator from $\text{SL}(2, \mathbb{Z})$ to $\text{GL}(2, \mathbb{Q})^+$, or $\text{GL}(2, \mathbb{R})$, does not necessarily need to be as defined above. Usually, the exponent on the factor $\det \gamma$ is taken to be $\frac{k}{2}$ or $k - 1$ depending on the calculations one wishes to perform. For instance, in Section 6.3.3, we use the different normalization with the exponent of $\det \gamma$ as $\frac{k}{2}$.

We must also talk about double cosets in order to talk about Hecke operators, because Hecke operators are instances of double coset operators. Given $\alpha \in \mathrm{GL}(2, \mathbb{Q})^+$, and Γ_1, Γ_2 subgroups of $\mathrm{GL}(2, \mathbb{Q})^+$, a double coset in $\mathrm{GL}(2, \mathbb{Q})^+$ is the following set:

$$\Gamma_1 \alpha \Gamma_2 := \{\gamma_1 \alpha \gamma_2 : \gamma_1 \in \Gamma_1, \gamma_2 \in \Gamma_2\}.$$

We note that Γ_1 can act on $\Gamma_1 \alpha \Gamma_2$ from the left, splitting $\Gamma_1 \backslash \Gamma_1 \alpha \Gamma_2$ into a disjoint union of Γ_1 -orbits $\bigsqcup_j \Gamma_1 \beta_j$, for some choice of representatives β_j . In fact, Lemmas 5.1.1 and 5.1.2 in [DS05] imply that this disjoint union is finite.

Definition 4.3 (Definition 5.1.3 in [DS05]). Let Γ_1, Γ_2 be two congruence subgroups of $\mathrm{SL}(2, \mathbb{Z})$ and $\alpha \in \mathrm{GL}(2, \mathbb{Q})^+$. The *double coset operator of weight k for Γ_1 and Γ_2* is the operator $|_k \Gamma_1 \alpha \Gamma_2 : \mathcal{M}_k(\Gamma_1) \rightarrow \mathcal{M}_k(\Gamma_2)$ defined as

$$(f|_k \Gamma_1 \alpha \Gamma_2) = \sum_j f|_k \beta_j,$$

where $\Gamma_1 \alpha \Gamma_2 = \bigsqcup_j \Gamma_1 \beta_j$. Note that by the discussion above, this sum is finite.

Remark 4.4. Note that the double coset operator is a linear operator, since the slash k operator is a linear operator, as noted in Remark 2.7.

Proposition 4.5. *The operator $|_k \Gamma_1 \alpha \Gamma_2$ is well defined.*

Proof. Let us retain the notation in Definition 4.3. To show well-definedness, we need to show that this operator is independent of the coset representatives we choose and that the image is a modular form over Γ_2 . We prove the former first.

Let $\{\beta_j\}$ and $\{\beta'_j\}$ be distinct coset representatives, so $\Gamma_1 \alpha \Gamma_2 = \bigsqcup_j \Gamma_1 \beta_j = \bigsqcup_j \Gamma_1 \beta'_j$. After possibly re-indexing the set $\{\beta'_j\}$, we have that for each j , $\beta'_j = \gamma_j \beta_j$ for some $\gamma_j \in \Gamma_1$. Now, suppose that $f \in \mathcal{M}_k(\Gamma_1)$, so that $f|_k \gamma_j = f$ for all j . We have as follows:

$$\sum_j f|_k \beta'_j = \sum_j f|_k (\gamma_j \beta_j) = \sum_j f|_k \beta_j$$

Thus, the first claim is proven.

Next, we want to show that the image is a modular form over Γ_2 . Let $\gamma \in \Gamma_2$. We have as follows:

$$\left(\sum_j f|_k \beta_j \right) |_k \gamma = \sum_j f|_k (\beta_j \gamma)$$

Because $\Gamma_1 \alpha \Gamma_2 = \bigsqcup_j \Gamma_1 \beta_j$, right multiplication by γ permutes the right cosets $\Gamma_1 \beta_j$. Thus, $\beta_j \gamma = \delta_j \beta_{\sigma(j)}$, where σ is some permutation of the indices for $\{\beta_j\}$. Therefore,

$$\sum_j f|_k (\beta_j \gamma) = \sum_j f|_k (\delta_j \beta_{\sigma(j)}) = \sum_j f|_k \beta_{\sigma(j)} = \sum_j f|_k \beta_j.$$

Therefore, the image is invariant under the operator $|_k \gamma$, so it is a modular form over Γ_2 . To conclude, we note that the double coset operator is a finite sum of the slash k operators, which preserve the analytic conditions defining modular forms; that is, holomorphy and holomorphy at the cusps. Therefore, we've proven what is desired. $\square$

Definition 4.6. The *diamond operator* $\langle d \rangle : \mathcal{M}_k(\Gamma_1(N)) \rightarrow \mathcal{M}_k(\Gamma_1(N))$ is defined as

$$\langle d \rangle f = f|_k \Gamma_1(N) \alpha \Gamma_1(N),$$

where $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ with $\delta \equiv d \pmod{N}$.

Remark 4.7. We note that in Definition 4.6, $\langle d \rangle$ is independent of the choice of δ . We will prove this here. Suppose we have distinct matrices $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \alpha' = \begin{pmatrix} e & f \\ g & \delta' \end{pmatrix} \in \Gamma_0(N)$ such that $\delta \equiv \delta' \pmod{N}$. We shall show that $\Gamma_1(N) \alpha \Gamma_1(N) = \Gamma_1(N) \alpha' \Gamma_1(N)$. Since $c \equiv g \equiv 0 \pmod{N}$ and $\delta \equiv \delta' \pmod{N}$, it follows that $a \equiv e \pmod{N}$. Then, we can calculate that $\alpha' \alpha^{-1} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$

(mod N), so $\alpha'\alpha^{-1} \in \Gamma_1(N)$. Therefore, $\Gamma_1(N)\alpha\Gamma_1(N) = \Gamma_1(N)\alpha'\Gamma_1(N)$ as desired, so the diamond operators induced by α and α' are the same.

In general, if we have congruence subgroups Γ_1, Γ_2 of $\mathrm{SL}(2, \mathbb{Z})$ and $\alpha \in \mathrm{GL}(2, \mathbb{Q})^+$ such that $\alpha^{-1}\Gamma_1\alpha = \Gamma_2$, then the weight- k double coset operator $\Gamma_1\alpha\Gamma_2$ induces an isomorphism from $\mathcal{M}_k(\Gamma_1)$ to $\mathcal{M}_k(\Gamma_2)$. The reason for this is that $\Gamma_1\alpha\Gamma_2 = \Gamma_1\alpha$, so the double coset operator is just $f \mapsto f|_k\alpha$ with inverse $g \mapsto g|_k\alpha^{-1}$, for $f \in \mathcal{M}_k(\Gamma_1)$ and $g \in \mathcal{M}_k(\Gamma_2)$. In line with this, $\Gamma_1(N)$ is a normal subgroup of $\Gamma_0(N)$, so the diamond operator reduces to a much simpler form:

$$\langle d \rangle f = f|_k\alpha$$

where $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ with $\delta \equiv d \pmod{N}$.

An advantage of this description of the diamond operator is that it illuminates the fact

$$\mathcal{M}_k(N, \chi) = \{f \in \mathcal{M}_k(\Gamma_1(N)) : \langle d \rangle f = \chi(d)f, \text{ for all } d \in (\mathbb{Z}/N\mathbb{Z})^\times\}. \quad (23)$$

Here $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}$ is a Dirichlet character. This is because the value of χ depends on an element in $(\mathbb{Z}/N\mathbb{Z})^\times$, and the slash k operator of a matrix with bottom right entry congruent to that number associates with the diamond operator. Therefore, the diamond operator respects the decomposition as in Proposition 2.29.

Remark 4.8. Note that if $f \in \mathcal{M}_k(\Gamma_0(N))$, then the diamond operator acts trivially on f . This is because $\mathcal{M}_k(\Gamma_0(N)) = \mathcal{M}_k(N, \mathbf{1})$, so $\langle d \rangle f = f$ by Equation (23).

The second type of Hecke operator is as follows.

Definition 4.9. The operator $T_p : \mathcal{M}_k(\Gamma_1(N)) \rightarrow \mathcal{M}_k(\Gamma_1(N))$, where p is prime, is defined as

$$T_p f = f|_k\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N).$$

In contrast to the diamond operator, the operator T_p is slightly more difficult to understand because $\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N)$ has more than just one coset. In fact, an explicit representation depends on whether p divides N or not, meaning that the orbit (coset) representatives of $\Gamma_1(N) \backslash \Gamma_1(N)\alpha\Gamma_1(N)$ vary depending on this condition.

Proposition 4.10 (Proposition 5.2.1 in [DS05]). *Let N be a positive integer, p be a prime integer, $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, and $f \in \mathcal{M}_k(\Gamma_1(N))$. Then, the operator T_p defined above has the explicit formula given by*

$$T_p f = \begin{cases} \sum_{j=0}^{p-1} f|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} & \text{if } p \mid N, \\ \sum_{j=0}^{p-1} f|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} + f|_k \begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} & \text{if } p \nmid N, \text{ where } mp - nN = 1. \end{cases}$$

Proof. To prove this proposition, we just need to find the representatives of the orbits of $\Gamma_1(N) \backslash \Gamma_1(N)\alpha\Gamma_1(N)$. By Lemma 5.1.2 in [DS05], these can be found by left multiplying a set of coset representatives for $\Gamma' \backslash \Gamma_1(N)$ by α , where $\Gamma' = \alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N)$. Consider the groups

$$\Gamma^0(p) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & 0 \\ * & * \end{pmatrix} \pmod{p} \right\}$$

and

$$\Gamma_1^0(N, p) = \Gamma_1(N) \cap \Gamma^0(p).$$

We claim that $\Gamma' = \Gamma_1^0(N, p)$. Suppose $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma'$. Then, $\alpha^{-1}\gamma\alpha = \begin{pmatrix} a & b/p \\ pc & d \end{pmatrix}$. Hence, $\alpha^{-1}\gamma\alpha \in \Gamma_1(N)$ if and only if b is divisible by p . Therefore, $\gamma \in \Gamma_1^0(N, p)$. Conversely, if $\gamma \in \Gamma_1^0(N, p)$, then it is clearly in $\Gamma_1(N)$. Next, $\alpha\gamma\alpha^{-1} = \begin{pmatrix} a & b/p \\ pc & d \end{pmatrix}$ and we have $a \equiv d \equiv 1 \pmod{N}$ and $pc \equiv 0 \pmod{N}$ because $\gamma \in \Gamma_1(N)$. Therefore, $\gamma \in \alpha^{-1}\Gamma_1(N)\alpha$ as well.

Now that we have the claim in hand, we see that Γ' is just $\Gamma_1(N)$ but with the top right entries congruent to 0 modulo p . Natural candidates are $\gamma_{2,j} = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}$, for $0 \leq j \leq p-1$. Given $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$, γ is in the coset $\Gamma'\gamma_{2,j}$ if $\gamma\gamma_{2,j}^{-1} \in \Gamma'$. Using the claim, this occurs if and only if $\gamma\gamma_{2,j}^{-1} \in \Gamma_1(N) \cap \Gamma^0(p)$. While it is immediate that $\gamma\gamma_{2,j}^{-1} \in \Gamma_1(N)$, we need the top right entry of $\gamma\gamma_{2,j}^{-1}$ to be divisible by p . We can calculate that the top right entry of $\gamma\gamma_{2,j}^{-1}$ is $b - ja$, so we want $b \equiv ja \pmod{p}$. We also note that for $j, \ell \in \{0, \dots, p-1\}$, $\gamma_{2,j}\gamma_{2,\ell}^{-1} = \begin{pmatrix} 1 & j-\ell \\ 0 & 1 \end{pmatrix}$, so this is in Γ' if and only if $j = \ell$, proving disjointness.

If $p \mid N$, then $p \mid c$, so we cannot have $p \mid a$ again by the determinant. It follows that a is invertible modulo p , so $j \equiv ba^{-1} \pmod{p}$ works. Therefore, in this case, there is a unique $j \in \{0, \dots, p-1\}$ such that $j \equiv ba^{-1} \pmod{p}$. Consequently, the p representatives $\gamma_{2,j}$ for $0 \leq j \leq p-1$, suffice in this case.

Next, suppose $p \nmid N$. If $p \nmid a$, then the representatives $\gamma_{2,j}$ suffice, but it is also possible for $p \mid a$. In this case, there exists no solution j to the congruence $j \equiv ba^{-1} \pmod{p}$ because p would also divide b due to the congruence $b - ja \equiv 0 \pmod{p}$. In turn, this implies that $p \mid ad - bc = 1$, which is a contradiction. Therefore, when $p \nmid N$, $\{\gamma_{2,j} : j = 0, \dots, p-1\}$ fails to represent $\Gamma' \backslash \Gamma_1(N)$. To complete the set of coset representatives in this case, we require $\gamma_{2,\infty} := \begin{pmatrix} mp & n \\ N & 1 \end{pmatrix}$, where $mp - nN = 1$. To prove this is a complete set of representatives, we must first show that $\Gamma_1(N)\gamma_{2,\infty}$ is a coset, and that $\Gamma_1(N)\gamma_{2,\infty}$ is disjoint from $\Gamma_1(N)\gamma_{2,j}$. To prove the former, we will show that $\gamma\gamma_{2,\infty}^{-1} \in \Gamma'$, for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$ with $p \mid a$. We calculate that $\gamma\gamma_{2,\infty}^{-1} = \begin{pmatrix} a-bN & -an+bmN \\ c-dN & -cn+dmN \end{pmatrix}$. We will show here that $-cn + dmp \equiv 1 \pmod{N}$, since the congruences for the top left and bottom left entries are straightforward. In particular, $mp - nN = 1$, so $mp \equiv 1 \pmod{N}$. If we also use the congruences for c and d modulo N , we attain that $-cn + dmp \equiv 1 \pmod{N}$. Consequently, $\gamma\gamma_{2,\infty}^{-1} \in \Gamma_1(N)$. To show that this is contained in $\alpha^{-1}\Gamma_1(N)\alpha$, we show that $\alpha\gamma\gamma_{2,\infty}^{-1}\alpha^{-1}$ lies in $\Gamma_1(N)$. Calculating $\alpha\gamma\gamma_{2,\infty}^{-1}\alpha^{-1} = \begin{pmatrix} a-bN & p^{-1}(-an+bmN) \\ p(c-dN) & -cn+dmN \end{pmatrix}$, this is indeed true. Next, we show disjointness. For this, we shall show that $\gamma_{2,\infty}\gamma_{2,j}^{-1} \notin \Gamma'$. We calculate that $\gamma_{2,\infty}\gamma_{2,j}^{-1} = \begin{pmatrix} mp & j^{-1}mp+n \\ N & j^{-1}N+1 \end{pmatrix}$. This is not in Γ' , because the top right entry is not divisible by p . In particular, $p \nmid n$ because $mp - nN = 1$. Hence, we are done.

To conclude, we deduce that the orbit representatives for $\Gamma_1(N) \backslash \Gamma_1(N)\alpha\Gamma_1(N)$ are given by $\beta_j = \alpha\gamma_{2,j}$, which provides the formulae as desired. $\square$

These explicit formulae are especially useful in proving Proposition 5.2.2 in [DS05], which provides a concrete formula of the effect of T_p on the Fourier coefficients of a modular form. Explicitly, if $f \in \mathcal{M}_k(\Gamma_1(N))$ with Fourier expansion $f(\tau) = \sum_{n=0}^{\infty} a_n(f)q^n$, where $q = e^{2\pi i\tau}$ and $a_n(f)$ is the n -th Fourier coefficient of f , then

$$a_n(T_p f) = a_{np}(f) + \mathbf{1}(p)p^{k-1}a_{\frac{n}{p}}(\langle p \rangle f), \quad (24)$$

where $\mathbf{1}$ denotes the trivial character modulo N . Moreover, if χ is a nontrivial character modulo N and $f \in \mathcal{M}_k(N, \chi)$, then

$$a_n(T_p f) = a_{np}(f) + \chi(p)p^{k-1}a_{\frac{n}{p}}(f). \quad (25)$$

Moreover, these expansions help with proving the second and third relations for the Hecke operators in the following proposition.

Proposition 4.11 (Proposition 5.2.4 in [DS05]). *If $d, e \in (\mathbb{Z}/N\mathbb{Z})^\times$ and p, q are primes (which are not necessarily distinct), then*

- (1) $\langle d \rangle T_p = T_p \langle d \rangle$,
- (2) $\langle d \rangle \langle e \rangle = \langle e \rangle \langle d \rangle = \langle de \rangle$,

$$(3) \ T_p T_q = T_q T_p.$$

Proof. We shall prove just the first of these listed properties, and refer the remaining properties to Proposition 5.2.4 in [DS05]. Continuing in the notation of Proposition 4.10, let $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. Then, we note that if $\gamma \in \Gamma_1(N)$, it follows that $\gamma\alpha\gamma^{-1} \equiv \begin{pmatrix} 1 & * \\ 0 & p \end{pmatrix} \pmod{N}$. In other words, α and $\gamma\alpha\gamma^{-1}$ lie in the same $\Gamma_1(N)$ -coset as α . Using the fact that $\Gamma_1(N)$ is normal in $\Gamma_0(N)$, we can deduce that $\Gamma_1(N)\alpha\Gamma_1(N) = \gamma\Gamma_1(N)\alpha\Gamma_1(N)\gamma^{-1}$. Therefore, if we write $\Gamma_1(N)\alpha\Gamma_1(N) = \bigsqcup_j \Gamma_1(N)\beta_j$, then we deduce that

$$\bigsqcup_j \Gamma_1(N)\beta_j = \bigsqcup_j \Gamma_1(N)\gamma\beta_j\gamma^{-1}.$$

In other words,

$$\bigsqcup_j \Gamma_1(N)\beta_j\gamma = \bigsqcup_j \Gamma_1(N)\gamma\beta_j.$$

Therefore, for any $f \in \mathcal{M}_k(\Gamma_1(N))$ and $\gamma \in \Gamma_0(N)$ with lower right entry congruent to d modulo N ,

$$\langle d \rangle T_p f = \sum_j f|_k(\beta_j\gamma) = \sum_j f|_k(\gamma\beta_j) = T_p \langle d \rangle f.$$

□

Chapter 5.3 in [DS05] then generalizes these Hecke operators so that they are defined over all of the positive integers, not just primes.

Definition 4.12. Let n be a positive integer. We define the Hecke operators $\langle n \rangle$ and T_n as follows.

$$\langle n \rangle = \begin{cases} |_k \alpha & \text{if } (n, N) = 1, \\ 0 & \text{if } (n, N) > 1, \end{cases}$$

and

$$T_n = \begin{cases} 1 \text{ (the identity)} & \text{if } n = 1, \\ T_p & \text{if } n = p, \\ T_p T_{p^{r-1}} - p^{k-1} \langle p \rangle T_{p^{r-2}} & \text{if } n = p^r \text{ and } r \geq 2, \\ \prod T_{p_i^{r_i}} & \text{if } n = \prod p_i^{r_i}. \end{cases}$$

The commutativity of the Hecke operators and the recursive definition of T_{p^r} tell us $T_n T_m = T_m T_n$, for all n, m . Further, in the case that $(n, m) = 1$, we have that $T_{nm} = T_n T_m$. Moreover, Proposition 5.3.1 in [DS05] generalizes Equations (24) and (25) to the following statement. Writing

$$(T_n f)(\tau) = \sum_{m=0}^{\infty} a_m(T_n f) q^m,$$

we have

$$a_m(T_n f) = \sum_{d|(m,n)} d^{k-1} a_{mn/d^2}(\langle d \rangle f). \quad (26)$$

Analogously, if $f \in \mathcal{M}_k(N, \chi)$ then

$$a_m(T_n f) = \sum_{d|(m,n)} \chi(d) d^{k-1} a_{mn/d^2}(f). \quad (27)$$

4.2. Eigenforms. We now wish to define modular forms which act as common eigenvectors to the Hecke operators, which we refer to as *eigenforms*. To get such eigenforms, we do this via applying the spectral theorem from linear algebra by showing that the Hecke operators are normal operators; that is, that they commute with their adjoint operators. To even get started with this, we must first endow the space of cusp forms $\mathcal{S}_k(\Gamma_1(N))$ with an inner product.

4.2.1. The Petersson Inner Product.

Definition 4.13. The *hyperbolic measure* on $\mathbb{H}$ is defined as

$$d\mu(\tau) := \frac{dx dy}{y^2}, \quad \tau = x + iy \in \mathbb{H}.$$

As we ultimately wish to integrate over $\mathbb{H}^*$ (as defined in Definition 3.5), we must show that this is the correct measure to consider. For this, we should verify that the hyperbolic measure is invariant under the $\mathrm{SL}(2, \mathbb{Z})$ -action.

Lemma 4.14 (Exercise 5.1.4 (a) in [DS05]). *The hyperbolic measure is $\mathrm{SL}(2, \mathbb{Z})$ -invariant.*

Proof. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ and $\tau = x + iy$. Write $\gamma \cdot \tau = u + iv$. We have as follows:

$$d\mu(\gamma \cdot \tau) = \frac{du dv}{v^2} = \frac{\frac{dx dy}{|c\tau + d|^4}}{\left(\frac{y}{|c\tau + d|}\right)^2} = \frac{dx dy}{y^2}$$

Note that in the second equality, we used that $v = \frac{y}{|c\tau + d|}$ and the Jacobian $du dv = \frac{dx dy}{|c\tau + d|^4}$. $\square$

Recall from Section 3.1.2 that $X(1)$ had only one cusp at infinity. We shall denote by $\mathcal{D}^*$ as the fundamental domain of $\mathbb{H}^*$ under the $\mathrm{SL}(2, \mathbb{Z})$ -action:

$$\mathcal{D}^* := \{\tau \in \mathbb{H} : -\frac{1}{2} \leq \mathrm{Re} \tau \leq \frac{1}{2}, |\tau| \geq 1\} \cup \{\infty\}.$$

Lemma 4.15 (Exercise 5.4.1 (b) in [DS05]). *Let $\alpha \in \mathrm{SL}(2, \mathbb{Z})$. Then, the integral*

$$\int_{\mathcal{D}^*} \varphi(\alpha(\tau)) d\mu(\tau)$$

converges for continuous bounded functions $\varphi: \mathbb{H} \rightarrow \mathbb{C}$.

Proof. Since $\varphi(\alpha(\tau))$ is continuous and bounded, it suffices to show that $\int_{\mathcal{D}^*} d\mu(\tau) < \infty$. To show this, we can consider the following superset of $\mathcal{D}^*$:

$$\mathcal{D}^* \subseteq S := \{\tau \in \mathbb{H} : -\frac{1}{2} \leq \mathrm{Re} \tau \leq \frac{1}{2}, \mathrm{Im} \tau \geq \frac{\sqrt{3}}{2}\}$$

Using this, we calculate the following:

$$\int_S d\mu(\tau) = \int_{x=-\frac{1}{2}}^{\frac{1}{2}} \int_{y=\frac{\sqrt{3}}{2}}^{\infty} \frac{dx dy}{y^2} = \frac{2}{\sqrt{3}} < \infty.$$

Therefore, if we suppose that $|\varphi(\alpha(\tau))| \leq M$ for some real constant M , the result follows by the calculation below.

$$\left| \int_{\mathcal{D}^*} \varphi(\alpha(\tau)) d\mu(\tau) \right| \leq \int_{\mathcal{D}^*} |\varphi(\alpha(\tau))| d\mu(\tau) \leq M \int_{\mathcal{D}^*} d\mu(\tau) < \infty$$

$\square$

Given a congruence subgroup Γ , we know that we can write the coset space $\Gamma\{\pm I\} \backslash \mathrm{SL}(2, \mathbb{Z})$ as a disjoint union using $\mathrm{SL}(2, \mathbb{Z}) = \bigsqcup_j \Gamma\{\pm I\}\alpha_j$, where $\alpha_j \in \mathrm{SL}(2, \mathbb{Z})$ are representatives. Since the union $\bigsqcup_j \alpha_j(\mathcal{D}^*)$ represents the modular curve $X(\Gamma)$, we can therefore define an integral over $X(\Gamma)$. Explicitly, we define

$$\int_{X(\Gamma)} \varphi(\tau) d\mu(\tau) := \sum_j \int_{\mathcal{D}^*} \varphi(\alpha_j(\tau)) d\mu(\tau).$$

Definition 4.16. We define the *volume* of the modular curve $X(\Gamma)$ as $V_\Gamma := \int_{X(\Gamma)} d\mu(\tau)$.

The point of introducing this was to get the following definition.

Definition 4.17 (Definition 5.4.1 in [DS05]). Let Γ be a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$. The *Petersson inner product* $\langle \cdot, \cdot \rangle: \mathcal{S}_k(\Gamma) \times \mathcal{S}_k(\Gamma) \rightarrow \mathbb{C}$ is defined by

$$\langle f, g \rangle_\Gamma = \frac{1}{V_\Gamma} \int_{X(\Gamma)} f(\tau) \overline{g(\tau)} (\mathrm{Im}(\tau))^k d\mu(\tau).$$

Proposition 4.18. *The Petersson inner product is a well-defined inner product on cusp forms. Moreover, the Petersson inner product converges as long as one of its entries is a cusp form.*

Proof. We refer to page 182 in [DS05]. $\square$

The key lemma in finding the adjoints of Hecke operators is the following result.

Lemma 4.19 (Proposition 5.5.2 in [DS05]). *Let Γ be a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$, and $\alpha, \alpha' \in \mathrm{GL}(2, \mathbb{Q})^+$ such that $\alpha' = \det(\alpha)\alpha^{-1}$. Then, the following holds.*

(1) *If $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}(2, \mathbb{Z})$ then for all $f \in S_k(\Gamma)$ and $g \in S_k(\alpha^{-1}\Gamma\alpha)$,*

$$\langle f|_k\alpha, g \rangle_{\alpha^{-1}\Gamma\alpha} = \langle f, g|_k\alpha' \rangle_{\Gamma}.$$

(2) *For all $f, g \in S_k(\Gamma)$,*

$$\langle f|_k\Gamma\alpha\Gamma, g \rangle_{\Gamma} = \langle f, g|_k\Gamma\alpha'\Gamma \rangle_{\Gamma}.$$

Using this lemma, we can find the adjoints of both Hecke operators for primes p such that $p \nmid N$. Since $\Gamma_1(N)$ is normal in $\Gamma_0(N)$, it is immediate that the adjoint $\langle p \rangle^* = \langle p \rangle^{-1}$ from the first part of Lemma 4.19. For T_p^* , we use the second part of Lemma 4.19 and analyze the action of $[\Gamma_1(N) \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma_1(N)]$. We can write this double coset space as

$$\Gamma_1(N) \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma_1(N) = \Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N) \begin{pmatrix} p & n \\ N & m \end{pmatrix},$$

where p, n, N, m are integers which satisfy $pm - nN = 1$. We note then that $m \equiv p^{-1} \pmod{N}$. Since there is the double coset space for T_p appearing and the matrix on the right is the matrix corresponding to the action of $\langle p \rangle^{-1}$, we attain that $T_p^* = \langle p \rangle^{-1}T_p$.

Theorem 4.20 (Theorem 5.5.4 in [DS05]). *The space $\mathcal{S}_k(\Gamma_1(N))$ has an orthogonal basis of simultaneous eigenvectors of modular forms for the Hecke operators $\{\langle n \rangle, T_n : (n, N) = 1\}$. We call such a nonzero common eigenvector (which is a modular form) an eigenform.*

Proof. From the discussion preceding this theorem, we explicitly know the adjoints of the Hecke operators for primes p . In particular, we see that each Hecke operator clearly commutes with its adjoint. Furthermore, we get that $\langle n \rangle$ and T_n commute with their adjoints, as long as n and N are coprime. Therefore, we may apply the spectral theorem to deduce that there's an orthogonal basis of simultaneous eigenvectors for these operators. $\square$

Definition 4.21. Given a Hecke eigenform $f(\tau) = \sum_{n=0}^{\infty} a_n(f)q^n$, we say that it is *normalized* if $a_1(f) = 1$.

Before continuing, we shall briefly discuss the relation of the Fourier coefficients of f and the eigenvalues of f with respect to the Hecke operators. Let $f \in \mathcal{S}_k(\Gamma_1(N))$ be an eigenform for all T_n and $\langle n \rangle$, where $(n, N) = 1$. Then, suppose we have $T_n f = c_n f$ and $\langle n \rangle f = d_n f$ for all n , for some $c_n, d_n \in \mathbb{C}$. It is clear that

$$a_1(T_n f) = c_n \cdot a_1(f) \tag{28}$$

by Equation (27). We can also check that the map $n \mapsto d_n$ defines a Dirichlet character $\chi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$. This is because $\langle 1 \rangle f = f$ and the second part of Proposition 4.11 states that $\langle de \rangle = \langle d \rangle \langle e \rangle$. This implies that f lies in the χ -eigenspace $\mathcal{S}_k(N, \chi)$. Via (27), we attain that

$$a_1(T_n f) = a_n(f), \tag{29}$$

for all positive integers n . If we equate Equations (28) and (29), we get that

$$a_n(f) = c_n \cdot a_1(f). \tag{30}$$

In particular, if f is normalized, we get that $a_n(f)$ is the eigenvalue of the Hecke operator T_n . We conclude this subsection by citing Proposition 5.8.5 in [DS05], which provides an equivalent characterization of when $f \in \mathcal{M}_k(N, \chi)$ is a normalized eigenform.

Remark 4.22. Note that Equation (30) implies that $a_n(f) = 0$ for all positive integers n if $a_1(f) = 0$, which contradicts f being nonzero. Thus, eigenforms have $a_1(f) \neq 0$.

Theorem 4.23 (Proposition 5.8.5 in [DS05]). *Let $f \in \mathcal{M}_k(N, \chi)$. Then f is a normalized eigenform if and only if its Fourier coefficients $a_n(f)$ satisfy the following conditions:*

- $a_1(f) = 1$
- $a_{p^r}(f) = a_p(f)a_{p^{r-1}}(f) - \chi(p)p^{k-1}a_{p^{r-2}}(f)$ for all p prime and $r \geq 2$
- $a_{mn}(f) = a_m(f)a_n(f)$ when $(m, n) = 1$

4.3. Oldforms and Newforms. In this subsection, we focus on how we can move from one level to another. A natural first way to relate modular forms of different levels is through the canonical inclusion $\mathcal{M}_k(\Gamma_1(M)) \hookrightarrow \mathcal{M}_k(\Gamma_1(N))$, where $M \mid N$. Note that we have this inclusion because if $f \in \mathcal{M}_k(\Gamma_1(M))$, then $f|_k \gamma = f$ for all $\gamma \in \Gamma_1(M)$. Since $\Gamma_1(N) \subseteq \Gamma_1(M)$, the claim holds. Another way to move between levels is motivated by the following lemma:

Lemma 4.24 (Exercise 1.2.11 (d) in [DS05]). *Let N, d be positive integers and $f \in \mathcal{M}_k(\Gamma_0(N))$. If $g(\tau) := f(d\tau)$ then $g \in \mathcal{M}_k(\Gamma_0(dN))$. Further, if $f \in \mathcal{M}_k(\Gamma_1(N))$ then $g \in \mathcal{M}_k(\Gamma_1(dN))$. The analogous statements hold for cusp forms as well.*

Proof. It suffices to check the invariance property. We shall show this for the case of $\Gamma_0(N)$ since the similar calculations hold for $\Gamma_1(N)$. Let $\alpha_d = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix}$. First, suppose $f \in \mathcal{M}_k(\Gamma_0(N))$. Let $\gamma = \begin{pmatrix} a & b \\ c & e \end{pmatrix} \in \Gamma_0(dN)$. Then $c \equiv 0 \pmod{dN}$. Then, $\alpha_d \gamma \alpha_d^{-1} = \begin{pmatrix} a & db \\ c/d & e \end{pmatrix}$. Since $c \equiv 0 \pmod{dN}$, we have $\frac{c}{d} \equiv 0 \pmod{N}$, so $\alpha_d \gamma \alpha_d^{-1} \in \Gamma_0(N)$. Therefore,

$$g|_k \gamma = f|_k \alpha_d \gamma = f|_k (\alpha_d \gamma \alpha_d^{-1}) \alpha_d = f|_k \alpha_d = g,$$

so $g \in \mathcal{M}_k(\Gamma_0(dN))$.

Finally, the same argument holds for cusp forms. Indeed, if f vanishes at every cusp, then $f|_k \alpha_d$ also vanishes at every cusp, since α_d sends cusps to cusps. Therefore, $f \in S_k(\Gamma_0(N))$ implies that $g \in S_k(\Gamma_0(dN))$. $\square$

Motivated by this lemma and the initial discussion, it is natural to consider cusp forms coming from lower levels and their complements.

Definition 4.25. Let $d \mid N$. Let $\iota_d: (\mathcal{S}_k(\Gamma_1(\frac{N}{d})))^2 \rightarrow \mathcal{S}_k(\Gamma_1(N))$ be the map $(f, g) \mapsto f + g|_k \alpha_d$, where $\alpha_d = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix}$.

- The *oldforms at level N* form the subspace of $\mathcal{S}_k(\Gamma_1(N))$

$$\mathcal{S}_k(\Gamma_1(N))^{\text{old}} := \sum_{\text{prime } p \mid N} \text{Im}(\iota_p).$$

Intuitively, oldforms are just forms which arise from a lower level.

- Define the space $\mathcal{S}_k(\Gamma_1(N))^{\text{new}} := (\mathcal{S}_k(\Gamma_1(N))^{\text{old}})^\perp$; that is, the orthogonal complement of oldforms at level N with respect to the Petersson inner product. A *newform* is then a normalized eigenform in $\mathcal{S}_k(\Gamma_1(N))^{\text{new}}$. Intuitively, newforms are forms that are contained in the new subspace and not contained in lower levels.

In Chapter 5.6 of [DS05], it is shown that the spaces of oldforms and newforms at level N are stable under the Hecke actions T_n and $\langle n \rangle$, for all positive integers n . Moreover, this implies that these two spaces have orthogonal bases of eigenforms for the Hecke operators $T_n, \langle n \rangle$, for $(n, N) = 1$. In the case of newforms, we can say more:

Theorem 4.26 (Theorem 5.8.2 of [DS05]). *Let $f \in \mathcal{S}_k(\Gamma_1(N))^{\text{new}}$ be an eigenform for the Hecke operators $T_n, \langle n \rangle$, for $(n, N) = 1$.*

- (1) *f is an eigenform for T_n and $\langle n \rangle$, for all positive integers n .*
- (2) *If g is another function satisfying the same conditions as f with the same eigenvalues for the T_n -Hecke operator, then $g = c \cdot f$, for some constant c .*

Proof. This essentially follows from the lemma below. For a proof, check pages 195-196 of [DS05]. $\square$

Lemma 4.27 (Theorem 5.7.1 in [DS05]). *If $f \in S_k(\Gamma_1(N))$ has the Fourier expansion $f(\tau) = \sum a_n(f)q^n$ with $a_n(f) = 0$ for all $(n, N) = 1$, then f takes the form*

$$f = \sum_{p|N} \iota_p f_p,$$

with each $f_p \in S_k(\Gamma_1(\frac{N}{p}))$. Here, ι_p is the map used in Definition 4.25.

Remark 4.28. The second statement of Theorem 4.26 is of great significance, and is called the *Multiplicity One* property of newforms. That is, it says that each newform spans a one-dimensional eigenspace for the T_n operators. By a deeper result called the *Strong Multiplicity One* (Theorem 4.6.19 in [Miy06]), it can be shown that the set

$$\{f(n\tau) : f \text{ a newform of level } M \text{ and } nM \mid N\}$$

acts as a basis for $S_k(\Gamma_1(N))$. This is Theorem 5.8.3 in [DS05] or Corollary 4.6.20 in [Miy06].

5. THE ADELIC FRAMEWORK

We will now shift gears and aim to generalize the notion of a modular form, or automorphic form, over the general linear group of the so-called *adele* ring. This group makes no sense at the moment, so our main objective of this section is to define this group and understand the topology on it. Before even discussing this group, we must talk about the adele ring and its topology, which is done after discussing the basic notions of an absolute value and the topology it induces.

Before jumping into absolute values, I will motivate the strength of the adelic viewpoint by briefly introducing some of its properties. As we will see, $\mathbb{Q}$ can be completed at many places and it is often of interest to number theorists what occurs at these completions. Instead of switching between these completions, the adele ring entails all the local data into one ring. Moreover, we will see that the adele ring contains the global data of $\mathbb{Q}$ inside it as well. The adele ring also possesses various topological properties that widen the scope of analytic and topological tools at our disposal.

5.1. Absolute Values. We shall first discuss about absolute values. Before I begin, I will highlight that most of the constructions presented here can be done more generally for number fields and function fields, but I have only stated them for $\mathbb{Q}$ because that is all we require.

Definition 5.1 (Definition 1.2 in [Sut21]). An *absolute value* on $\mathbb{Q}$ is a function $|\cdot| : \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$ subject to the following conditions: for all $x, y \in \mathbb{Q}$,

- $|x| = 0$ if and only if $x = 0$,
- $|xy| = |x||y|$,
- $|x + y| \leq |x| + |y|$ (Triangle Inequality).

If the stronger condition $|x + y| \leq \max\{|x|, |y|\}$ holds instead of the triangle inequality, then $|\cdot|$ is called *non-Archimedean*. If otherwise, we call the absolute value *Archimedean*.

Remark 5.2. Note that if $|\cdot|$ is non-Archimedean, $|x + y| \leq \max\{|x|, |y|\} \leq |x| + |y|$, so it is a stronger condition.

Example 5.3. If $|x| = 1$ for all nonzero $x \in \mathbb{Q}$ and $|0| = 0$, then this is an absolute value; it is called the *trivial absolute value*.

Example 5.4. The usual absolute value

$$|x|_\infty = \begin{cases} 0 & \text{if } x = 0, \\ x & \text{if } x > 0, \\ -x & \text{if } x < 0 \end{cases}$$

is an absolute value. We shall refer to this as the *infinity norm* or *absolute value at infinity*.

Example 5.5. Let p be a prime. The p -adic absolute value $|\cdot|_p$ is defined as $|x|_p = p^{-r}$, where $x = p^r \frac{a}{b}$, with $a, b \in \mathbb{Z}$, $p \nmid a, b$, and $b \neq 0$. Let us verify that it is not just an Archimedean absolute value, but a non-Archimedean absolute value. It is clear that $|x|_p = 0$ if and only if $x = 0$ and that $|xy| = |x||y|$ by unique factorization of $\mathbb{Q}$. Now, consider $p^r \frac{a}{b}$ and $p^s \frac{c}{d}$, where $a, b, c, d \in \mathbb{Z}$, $p \nmid a, b, c, d$, and $b, d \neq 0$. Assume without loss of generality that $r < s$. We have as follows:

$$\left| p^r \frac{a}{b} + p^s \frac{c}{d} \right|_p = p^{-r} \left| \frac{a}{b} + p^{s-r} \frac{c}{d} \right|_p \leq p^{-r} \left| \frac{a + p^{s-r}c}{bd} \right|_p = p^{-r}.$$

Therefore, we have shown that $|\cdot|_p$ is indeed a non-Archimedean norm.

Definition 5.6 (Definition 1.6 in [Sut21]). We say that two absolute values $|\cdot|, |\cdot|': \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$ are *equivalent* if there exists some positive real α such that $|x|' = |x|^\alpha$, for all $x \in \mathbb{Q}$.

Definition 5.7. A *place* of $\mathbb{Q}$ is an equivalence class of nontrivial absolute values on $\mathbb{Q}$. We denote the set of places of $\mathbb{Q}$ as $M_{\mathbb{Q}}$.

Definition 5.8. We shall refer to the absolute values introduced in Examples 5.4 and 5.5 as the *normalized absolute values*, as they will be our usual choice of representative.

The following theorem says that Examples 5.4 and 5.5 represent the equivalence classes of nontrivial absolute values on $\mathbb{Q}$.

Theorem 5.9 (Ostrowski; Theorem 7.12 in [Mil20]). *Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or $|\cdot|_p$, for some prime p . That is, the set of places of $\mathbb{Q}$ is represented by the infinite place and all primes.*

Proof. We refer to Theorem 7.12 in [Mil20]. □

5.2. The Topology from an Absolute Value. Given an absolute value $|\cdot|: \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$, we can induce a metric on $\mathbb{Q}$, $d: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$ by defining $d(x, y) = |x - y|$. This makes $\mathbb{Q}$ a metric space, and so a topological space, with a basis of the topology generated by open balls which look like

$$B_r(x) = \{y \in \mathbb{Q}: d(y, x) < r\}.$$

Given a metric space, it is natural to want to perform analysis, so the next step is to study completions of $\mathbb{Q}$. For this, we shall introduce what it means for two Cauchy sequences to be equivalent.

Definition 5.10 (Definition 8.3 in [Sut21]). Given two Cauchy sequences (x_n) and (y_n) in $\mathbb{Q}$ with respect to a metric d , they are *equivalent* if $d(x_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$.

Definition 5.11 (Definition 8.3 in [Sut21]). Let $|\cdot|$ be an absolute value on $\mathbb{Q}$ and d be the induced metric from $|\cdot|$. The *completion* of $\mathbb{Q}$ with respect to $|\cdot|$ is the set of elements that are equivalence classes of Cauchy sequences with the metric induced by d as

$$d([(x_n)], [(y_n)]) := \lim_{n \rightarrow \infty} d(x_n, y_n),$$

where $[(x_n)]$ and $[(y_n)]$ are elements inside the completion.

Remark 5.12. We shall note some properties of a completion of $\mathbb{Q}$, which we denote by $\hat{\mathbb{Q}}$ here. We shall refer to [Sut21] for proofs of these statements. We first note that $\hat{\mathbb{Q}}$ is a complete metric space with respect to the metric d as described in Definition 5.11. There also exists a canonical embedding $\mathbb{Q} \hookrightarrow \hat{\mathbb{Q}}$ defined by sending $x \in \mathbb{Q}$ to the constant sequence $(x, x, x, \dots)$. This has dense image in $\hat{\mathbb{Q}}$.

Example 5.13. $\mathbb{Q}$ completed with respect to $|\cdot|_\infty$ is $\mathbb{R}$.

Example 5.14. $\mathbb{Q}$ completed with respect to $|\cdot|_p$, for some prime p , is $\mathbb{Q}_p$, the ring of p -adic numbers. We define $\mathbb{Z}_p$ to be the subring of $\mathbb{Q}_p$ defined as $\mathbb{Z}_p := \{x \in \mathbb{Q}_p: |x|_p \leq 1\}$.

While we have defined completions and the topology on $\mathbb{Q}$ induced by an absolute value individually, we can form a correspondence between places and the induced topologies. This will be described in Proposition 5.16, but to prove this, we will require the following result.

Proposition 5.15 (Weak Approximation; Theorem 8.5 in [Sut21]). *Let $|\cdot|_1, \dots, |\cdot|_n$ be pairwise inequivalent nontrivial absolute values on $\mathbb{Q}$. Let $a_1, \dots, a_n \in \mathbb{Q}$ and $\epsilon_1, \dots, \epsilon_n > 0$ be real. Then there exists an element $x \in \mathbb{Q}$ such that $|x - a_i|_i < \epsilon_i$, for all $1 \leq i \leq n$.*

Proof. We refer to Theorem 8.5 in [Sut21]. $\square$

Proposition 5.16 (Corollary 8.6 in [Sut21]). *Let $|\cdot|, |\cdot|'$ be two absolute values on $\mathbb{Q}$. Then, the topologies induced by $|\cdot|$ and $|\cdot|'$ are the same if and only if $|\cdot|$ and $|\cdot|'$ are equivalent.*

Proof. First suppose that $|\cdot|$ and $|\cdot|'$ are equivalent, so $|\cdot|' = |\cdot|^\alpha$, for some positive α . It is then clear that the basis of open balls remain the same.

Assume, for the sake of contradiction, that $|\cdot|$ and $|\cdot|'$ are not equivalent. By weak approximation (Proposition 5.15), for each $n > 0$, there exists some rational x_n such that

$$|x_n| < \frac{1}{n}, \quad |x_n - 1|' < \frac{1}{n}.$$

Then, as n grows large, x_n tends to 0 with respect to $|\cdot|$ but tends to 1 with respect to $|\cdot|'$. By uniqueness of limits in metric spaces, it follows that the topologies cannot be the same because the sequence (x_n) tends to distinct values. $\square$

This proposition tells us that, topologically speaking, we should really only care about the places of $\mathbb{Q}$ instead of absolute values. Moreover, Ostrowski's theorem (Theorem 5.9) tells us that we know what they look like as well.

5.3. The Adeles. Here, we introduce the adèle ring of $\mathbb{Q}$. I will note at the start that many of these constructions can again be repeated for number fields and function fields.

Notation 5.17. Let $v \in M_{\mathbb{Q}}$. We denote $\mathbb{Q}_v$ to be $\mathbb{R}$ and $\mathcal{O}_v$ to be $\mathbb{R}$ if v is the infinite place, which we usually denote with ∞ . If v is a finite place, v is a prime, and $\mathbb{Q}_v$ denotes the usual v -adic numbers and $\mathcal{O}_v := \mathbb{Z}_v$.

Remark 5.18. More generally, we note that in the setting of number fields or function fields, the notions of infinite and finite places correspond to Archimedean and non-Archimedean places, respectively.

Definition 5.19 (Definition 25.7 in [Sut21]). The *adele ring* of $\mathbb{Q}$ is the topological ring

$$\mathbb{A}_{\mathbb{Q}} := \prod_{v \in M_{\mathbb{Q}}} (\mathbb{Q}_v, \mathcal{O}_v) := \{(a_v) \in \prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v : a_v \in \mathcal{O}_v \text{ for all but finitely many } v\}$$

with the basis of open sets given by $\prod_{v \in M_{\mathbb{Q}}} U_v$, where the $U_v \subseteq \mathbb{Q}_v$ are open for all places v and $U_v = \mathcal{O}_v$ for all but finitely many places v .

We may view $\mathbb{Q}$ as a subring of $\mathbb{A}_{\mathbb{Q}}$ through the diagonal embedding $\mathbb{Q} \hookrightarrow \mathbb{A}_{\mathbb{Q}}$, and we refer to $\mathbb{Q}$ as the *principal adeles*. Explicitly, this is the map defined by

$$x \mapsto (x, x, x, \dots), \quad x \in \mathbb{Q}.$$

This map is often called the *global embedding*. Here, at each place v , we are viewing x as an element in $\mathbb{Q}_v$ via the canonical inclusion $\mathbb{Q} \hookrightarrow \mathbb{Q}_v$.

Remark 5.20. For $x \in \mathbb{Q}$, $(x, x, x, \dots)$ is a well-defined element in $\mathbb{A}_{\mathbb{Q}}$. By unique factorization into irreducible elements of $\mathbb{Q}$, there are only finitely many primes that arise as a denominator in x . Consequently, there are only finitely many finite places v such that $|x|_v > 1$. In other words, for all but finitely many places, $x \in \mathcal{O}_v$.

Remark 5.21. The topology on $\mathbb{A}_{\mathbb{Q}}$ is much finer than the product topology on $\prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v$. Thus, to avoid confusion we will generally call this topology the *restricted product topology*. As an example of an open set in $\mathbb{A}_{\mathbb{Q}}$ not open in $\prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v$, we consider the following set:

$$U := \mathbb{R} \times \prod_{p \text{ prime}} \mathbb{Z}_p.$$

This is certainly open in the restricted product topology, since all but finitely many factors are given by $\mathbb{Z}_p$. We claim this is not open in the product topology. Consider the element $(0, 0, 0, \dots) \in U$. If U is open in the product topology of $\prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v$, then it contains a basic open set of $(0, 0, 0, \dots)$ in the product topology of the form

$$U_{\infty} \times \prod_{v \in S} U_v \times \prod_{v \notin S} \mathbb{Q}_v,$$

where U_v is open in $\mathbb{Q}_v$ and S is a finite set of primes. Call this set W ; let ℓ be a prime not in S , and consider the element $x_{\ell} := (0, 0, \dots, 0, \ell^{-1}, 0, \dots)$, where the only nonzero coordinate is ℓ^{-1} at the place ℓ . Then, $x_{\ell} \in W$, but $x_{\ell} \notin U$ because at the place ℓ , $\ell^{-1} \notin \mathbb{Z}_{\ell}$. For any set S , we could've constructed such an element x_{ℓ} , so it follows that U is not open in the product topology.

There are some subrings of the adeles which we can also define and will become important later.

Definition 5.22. The ring of *finite adeles* $\mathbb{A}_{\mathbb{Q}}^f$ is the following

$$\mathbb{A}_{\mathbb{Q}}^f := \prod_{p \text{ prime}} (\mathbb{Q}_p, \mathbb{Z}_p).$$

As one might expect from Definition 5.19, we have the following result.

Lemma 5.23. *We have the isomorphism of topological rings $\mathbb{A}_{\mathbb{Q}} \cong \mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^f$.*

Proof. Consider the maps of topological rings $\varphi: \mathbb{A}_{\mathbb{Q}} \rightarrow \mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^f$ defined by $(x_v) \mapsto (x_{\infty}, (x_p))$ and $\psi: \mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^f \rightarrow \mathbb{A}_{\mathbb{Q}}$ defined by $(x_{\infty}, (x_p)) \mapsto (x_{\infty}, x_2, x_3, \dots)$. These maps provide the desired isomorphism. This is clearly an isomorphism of rings, so we shall just check that this is a homeomorphism. An open set in $\mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^f$ is of the form $U_{\infty} \times \left(\prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p \right)$, where S is a finite set of primes and $U_p \subseteq \mathbb{Q}_p$ is open for all $p \in S$. Applying φ^{-1} , this becomes the open set $U_{\infty} \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p$, which is clearly open in $\mathbb{A}_{\mathbb{Q}}$. Thus, φ is continuous. Now, consider an open set $U_{\infty} \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p$ of $\mathbb{A}_{\mathbb{Q}}$, where again S is a finite set of primes and $U_p \subseteq \mathbb{Q}_p$ is open for all $p \in S$. Through φ , this gets sent to the set $U_{\infty} \times \left(\prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p \right)$, which is clearly open in $\mathbb{R} \times \mathbb{A}_{\mathbb{Q}}^f$. Therefore, φ is also an open map, so φ is a homeomorphism as desired. $\square$

Definition 5.24. Let S be a finite set of primes of $\mathbb{Q}$. The subring of $\mathbb{A}_{\mathbb{Q}}$ of *S-adeles* is the following direct product of topological rings endowed with the subspace topology of $\mathbb{A}_{\mathbb{Q}}$.

$$\mathbb{A}_{\mathbb{Q},S} := \mathbb{R} \times \prod_{p \in S} \mathbb{Q}_p \times \prod_{p \notin S} \mathbb{Z}_p$$

Remark 5.25. Note that $\mathbb{A}_{\mathbb{Q},S}$ is open in $\mathbb{A}_{\mathbb{Q}}$, since all but finitely many factors are $\mathbb{Z}_p$.

Another natural way to define a topology on the underlying set of $\mathbb{A}_{\mathbb{Q},S}$ is using the product topology. While these two topologies generally look very different on $\mathbb{A}_{\mathbb{Q}}$, as seen in Remark 5.21, the following result says that these two topologies agree for the S -adeles.

Proposition 5.26. *Let us continue with the notation in Definition 5.24. The topology on $\mathbb{A}_{\mathbb{Q},S}$ agrees with the product topology endowed on the same underlying set.*

Proof. Let us start by showing an open set in $\mathbb{A}_{\mathbb{Q},S}$ is open in the product topology. First note that a basic open set in the product topology restricts only finitely many factors, while all remaining factors are the whole factor. A basic open set in the restricted product topology looks like

$$\prod_{v \in T} U_v \times \prod_{v \notin T} \mathcal{O}_v,$$

where T is a finite set of places containing the infinite place and each $U_v \subseteq \mathbb{Q}_v$ is open. If we intersect this with $\mathbb{A}_{\mathbb{Q},S}$, we get the following set:

$$\prod_{v \in S \cap T} U_v \times \prod_{v \in S \setminus T} \mathbb{Q}_v \times \prod_{v \in T \setminus S} (U_v \cap \mathcal{O}_v) \times \prod_{v \notin S \cup T} \mathcal{O}_v.$$

Since this has all but finitely many factors as $\mathcal{O}_v$, this is open in the product topology endowed on the underlying set of $\mathbb{A}_{\mathbb{Q},S}$.

Conversely, a basic open set in the product topology of the underlying set of $\mathbb{A}_{\mathbb{Q},S}$ looks like

$$\prod_{v \in S \cap T} U_v \times \prod_{v \in S \setminus T} \mathbb{Q}_v \times \prod_{v \in T \setminus S} U_v \times \prod_{v \notin S \cup T} \mathcal{O}_v, \quad (31)$$

where for $v \in T \setminus S$, $U_v \subseteq \mathcal{O}_v$ is open in $\mathcal{O}_v$. Now, for each $v \in T \setminus S$, U_v being open in $\mathcal{O}_v$ implies that there exists some open set $W_v \subseteq \mathbb{Q}_v$ such that $U_v = W_v \cap \mathcal{O}_v$. Using these W_v , we can define an open set in the subspace topology from $\mathbb{A}_{\mathbb{Q}}$:

$$U := \prod_{v \in S \cap T} U_v \times \prod_{v \in S \setminus T} \mathbb{Q}_v \times \prod_{v \in T \setminus S} W_v \times \prod_{v \notin S \cup T} \mathcal{O}_v$$

Then, our open set in Equation (31) is equal to $U \cap \mathbb{A}_{\mathbb{Q},S}$. Thus, our original open set in the product topology is open in the subspace topology from $\mathbb{A}_{\mathbb{Q}}$. $\square$

Remark 5.27. While the restricted product topology is odd at first glance, Proposition 5.26 gives us a lead into how this arises naturally. As in Proposition 25.5 of [Sut21], we can write $\mathbb{A}_{\mathbb{Q}}$ as a direct limit of the product topologies of $\mathbb{A}_{\mathbb{Q},S}$: $\mathbb{A}_{\mathbb{Q}} \cong \varinjlim \mathbb{A}_{\mathbb{Q},S}$. Here, the limit is taken over finite sets of finite places and the transition maps are the inclusions $\mathbb{A}_{\mathbb{Q},S} \hookrightarrow \mathbb{A}_{\mathbb{Q},T}$, where $S \subseteq T$. This is an isomorphism of topological rings, whose proof we refer to Proposition 25.5 of [Sut21].

Another reason why the S -adeles are worth mentioning is because of the following lemma:

Lemma 5.28. *The S -adeles cover $\mathbb{A}_{\mathbb{Q}}$.*

Proof. Given an adele $x = (x_{\infty}, x_2, x_3, \dots) \in \mathbb{A}_{\mathbb{Q}}$, only finitely many of the x_p (for p prime) don't lie in $\mathbb{Z}_p$. If we let S denote the set of primes for which $x_p \notin \mathbb{Z}_p$, then, $x \in \mathbb{A}_{\mathbb{Q},S}$. Thus, every element lies in some S -adele. $\square$

Definition 5.29. A topological space X is *locally compact* if every point has a compact neighborhood; that is, for every $x \in X$, there exists a compact set K of X and an open set U containing x such that $U \subseteq K$.

Example 5.30. $\mathbb{R}$ is locally compact. If $x \in \mathbb{R}$, then $U := (x - \epsilon, x + \epsilon)$ is an open subset of $\mathbb{R}$ containing x , for real $\epsilon > 0$. Then, $[x - \epsilon, x + \epsilon]$ is compact and contains U .

Theorem 5.31 (Proposition 25.9 in [Sut21]). *$\mathbb{A}_{\mathbb{Q}}$ is locally compact and Hausdorff.*

Proof. Let $x \in \mathbb{A}_{\mathbb{Q}}$. Since the S -adeles cover $\mathbb{A}_{\mathbb{Q}}$ by Lemma 5.28, we can find a suitable finite set of primes S such that $x \in \mathbb{A}_{\mathbb{Q},S}$. Then, it suffices to show that $\mathbb{A}_{\mathbb{Q},S}$ is locally compact as a subspace of $\mathbb{A}_{\mathbb{Q}}$. This is because $\mathbb{A}_{\mathbb{Q},S}$ is open in $\mathbb{A}_{\mathbb{Q}}$; if x admits an open neighborhood $U \subseteq \mathbb{A}_{\mathbb{Q},S}$ contained in a compact subset $K \subseteq \mathbb{A}_{\mathbb{Q},S}$, then U remains open and K remains compact when regarded as subsets of $\mathbb{A}_{\mathbb{Q}}$. To prove that $\mathbb{A}_{\mathbb{Q},S}$ is locally compact as a subspace of $\mathbb{A}_{\mathbb{Q}}$, we will show that $\mathbb{A}_{\mathbb{Q},S}$ is locally compact with respect to the product topology, because Proposition 5.26 tells us that the product topology on $\mathbb{A}_{\mathbb{Q},S}$ and the subspace topology coincide. By Tychonoff's theorem, $\prod_{v \notin S} \mathcal{O}_v$ is a product of compact spaces, so is compact. The remaining components in the product are each locally compact. Since a finite product of

locally compact spaces is locally compact, and a product of a locally compact space with a compact space is locally compact, we conclude that $\mathbb{A}_{\mathbb{Q},S}$ is locally compact with the product topology.

For Hausdorff-ness, we first note that the product space $\prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v$ is Hausdorff since each $\mathbb{Q}_v$ is Hausdorff. Then, the topology on $\mathbb{A}_{\mathbb{Q}} \subseteq \prod_{v \in M_{\mathbb{Q}}} \mathbb{Q}_v$ is finer than the subspace topology, so we are done. $\square$

Remark 5.32. While we will not explore this topic in this thesis, one can perform integral analysis over $\mathbb{A}_{\mathbb{Q}}$. Since $\mathbb{A}_{\mathbb{Q}}$ is a locally compact topological ring, its underlying additive group forms a locally compact topological group. It follows that it admits a Haar measure on $\mathbb{A}_{\mathbb{Q}}$, which is unique up to scaling (for instance, as in Chapter 1.2 in [RV98]). This allows us to perform integration on $\mathbb{A}_{\mathbb{Q}}$ on appropriate functions on $\mathbb{A}_{\mathbb{Q}}$. As one might guess, such an integral is constructed using its local components. This leads to notions such as the Fourier transform, which, in turn, provide summation formulae analogous to the Poisson summation formula in the real case. Note that for the same reason, we could perform integral analysis on the rings $\mathbb{Q}_p$, and integration on $\mathbb{A}_{\mathbb{Q}}$ is constructed using the integral at the local places. This is expanded much more in Chapter 1 of [GH11].

5.4. The Group $GL(2, \mathbb{A}_{\mathbb{Q}})$. Now that we are equipped with the adèle ring, we are ready to define the group $GL(2, \mathbb{A}_{\mathbb{Q}})$ which is the group we shall define automorphic forms over. Firstly, $GL(2, \mathbb{A}_{\mathbb{Q}})$ is the set of 2×2 -invertible matrices with entries in $\mathbb{A}_{\mathbb{Q}}$, but as was done in the adèle ring, we want to endow this with a topology. The first instinct is to view this as the multiplicative subgroup of $M(2, \mathbb{A}_{\mathbb{Q}})$ and take the subspace topology. While there are issues with this idea, we shall come back to it after studying the topology on $M(2, \mathbb{A}_{\mathbb{Q}})$ briefly.

$M(2, \mathbb{A}_{\mathbb{Q}})$ is just the set of matrices with entries in $\mathbb{A}_{\mathbb{Q}}$. Canonically, we can endow $M(2, \mathbb{A}_{\mathbb{Q}})$ with the product topology coming from its four matrix entries; that is, we endow $M(2, \mathbb{A}_{\mathbb{Q}})$ with the topology which makes the following identification a homeomorphism:

$$\begin{aligned} M(2, \mathbb{A}_{\mathbb{Q}}) &\xrightarrow{\sim} (\mathbb{A}_{\mathbb{Q}})^4, \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\longmapsto (a, b, c, d), \end{aligned} \tag{32}$$

Seeing the restricted topology appear for $\mathbb{A}_{\mathbb{Q}}$, it is also natural to see if we can define a topology on $M(2, \mathbb{A}_{\mathbb{Q}})$ through a restricted product. The answer is yes, and it actually coincides with the product topology.

Proposition 5.33. *The product topology on $M(2, \mathbb{A}_{\mathbb{Q}})$ and the restricted product topology coincide, where the restricted product is taken as follows:*

$$M(2, \mathbb{R}) \times \prod_p (M(2, \mathbb{Q}_p), M(2, \mathbb{Z}_p))$$

That is, we have an isomorphism of topological rings $M(2, \mathbb{A}_{\mathbb{Q}}) \cong M(2, \mathbb{R}) \times \prod_p (M(2, \mathbb{Q}_p), M(2, \mathbb{Z}_p))$.

Proof. We first show an open set in the product topology is open in the restricted product topology. Note that an open set in $\mathbb{A}_{\mathbb{Q}}$ looks like

$$U_{\infty} \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p. \tag{33}$$

Thus, an open set in the product topology looks like a product of four such sets. Under the identification as in Equation (32), we can write such a product in the form

$$W_{\infty} \times \prod_{p \in T} W_p \times \prod_{p \notin T} M(2, \mathbb{Z}_p),$$

where now $W_{\infty} \subseteq M(2, \mathbb{R})$ and $W_p \subseteq M(2, \mathbb{Q}_p)$ are open. The set T here is the union of four finite sets of places, which is taken from the product of four sets as in Equation (33), we still have that for all but finitely many primes, the opens are $\mathbb{Z}_p$. This is open in the restricted

product topology, so this implication is proven.

Conversely, take an open set in the restricted product topology:

$$U := U_\infty \times \prod_{p \in S} U_p \times \prod_{p \notin S} M(2, \mathbb{Z}_p),$$

where S is a finite set of primes, $U_\infty \subseteq M(2, \mathbb{R})$ is open, and $U_p \subseteq M(2, \mathbb{Q}_p)$ are open for each p . We want to show that this is open in the product topology. We note that for $p \in S$, U_p need not be identified with a product of four open sets. However, since U_p is open, it is a union of products of open sets. Therefore, for every matrix $g_p \in U_p$, there exist open subsets

$$U_{p,11}, U_{p,12}, U_{p,21}, U_{p,22} \subseteq \mathbb{Q}_p$$

such that

$$g_p \in U_{p,11} \times U_{p,12} \times U_{p,21} \times U_{p,22} \subseteq U_p.$$

Similarly, every point of $U_\infty \subseteq M(2, \mathbb{R}) \cong \mathbb{R}^4$ has an open neighborhood given by products

$$U_{\infty,11} \times U_{\infty,12} \times U_{\infty,21} \times U_{\infty,22} \subseteq U_\infty.$$

It then follows that U is a union of sets of the form

$$W := W_\infty \times \prod_{p \in S} W_p \times \prod_{p \notin S} M(2, \mathbb{Z}_p),$$

where each W_v is a product: $W_v = U_{v,11} \times U_{v,12} \times U_{v,21} \times U_{v,22}$. Under the identification $M(2, \mathbb{A}_\mathbb{Q}) \cong \mathbb{A}_\mathbb{Q}^4$ in Equation (32), such a set W becomes

$$\prod_{1 \leq i, j \leq 2} \left(U_{\infty, ij} \times \prod_{p \in S} U_{p, ij} \times \prod_{p \notin S} \mathbb{Z}_p \right).$$

For each pair (i, j) , $U_{\infty, ij} \times \prod_{p \in S} U_{p, ij} \times \prod_{p \notin S} \mathbb{Z}_p$ is open in $\mathbb{A}_\mathbb{Q}$, so W is open in the product topology. Since U is a union of such open sets W , it follows that U is open in the product topology. $\square$

Now that we understand the topology on $M(2, \mathbb{A}_\mathbb{Q})$, let us return to the issue of endowing $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ with the subspace topology from $M(2, \mathbb{A}_\mathbb{Q})$. The issue is not unique to the adèle ring or the ring of matrices, but an issue with making $R^\times$ a topological group when R is a topological ring. R being a topological ring means that R is an additive topological group with a continuous multiplication operation. Nowhere in this definition does it talk about the continuity of the multiplicative inverse, so there should be no reason that $R^\times$ should be a topological group under multiplication. We maneuver around this by endowing $R^\times$ with the weakest topology which does make it a topological group. That is, we embed $R^\times$ into $R \times R$ via the map $r \mapsto (r, r^{-1})$ and declare this to be a homeomorphism onto its image. We shall refer to this topology on $R^\times$ as the *unit topology*. If we repeat this construction for $M(2, \mathbb{A}_\mathbb{Q})$, we reach the following definition, which is now more sensible.

Definition 5.34. The group $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ is the group of units of $M(2, \mathbb{A}_\mathbb{Q})$ endowed with the unit topology.

Again, we can define the topology on $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ as a restricted product. This is contained in the following lemma.

Proposition 5.35. *The unit topology on $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ coincides with the restricted product topology on the underlying set of $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$; that is, we have an isomorphism of topological groups*

$$\mathrm{GL}(2, \mathbb{A}_\mathbb{Q}) \cong \mathrm{GL}(2, \mathbb{R}) \times \prod_p (\mathrm{GL}(2, \mathbb{Q}_p), \mathrm{GL}(2, \mathbb{Z}_p)).$$

Here, the notation $\prod$ is the restricted product from Definition 5.19. Explicitly, an element $g = (g_v)_v \in \mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ is given by

$$g = (g_v) = (g_\infty, g_2, \dots, g_p, \dots),$$

where $g_\infty \in \mathrm{GL}(2, \mathbb{R})$ and $g_p \in \mathrm{GL}(2, \mathbb{Z}_p)$, for all but finitely many primes p .

Proof. Let $\iota: \mathrm{GL}(2, \mathbb{A}_\mathbb{Q}) \rightarrow M(2, \mathbb{A}_\mathbb{Q}) \times M(2, \mathbb{A}_\mathbb{Q})$ denote the map sending $g \mapsto (g, g^{-1})$ from above which induces the unit topology on $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$. We first show that a basic open set in the restricted product topology is open in the topology in Definition 5.34. A basic open set in the restricted product topology is of the form

$$U_\infty \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathrm{GL}(2, \mathbb{Z}_p),$$

where S is a finite set of primes, and $U_\infty \subseteq \mathrm{GL}(2, \mathbb{R})$ and $U_p \subseteq \mathrm{GL}(2, \mathbb{Q}_p)$ for $p \in S$ are open. First, there exists $W_\infty \subseteq M(2, \mathbb{R})$ open such that $U_\infty = W_\infty \cap \mathrm{GL}(2, \mathbb{R})$. Further, for every prime $p \in S$, U_p being open implies that there exists some open $W_p \subseteq M(2, \mathbb{Q}_p)$ such that $U_p = W_p \cap \mathrm{GL}(2, \mathbb{Q}_p)$. Now, define the following subsets of $M(2, \mathbb{A}_\mathbb{Q})$.

$$W := W_\infty \times \prod_{p \in S} W_p \times \prod_{p \notin S} M(2, \mathbb{Z}_p), \quad V := M(2, \mathbb{R}) \times \prod_{p \in S} M(2, \mathbb{Q}_p) \times \prod_{p \notin S} M(2, \mathbb{Z}_p)$$

Now consider the set

$$\iota^{-1}(W \times V) = \{g \in \mathrm{GL}(2, \mathbb{A}_\mathbb{Q}) : g \in W, g^{-1} \in V\}.$$

Taking $g = (g_v) \in \iota^{-1}(W \times V)$, if $p \notin S$, it follows that $g_p, g_p^{-1} \in M(2, \mathbb{Z}_p)$, so $g_p \in \mathrm{GL}(2, \mathbb{Z}_p)$. As for $p \in S$, $g_p \in W_p \subset M(2, \mathbb{Q}_p)$ and $g_p^{-1} \in M(2, \mathbb{Z}_p)$, so g_p lies in an open subset of $\mathrm{GL}(2, \mathbb{Q}_p)$. At the infinite place, g_∞ lies in an open subset of $\mathrm{GL}(2, \mathbb{R})$ by the same argument as that for g_p for $p \in S$. Thus any open set in the restricted product topology is open in $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$.

Now take some open set in $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$. This is of the form

$$U := \{g \in \mathrm{GL}(2, \mathbb{A}_\mathbb{Q}) : g \in W, g^{-1} \in V\}.$$

Here, $W, V \subseteq M(2, \mathbb{A}_\mathbb{Q})$ are basic open sets in $M(2, \mathbb{A}_\mathbb{Q})$. Therefore, we may write $W = \prod_{v \in T} W_v \times \prod_{v \notin T} M(2, \mathbb{Z}_v)$ and $V = \prod_{v \in S} V_v \times \prod_{v \notin S} M(2, \mathbb{Z}_v)$. Here, $W_v, V_v \subseteq M(2, \mathbb{Q}_v)$ are open, and S, T are finite sets of places containing the infinite place. For each place v , we shall define the set

$$U'_v := \{x \in \mathrm{GL}(2, \mathbb{Q}_v) : x \in V'_v, x^{-1} \in W'_v\},$$

where

$$V'_v := \begin{cases} V_v & v \in S, \\ \mathrm{GL}(2, \mathbb{Z}_v) & v \notin S \end{cases}, \quad W'_v := \begin{cases} W_v & v \in T, \\ \mathrm{GL}(2, \mathbb{Z}_v) & v \notin T. \end{cases}$$

Then, each U'_v is open in $\mathrm{GL}(2, \mathbb{Q}_v)$ because $U'_v = (V'_v \cap \mathrm{GL}(2, \mathbb{Q}_v)) \cap \iota_2^{-1}(W'_v \cap \mathrm{GL}(2, \mathbb{Q}_v))$, where ι_2 is the inverse map $g \mapsto g^{-1}$. Here, ι_2 is continuous by definition, and so U'_v is an intersection of open sets. We also have that for any place $v \notin S \cup T$, $U'_v = \mathrm{GL}(2, \mathbb{Z}_v)$, so $U'_v = \mathrm{GL}(2, \mathbb{Z}_v)$ for all but finitely many places. Therefore, $U = \prod_{v \in M_\mathbb{Q}} U'_v$, which is open in the restricted product topology. $\square$

Remark 5.36. Note that the multiplicative subgroup of $\mathbb{A}_\mathbb{Q}$ can be endowed with a topology analogous to $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ which makes the multiplicative inverse a continuous map. This is the group of *ideles*, denoted as $\mathbb{I}_\mathbb{Q}$. Then, the analogous statement of Proposition 5.35 holds for $\mathbb{I}_\mathbb{Q}$ as well. While it is more natural to discuss the idele group first, then come to this discussion for $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$, I have omitted the discussion of $\mathbb{I}_\mathbb{Q}$ as we will not require it deeply in this thesis. Another reason why is because the topological technicalities for $\mathbb{I}_\mathbb{Q}$ are similar to those for $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$. Naturally, we may view $\mathbb{I}_\mathbb{Q}$ as $\mathrm{GL}(1, \mathbb{A}_\mathbb{Q})$ and study automorphic forms on $\mathrm{GL}(1, \mathbb{A}_\mathbb{Q})$. Historically, this case was developed first in Tate's thesis (contained in Chapter 15 of [CFSU10]) and may be regarded as the $\mathrm{GL}(1)$ -analogue of the theory of automorphic forms on $\mathrm{GL}(2, \mathbb{A}_\mathbb{Q})$ considered in this thesis.

As in the case for $\mathbb{A}_{\mathbb{Q}}$, we may view $\mathrm{GL}(2, \mathbb{Q})$ as a subgroup of $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ via the global mapping ι_{diag} defined as

$$\begin{aligned} \iota_{\mathrm{diag}}: \mathrm{GL}(2, \mathbb{Q}) &\longrightarrow \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \\ \gamma &\longmapsto (\gamma, \gamma, \gamma, \dots) \end{aligned} \quad (34)$$

We can also define the local embeddings

$$\begin{aligned} \iota_v: \mathrm{GL}(2, \mathbb{Q}_v) &\longrightarrow \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \\ \alpha &\longmapsto (I, \dots, \alpha, I, \dots), \end{aligned} \quad (35)$$

where $I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ denotes the identity matrix and α is in the v -th place. Note that for the local embeddings, v can be any place.

Notation 5.37. Moving forward, Proposition 5.35 allows us to adapt the following notation. Whenever we write $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ with $a = (a_v)_v, b = (b_v)_v, c = (c_v)_v, d = (d_v)_v \in \mathbb{A}_{\mathbb{Q}}$, we can interpret this as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \left(\begin{pmatrix} a_{\infty} & b_{\infty} \\ c_{\infty} & d_{\infty} \end{pmatrix}, \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}, \dots \right).$$

Proposition 5.38. ι_{diag} and ι_v are well-defined; that is, their images lie in $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$.

Proof. We prove the proposition for the global embedding first. Let $\gamma \in \mathrm{GL}(2, \mathbb{Q})$. Then, we need to check if $\gamma \in \mathrm{GL}(2, \mathbb{Z}_p)$ for all but finitely many primes p . Let us write $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, for $a, b, c, d \in \mathbb{Q}$. Then, for all but finitely many primes, a, b, c, d are all contained in $\mathbb{Z}_p$. Further, $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, so $\det(\gamma) \in \mathbb{Q}^{\times}$. Therefore, $\det(\gamma) \in \mathbb{Z}_p^{\times}$ for all but finitely many primes. It follows then that $\gamma \in \mathrm{GL}(2, \mathbb{Z}_p)$ for all but finitely many primes.

As for the local embedding, this is immediate. The only nontrivial entry of $\iota_v(\alpha)$, for some place v and $\alpha \in \mathrm{GL}(2, \mathbb{Q}_v)$, is at the v -th place. Thus, $\iota_v(\alpha) \in \mathrm{GL}(2, \mathbb{Z}_p)$ for all but finitely many primes p . $\square$

6. AUTOMORPHIC FORMS FOR $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$

We are finally ready to introduce the definition of an automorphic form over the group $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. Comparing this with how we defined classical modular forms or classical automorphic forms, this may seem strange at the beginning because we defined those objects on a concrete space—the upper half plane. However, we can view classical automorphic forms as being defined on a group as well, although this group-theoretic formulation is less explicit. By the orbit-stabilizer theorem or Iwasawa decomposition (as in Chapter 1.2 in [Gol06]), we have the bijection of sets

$$\mathbb{H} \cong \mathrm{GL}(2, \mathbb{R}) / (\mathrm{O}(2, \mathbb{R}) \cdot \mathbb{R}^{\times}).$$

Therefore, we may view classical modular forms or classical automorphic forms as functions defined on $\mathrm{GL}(2, \mathbb{R})$ satisfying some invariance properties with respect to $\mathrm{O}(2, \mathbb{R}) \cdot \mathbb{R}^{\times}$ and extra analytic conditions.

The main objective of this section will be to define automorphic forms over $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. Throughout the section, we will give analogies between the definitions in the classical case to observe what notions we are generalizing. To finish the chapter, we will give an example of an adelic automorphic form by lifting a classical modular form to the adelic setting. This will show us that automorphic forms over $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ are indeed a generalization of what we had in the classical case.

6.1. Some Lie Theory. It is now the right time to introduce some Lie theory and discuss the universal enveloping algebra. The reason for introducing these objects is to provide an isomorphism between the universal enveloping algebra of $\mathfrak{gl}(2, \mathbb{C})$ and the algebra of relevant differential operators which act on the automorphic forms. We note that these differential operators are necessary in generalizing the notion of holomorphy.

6.1.1. *The Universal Enveloping Algebra.* In this subsection, let F be a field.

Definition 6.1. Let L be a Lie algebra over F with Lie bracket $[\cdot, \cdot]$. We define the *tensor algebra* on L as

$$T(L) := \bigoplus_{k=0}^{\infty} T^k L,$$

where $T^k L = L \otimes \cdots \otimes L$ (k times). It is endowed with the associative product defined on homogeneous generators of $T(L)$ as $(v_1 \otimes \cdots \otimes v_k)(w_1 \otimes \cdots \otimes w_m) := v_1 \otimes \cdots \otimes v_k \otimes w_1 \otimes \cdots \otimes w_m$.

Remark 6.2. We note that $T^0(L) = F$, and $T^1 L = L$. Therefore, elements of F and L are contained in $T(L)$.

Following the remark above, $T(L)$ is a very big tensor product; it contains finite sums of tensor products of different length. That is, elements of $T(L)$ look like

$$c + v + \sum_i a_i v_i \otimes w_i + \sum_j b_j v_j \otimes w_j \otimes x_j + \dots,$$

where only finitely many of the coefficients are nonzero. Here, c, a_i, b_j are elements in the base field F and $v, v_i, w_i, v_j, w_j, x_j$ are elements of L . Note that the product of two elements in $T(L)$ gives it a graded structure. Furthermore, $T(L)$ is an associative noncommutative algebra, which follows from the tensor product being associative and, generally, noncommutative.

Remark 6.3. As noted in Chapter 17 of [Hum72], the object $T(L)$ admits the following universal property: for any associative F -algebra A with identity and any F -linear map $\phi: L \rightarrow A$, there exists a unique homomorphism of F -algebras $\psi: T(L) \rightarrow A$ such that $\psi(1) = 1$ and $\phi = \psi \circ i$, where $i: L \rightarrow T(L)$ is the canonical inclusion.

Lemma 6.4. $T(L)$ admits a two-sided ideal generated by the tensors $x \otimes y - y \otimes x - [x, y]$, which we denote by $I(L)$.

Proof. It is built into the definition that $I(L)$ is closed under addition. It is also clearly closed under scalars. We now show that $I(L)$ is closed under left multiplication. Consider the finite sum $\sum_i a_i s_i b_i \in I(L)$, where $a_i, b_i \in T(L)$ and s_i is a tensor of the form $x \otimes y - y \otimes x - [x, y]$. Let $z \in T(L)$. Then, $z \sum_i a_i s_i b_i = \sum_i (z a_i) s_i b_i$ by associativity, so it is closed by multiplication on the left. Analogously $I(L)$ is closed by multiplication on the right. We are done. $\square$

Definition 6.5. Let L be a Lie algebra over a field F . A *universal enveloping algebra* of a Lie algebra L is a pair (U, ι) , where U is an associative F -algebra with identity and $\iota: L \rightarrow U$ is a linear map which satisfies

$$\iota([x, y]) = \iota(x) \cdot \iota(y) - \iota(y) \cdot \iota(x) \quad (36)$$

for $x, y \in L$ with the following universal property: for any associative F -algebra A with identity and any linear map $j: L \rightarrow A$ satisfying (36), there exists a unique homomorphism of algebras $\phi: U \rightarrow A$ such that $\phi \circ \iota = j$.

Proposition 6.6. $U(L) := T(L)/I(L)$, with multiplication defined as $xy = x \otimes y \bmod I(L)$, is a universal enveloping algebra.

Proof. First, we have a map $\iota: L \rightarrow U(L)$ by composing the natural maps $L \hookrightarrow T(L) \rightarrow U(L)$. We note that ι is a Lie algebra homomorphism from L to $U(L)$ since, for $x, y \in L$,

$$\iota(x)\iota(y) - \iota(y)\iota(x) = \iota(x \otimes y - y \otimes x) = \iota([x, y]).$$

We now check that $U(L)$ has the universal property specified above. Let A be an associative F -algebra with identity, and let $j: L \rightarrow A$ be a Lie algebra homomorphism. Note that we may view A as a Lie algebra under the bracket $[a, b] = ab - ba$, for $a, b \in A$. By the universal property of the tensor algebra (Remark 6.3), there exists a unique F -algebra homomorphism $\tilde{j}: T(L) \rightarrow A$ such that $\tilde{j} \circ i = j$, where $i: L \rightarrow T(L)$ is the natural inclusion. $\tilde{j}$ is a Lie algebra homomorphism, so for all $x, y \in L$, $\tilde{j}([x, y]) = \tilde{j}(x)\tilde{j}(y) - \tilde{j}(y)\tilde{j}(x)$. Therefore, $\tilde{j}(x \otimes y - y \otimes x - [x, y]) = 0$. It follows that $I(L) \subseteq \ker(\tilde{j})$, since every generator of $I(L)$ lies in the kernel. Consequently, $\tilde{j}$

descends uniquely to an algebra homomorphism $\bar{j}: U(L) \rightarrow A$, which satisfies $\bar{j} \circ \iota = j$. Thus, $U(L)$ satisfies the universal property of the universal enveloping algebra. $\square$

Remark 6.7. As with many universal properties, we can show that a universal enveloping algebra over L is unique up to isomorphism of algebras. By Proposition 6.6, we know that a universal enveloping algebra of L exists, so we may call it *the* universal enveloping algebra of L .

6.1.2. The Differential Operators. We will now define the relevant differential operators. While the setting we will ultimately use is $\mathfrak{gl}(2, \mathbb{C})$, we first focus on the situation of the Lie group $GL(2, \mathbb{R})$ and its Lie algebra $\mathfrak{gl}(2, \mathbb{R})$. We will then complexify $\mathfrak{gl}(2, \mathbb{R})$ so that the analogous statements hold for $\mathfrak{gl}(2, \mathbb{C})$.

Definition 6.8 (Definition 4.5.1 in [GH11]). Let $\alpha \in \mathfrak{gl}(2, \mathbb{R})$, and $F: GL(2, \mathbb{R}) \rightarrow \mathbb{C}$ be smooth (in the sense of a smooth function on a manifold). Then, we define a differential operator D_α which acts on F as follows.

$$D_\alpha F(g) := \left. \frac{\partial}{\partial t} F(g \cdot \exp(t\alpha)) \right|_{t=0}, \quad g \in GL(2, \mathbb{R}).$$

Remark 6.9. We note that only the first two terms in the Taylor expansion of $\exp(t\alpha)$ matter because we are taking the derivative with respect to t and evaluating at $t = 0$. Thus, we can just write D_α in a much simpler form:

$$D_\alpha F(g) = \left. \frac{\partial}{\partial t} F(g + t(g \cdot \alpha)) \right|_{t=0}$$

Notation 6.10. The operators D_α , with $\alpha \in \mathfrak{gl}(2, \mathbb{R})$, generate an associative algebra where multiplication is defined by composition of operators. We denote this algebra as $\mathcal{D}_{\mathbb{R}}^2$.

The proof of the following proposition illustrates how we actually apply D_α to $F(g)$ concretely.

Proposition 6.11 (Proposition 4.5.2 in [GH11]). *For $\alpha, \beta \in \mathfrak{gl}(2, \mathbb{R})$ and $r \in \mathbb{R}$, we have that*

$$\begin{aligned} D_{r\alpha+\beta} &= r \cdot D_\alpha + D_\beta, \\ D_{[\alpha, \beta]} &= D_\alpha \circ D_\beta - D_\beta \circ D_\alpha. \end{aligned}$$

Proof. First, note that we may identify a smooth function $F: GL(2, \mathbb{R}) \rightarrow \mathbb{C}$ as a function on 4 real variables. This is because we may identify $M(2, \mathbb{R})$ with $\mathbb{R}^4$, and $GL(2, \mathbb{R})$ is an open subset of $M(2, \mathbb{R})$. Therefore, writing $g = (g_{ij})_{ij} \in GL(2, \mathbb{R})$, F is a function of the g_{ij} . Using Remark 6.9 and the multivariate chain rule, we can calculate that

$$D_\alpha F(g) = \sum_{i,j=1}^2 (g\alpha)_{ij} \cdot \frac{\partial}{\partial g_{ij}} F(g). \quad (37)$$

The linearity $D_{r\alpha+\beta} = r \cdot D_\alpha + D_\beta$ is then immediate from Equation (37). As for the second identity of the proposition, if we apply D_α to Equation (37) for $D_\beta F(g)$, this gives the following:

$$D_\alpha(D_\beta F)(g) = \sum_{i,j=1}^2 (g\alpha\beta)_{ij} \cdot \frac{\partial F(g)}{\partial g_{ij}} + \sum_{k,\ell=1}^2 (g\alpha)_{k\ell} (\beta)_{ij} \cdot \frac{\partial^2 F(g)}{\partial g_{ij} \partial g_{k\ell}}.$$

If we subtract $D_\beta(D_\alpha F)(g)$ from this, the commutativity of second partials imply that the second order partials vanish, giving the following difference:

$$(D_\alpha D_\beta - D_\beta D_\alpha)F(g) = \sum_{i,j=1}^2 (g(\alpha\beta - \beta\alpha))_{ij} \cdot \frac{\partial F}{\partial g_{ij}}$$

Comparing this with the expression in Equation (37), we see that this is just $D_{[\alpha, \beta]}F(g)$, so we have proven the listed identities. $\square$

A key takeaway from Proposition 6.11 is that the map $\tilde{\delta}: \alpha \mapsto D_\alpha$ is a linear map from $\mathfrak{gl}(2, \mathbb{R})$ to $\text{End}(C^\infty(GL(2, \mathbb{R})))$. Now, by the universal property of the universal enveloping algebra, there exists a map

$$\delta: U(\mathfrak{gl}(2, \mathbb{R})) \rightarrow \text{End}(C^\infty(GL(2, \mathbb{R})))$$

such that $\delta(\alpha) = D_\alpha$ for every $\alpha \in \mathfrak{gl}(2, \mathbb{R})$. It can be shown that δ is injective (Lemma 4.5.4 in [GH11]), and its image is $\mathcal{D}_{\mathbb{R}}^2$. In other words, we have an isomorphism of $\mathbb{R}$ -algebras

$$\begin{aligned} \delta: U(\mathfrak{gl}(2, \mathbb{R})) &\xrightarrow{\sim} \mathcal{D}_{\mathbb{R}}^2 \\ \alpha &\longmapsto D_\alpha. \end{aligned} \quad (38)$$

Due to this isomorphism, $\mathcal{D}_{\mathbb{R}}^2$ is said to be a *realization* of the algebra $U(\mathfrak{gl}(2, \mathbb{R}))$.

Remark 6.12. It is worth noting that because $\mathcal{D}_{\mathbb{R}}^2$ is an associative algebra, we can make it into a Lie algebra by defining the Lie bracket to be $[D, D'] := DD' - D'D$. Then, this map δ is even a Lie algebra homomorphism.

Remark 6.13. Note that the definition above and discussion holds even for $\mathfrak{gl}(n, \mathbb{R})$; that is, nothing special was used about being in dimension 2.

Now, we want to extend the action of $\mathfrak{gl}(2, \mathbb{R})$ on $C^\infty(\mathrm{GL}(2, \mathbb{R}))$ to an action of $\mathfrak{gl}(2, \mathbb{C})$. We do this by complexifying $\mathfrak{gl}(2, \mathbb{R})$; that is, we tensor it with $\mathbb{C}$ to attain $\mathfrak{gl}(2, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} =: \mathfrak{gl}(2, \mathbb{C})$. This allows us to extend the differential operators linearly.

Definition 6.14 (Definition 4.5.5 in [GH11]). Let $\alpha, \beta \in \mathfrak{gl}(2, \mathbb{R})$, so that $\alpha + i\beta \in \mathfrak{gl}(2, \mathbb{C})$, and $F: \mathrm{GL}(2, \mathbb{R}) \rightarrow \mathbb{C}$ be smooth. Here, $i = \sqrt{-1}$. Then, we define a differential operator $D_{\alpha+i\beta}$ acting on F via the rule

$$D_{\alpha+i\beta}F(g) := (D_\alpha + iD_\beta)F(g).$$

Analogous to the real case as in Equation (38), we can identify every element of $U(\mathfrak{gl}(2, \mathbb{C}))$ with a differential operator via the mapping $\alpha + i\beta \mapsto D_{\alpha+i\beta}$. We introduced these operators because they preserve many invariance properties seen in the definition of an automorphic form. If f is left $\mathrm{GL}(2, \mathbb{Q})$ -invariant and transforms appropriately under the center (which will be made precise in Definition 6.25), then Df has the same invariance properties for $D \in U(\mathfrak{gl}(2, \mathbb{C}))$.

6.1.3. The Center $Z(U(\mathfrak{gl}(2, \mathbb{C})))$. To define automorphic forms for $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, we must also discuss the center of the universal enveloping algebra of $\mathfrak{gl}(2, \mathbb{C})$, $Z(U(\mathfrak{gl}(2, \mathbb{C})))$. By definition, the differential operators in the center are those which commute with any other differential operator in $U(\mathfrak{gl}(2, \mathbb{C}))$. Motivated by this, these differential operators will be the operators which act on automorphic forms by scalars.

In fact, by the Harish-Chandra isomorphism (Theorem 8.18 in [Kna01]), we can explicitly describe $Z(U(\mathfrak{gl}(2, \mathbb{C})))$. $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ is isomorphic to $\mathbb{C}[D_I, D_\Omega]$ as $\mathbb{C}$ -algebras. Here Ω is the *Casimir element* defined as $\Omega = \frac{1}{2}h^2 + ef + fe \in U(\mathfrak{gl}(2, \mathbb{C}))$, where

$$h := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

We note that the differential operator corresponding to the Casimir element D_Ω is often referred to as the *Laplace-Beltrami operator*, as in Chapter 2.2 of [Bum97]. To make this operator D_Ω more concrete, we may use the Iwasawa decomposition theorem, as stated below.

Proposition 6.15 (Iwasawa Decomposition Theorem for $\mathrm{GL}(2, \mathbb{R})$; Proposition 4.1.1 in [GH11]). *Every matrix in $\mathrm{GL}(2, \mathbb{R})$ can be uniquely written in the following form:*

$$g = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \pm 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix},$$

where $x, y, \theta, r \in \mathbb{R}$ with $y > 0$, $0 \leq \theta < 2\pi$, and $r > 0$.

In the language of Iwasawa decomposition, a function on $\mathrm{GL}(2, \mathbb{R})$ is a function depending on $x, y, r, \theta \in \mathbb{R}$ with the appropriate constraints as in Proposition 6.15. Using these coordinates, the Laplace-Beltrami operator can be written as follows, as explained in Proposition 2.2.5 of [Bum97]:

$$D_\Omega = -y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + y \frac{\partial^2}{\partial x \partial \theta}. \quad (39)$$

Example 6.16. Let us provide an example of how this calculation is carried out. We denote by $GL(2, \mathbb{R})^+$ to be the subgroup of $GL(2, \mathbb{R})$ containing matrices of positive determinant (note that we will continue with this notation throughout this section). Consider the function $F: GL(2, \mathbb{R})^+ \rightarrow \mathbb{C}$ defined as $F(g) = (\text{Im}(g \cdot i))^s$, for some positive real s , where $g \cdot i$ is the action of a matrix as in Equation (7). Iwasawa decomposition then tells us that we can write g as follows:

$$g = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix},$$

with $x, y, r, \theta \in \mathbb{R}$, with $y > 0$, $0 \leq \theta < 2\pi$, and $r > 0$. Then, the three matrices on the right fix i , so we have that

$$g \cdot i = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix} \cdot i = x + iy.$$

Therefore, using the coordinates x, y, θ, r , we have that

$$F(x, y, \theta, r) = (\text{Im}(x + iy))^s = y^s.$$

Applying the Laplace-Beltrami operator, we find that

$$D_\Omega F(g) = D_\Omega F(x, y, \theta, r) = -s(s-1)y^s = -s(s-1)F(g).$$

Therefore, F is an eigenfunction of D_Ω , which illustrates that D_Ω acts on F by scalar multiplication. To illustrate that this phenomenon doesn't occur for general differential operators in $U(\mathfrak{gl}(2, \mathbb{C}))$, we shall calculate $D_\alpha F(g)$, for $\alpha = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Note that D_α is not in the center, so we expect a non-scalar relation. If $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then we can calculate that $\text{Im}(g \cdot i) = \frac{\det(g)}{c^2 + d^2}$. Therefore,

$$F(g) = \left(\frac{\det(g)}{c^2 + d^2} \right)^s.$$

Next, we can calculate that $g + t(g\alpha) = \begin{pmatrix} a+tb & b \\ c+td & d \end{pmatrix}$. Calculating that $\det(g + t(g\alpha)) = \det(g)$, we have that

$$F(g + t(g\alpha)) = \left(\frac{\det(g)}{(c+td)^2 + d^2} \right)^s.$$

Differentiating F with respect to t and then evaluating t at 0, we find that

$$D_\alpha F(g) = -\frac{2scd \det(g)^s}{(c^2 + d^2)^{s+1}} = -\frac{2scd}{c^2 + d^2} F(g).$$

In particular, the coefficient $-\frac{2scd}{c^2 + d^2}$ depends on g , so F is not an eigenfunction of D_α .

6.2. The Definition. To get to the definition of an automorphic form, we need to go through a few more definitions and results. As we proceed, we will discuss the motivation behind each definition or create analogies to the classical case.

6.2.1. Analytic Conditions. This subsection will contain analogues and generalizations of the analytic conditions from the classical case. We shall first discuss smoothness of an adelic function. At the infinite place, there is a notion of smoothness and differentiability which we are used to. However, this breaks down at the finite places, because the topology on $\mathbb{Q}_p$ is totally disconnected. For the finite places, the appropriate substitutes are locally constant functions. For adelic smoothness, we wish for a definition containing both of these ideas. In particular, we want that, after restricting to some small enough open set U containing a point of $GL(2, \mathbb{A}_\mathbb{Q})$, varying the finite factors keeps the function constant, while the function varies smoothly depending on the infinite place on U .

Definition 6.17 (Definition 4.7.2 in [GH11]). A function $\phi: GL(2, \mathbb{A}_\mathbb{Q}) \rightarrow \mathbb{C}$ is *smooth* if for every fixed $g_0 \in GL(2, \mathbb{A}_\mathbb{Q})$, there exists an open set U of $GL(2, \mathbb{A}_\mathbb{Q})$ containing g_0 , and a smooth function $\phi_\infty^U: GL(2, \mathbb{R}) \rightarrow \mathbb{C}$ such that $\phi(g) = \phi_\infty^U(g_\infty)$ for all $g \in U$.

We next discuss the notion of moderate growth. This notion plays a similar role to meromorphy at ∞ for classical modular forms. As in Remark 3.12, if a function f on the upper half plane is meromorphic at ∞ , this means that its Laurent series on a punctured disk about 0 has only finitely many negative terms. In other words, the growth of f is of polynomial growth with respect to its entries. We wish for the growth of automorphic forms to be controlled in a similar polynomial fashion, which is what the following definition encapsulates.

Definition 6.18 (Definition 4.7.3 in [GH11]). Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, where $a = (a_v)_v, b = (b_v)_v, c = (c_v)_v, d = (d_v)_v \in \mathbb{A}_{\mathbb{Q}}$. Define the norm function

$$|g| := \prod_v \max\{|a_v|_v, |b_v|_v, |c_v|_v, |d_v|_v, |a_v d_v - b_v c_v|^{-1}\}.$$

A function $\phi: \mathrm{GL}(2, \mathbb{Q}) \backslash \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \rightarrow \mathbb{C}$ is of *moderate growth* if there exist real constants $C, B > 0$ such that $|\phi(g)|_{\mathbb{C}} < C|g|^B$ for all $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$.

Remark 6.19. In the above, we've denoted by $|\cdot|_{\mathbb{C}}$ to make it apparent that we are taking the complex absolute value of $\phi(g)$.

The norm function above measures the ‘height’ or ‘size’ of an element in $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. Thus, the moderate growth condition requires an automorphic form to grow at most polynomially in this ‘height’, ruling out uncontrolled growth such as exponential growth.

Remark 6.20. Note that the norm function above is well-defined. For all but finitely many places, $a_p, b_p, c_p, d_p \in \mathbb{Z}_p$, so we have that $|a_p|_p, |b_p|_p, |c_p|_p, |d_p|_p \leq 1$ and $|a_p d_p - b_p c_p|_p^{-1} = 1$ because $a_p d_p - b_p c_p \in \mathbb{Z}_p^{\times}$. Therefore, $\max\{|a_p|_p, |b_p|_p, |c_p|_p, |d_p|_p, |a_p d_p - b_p c_p|_p^{-1}\} = 1$ for all but finitely many primes, implying that the norm is actually a finite product.

Next, we want to control the action of the differential operators. In the classical setting, modular forms are not arbitrary smooth functions; they satisfy differential constraints arising from holomorphy. There are also a family of classical non-holomorphic automorphic forms, such as *Maass forms* (as defined in Section 2.1 of [Bum97]), which serve as eigenfunctions to some differential operators. The following definition is the adelic analogue of these classical differential conditions, requiring the space generated by the action of invariant differential operators on an automorphic form to be finite-dimensional.

Definition 6.21 (Definition 4.7.5 in [GH11]). A smooth function $\phi: \mathrm{GL}(2, \mathbb{Q}) \backslash \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \rightarrow \mathbb{C}$ is $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ -finite if the set $\{D\phi: D \in Z(U(\mathfrak{gl}(2, \mathbb{C})))\}$ generates a finite dimensional vector space.

6.2.2. Symmetry Conditions. We now discuss some conditions concerning symmetry. We will first introduce Hecke characters, which are the adelic analogues of Dirichlet characters for classical automorphic forms. As in Remark 2.29, automorphic forms decompose into χ -eigenspaces according to how they transform under certain symmetries. In the adelic setting, we will see (in Definition 6.25) that the relevant symmetry to consider is the center of $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, which is naturally identified with $\mathbb{A}_{\mathbb{Q}}^{\times}$.

Definition 6.22 (Definition 2.1.2 in [GH11]). A *Hecke character* of $\mathbb{A}_{\mathbb{Q}}^{\times}$ is a continuous homomorphism $\omega: \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{\times} \rightarrow \mathbb{C}^{\times}$. If all of its values have absolute value 1, then ω is called a *unitary Hecke character*.

Remark 6.23. In other words, a Hecke character is characterized by the following conditions:

- $\omega(gg') = \omega(g)\omega(g')$, for all $g, g' \in \mathbb{A}_{\mathbb{Q}}^{\times}$.
- $\omega(\gamma g) = \omega(g)$, for all $\gamma \in \mathbb{Q}^{\times}, g \in \mathbb{A}_{\mathbb{Q}}^{\times}$.
- ω is continuous at $(1, 1, 1, \dots)$.

The unitary condition says that $|\omega(g)|_{\mathbb{C}} = 1$, for all $g \in \mathbb{A}_{\mathbb{Q}}^{\times}$.

The remaining points of a classical modular form which we have not touched on yet are the weight and the congruence subgroup the modular form is defined over. Adelicly, this

information is encoded into a compact open subgroup K . In particular, the infinite place encodes the weight while the finite part encodes the level. This is exemplified more concretely in Remark 6.30 for the compact open subgroup $U_0(N)$ as defined in Definition 6.29. Ultimately, this leads to the following definition of right- K -finiteness.

Definition 6.24 (Definition 4.7.4 in [GH11]). Let K be a compact subgroup of $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. A function $\phi: \mathrm{GL}(2, \mathbb{Q}) \backslash \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \rightarrow \mathbb{C}$ is *right K -finite* if the set $\{\phi(gk): k \in K\}$ of all right translates of $\phi(g)$ generates a finite dimensional vector space.

6.2.3. An Automorphic Form with Central Character. We can finally state the definition of an automorphic form. This brings together Definitions 6.17, 6.18, 6.21, 6.22, and 6.24.

Definition 6.25 (Definition 4.7.6 in [GH11]). Let $\omega: \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{\times} \rightarrow \mathbb{C}^{\times}$ be a unitary character and K be a compact subgroup of $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. An *automorphic form* for $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ (or adelic automorphic form) with central character ω is a smooth function $\phi: \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \rightarrow \mathbb{C}$ which satisfies the following properties:

- $\phi(\gamma g) = \phi(g)$, for all $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$.
- $\phi(zg) = \omega(z)\phi(g)$, for all $z \in \mathbb{A}_{\mathbb{Q}}^{\times}$, $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$.
- ϕ is right K -finite.
- ϕ is $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ -finite.
- ϕ is of moderate growth.

6.3. The Adelic Lift. Let k, N be positive integers and χ be a Dirichlet character modulo N . This last subsection will show that a classical modular form in $\mathcal{M}_k(N, \chi)$ (as in Definition 2.27) can be lifted to an automorphic form as in Definition 6.25. Before doing this, we must introduce a few more preliminaries that will generalize the notions of a congruence subgroup or Dirichlet character in the classical setting.

6.3.1. The Groups $U_0(N)$.

Definition 6.26 (Definition 4.11.2 in [GH11]). Let p be a prime number. The *Iwahori subgroup* of $\mathrm{GL}(2, \mathbb{Z}_p)$, denoted $K_0(N)_p$, is defined to be

$$K_0(N)_p := \left\{ \begin{pmatrix} a & b \\ cN & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{Z}_p) : c \in \mathbb{Z}_p \right\}.$$

Remark 6.27. Note that in the case that $p \nmid N$, N is a p -adic unit. Therefore, $N\mathbb{Z}_p = \mathbb{Z}_p$, so the extra condition becomes redundant. Consequently, $K_0(N)_p = \mathrm{GL}(2, \mathbb{Z}_p)$ when $p \nmid N$.

Theorem 6.28 (Strong Approximation; Theorem 4.11.5 in [GH11]). *Let N have the prime decomposition as $N = \prod_{i=1}^r p_i^{e_i}$. Then, every $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ can be written as*

$$g = \iota_{\mathrm{diag}}(\gamma) \cdot \iota_{\infty}(g_{\infty}) \cdot k_f,$$

where $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g_{\infty} \in \mathrm{GL}(2, \mathbb{R})^+$, and $k_f \in \prod_p K_0(N)_p$. This is not necessarily unique.

Definition 6.29. We define $U_0(N) := \prod_p K_0(N)_p$, where $K_0(N)_p$ is the Iwahori subgroup defined above.

Remark 6.30. We note that $U_0(N)$ is the adelic version of the congruence subgroup $\Gamma_0(N)$ of $\mathrm{SL}(2, \mathbb{Z})$ as introduced in Example 2.3. In fact, if we embed $\Gamma_0(N)$ diagonally into $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, we find that

$$\Gamma_0(N) = \mathrm{GL}(2, \mathbb{Q}) \cap (\mathrm{GL}(2, \mathbb{R})^+ \times U_0(N)) \subseteq \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}).$$

We show here the containment $\Gamma_0(N) \supseteq \mathrm{GL}(2, \mathbb{Q}) \cap (\mathrm{GL}(2, \mathbb{R})^+ \times U_0(N))$, for the reverse inclusion will follow similarly. Take some $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{Q}) \cap (\mathrm{GL}(2, \mathbb{R})^+ \times U_0(N))$, then $\gamma_p \in K_0(N)_p \subseteq \mathrm{GL}(2, \mathbb{Z}_p)$ for every prime p . Therefore, $a, b, c, d \in \mathbb{Z}_p$ for all p , so $a, b, c, d \in \mathbb{Z}$. Furthermore, $\det(\gamma_p) \in \mathbb{Z}_p^{\times}$ for all p , so $\det(\gamma) = \pm 1$, and we must have $\det(\gamma) = 1$ because the real component is given by $\mathrm{GL}(2, \mathbb{R})^+$. To finish, we just need to show that $c \equiv 0 \pmod{N}$. Since $\gamma \in K_0(N)_p$ for all p , $c \in N\mathbb{Z}_p$, for all p . Therefore, $c \in N\mathbb{Z}$, implying that c is a multiple of N .

6.3.2. The Idelic Lift of a Dirichlet Character. Now, recall Proposition 2.29 which stated that the space of modular forms $\mathcal{M}_k(\Gamma_1(N))$ decomposed into χ -eigenspaces. In particular, if we wish to define the adelic lift of some $f \in \mathcal{M}_k(N, \chi)$, then we must also lift the Dirichlet character χ to the adelic setting.

Definition 6.31 (Idelic lift for prime-power modulus; Definition 2.1.7 in [GH11]). The *idelic lift* of a Dirichlet character χ modulo p^e is the unitary Hecke character $\chi_{\mathbb{A}}: \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{\times} \rightarrow \mathbb{C}^{\times}$ defined as

$$\chi_{\mathbb{A}}(g) := \chi_{\infty}(g_{\infty}) \cdot \chi_2(g_2) \cdot \chi_3(g_3) \cdots,$$

where $g = (g_{\infty}, g_2, g_3, \dots) \in \mathbb{A}_{\mathbb{Q}}^{\times}$,

$$\chi_{\infty}(g_{\infty}) = \begin{cases} 1 & \text{if } \chi(-1) = 1, \\ \text{sgn}(g_{\infty}) & \text{if } \chi(-1) = -1, \end{cases}$$

and for finite places v ,

$$\chi_v(g_v) = \begin{cases} \chi(v)^m & \text{if } g_v \in v^m \mathbb{Z}_v^{\times} \text{ and } v \neq p, \\ \chi(j)^{-1} & \text{if } g_v \in p^m(j + p^e \mathbb{Z}_p), j, m \in \mathbb{Z}, (j, p) = 1, \text{ and } v = p. \end{cases}$$

Definition 6.32 (Idelic lift for general modulus). Let N be an integer with prime decomposition $N = \prod_{i=1}^r p_i^{e_i}$. The *idelic lift* of a Dirichlet character χ modulo N is the unitary Hecke character $\chi_{\mathbb{A}}: \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{\times} \rightarrow \mathbb{C}^{\times}$ defined as

$$\chi_{\mathbb{A}} = \prod_{i=1}^r \chi_{\mathbb{A}}^{(i)}.$$

Here, $\chi_{\mathbb{A}}^{(i)}$ is the *idelic lift* of a Dirichlet character $\chi^{(i)}$ of conductor $p_i^{e_i}$ (as in Definition 2.25) appearing in the factorization

$$\chi = \prod_{i=1}^r \chi^{(i)},$$

which exists.

Remark 6.33. Note that $\chi_{\mathbb{A}}$ is actually a well-defined unitary Hecke character. The verification is described in Chapter 2.1 of [GH11].

We discuss how $\chi_{\mathbb{A}}$ is a lift of a Dirichlet character χ modulo N . Retaining the notation in Definition 6.32, let n be an integer such that $\gcd(n, N) = 1$. Consider the adele $u = (u_v)$, where

$$u_v = \begin{cases} 1 & \text{if } v = \infty, \\ n & \text{if } v \mid N, \\ 1 & \text{if } v \nmid N. \end{cases}$$

We have that $\chi_{\mathbb{A}}(u) = \prod_{i=1}^r \chi_{\mathbb{A}}^{(i)}(u)$, so we must determine $\chi_{\mathbb{A}}^{(i)}(u)$, for each i . Now fix i . In the notation of Definition 6.31, it suffices to calculate $\chi_v^{(i)}(u_v)$, for all places v . If v is either the infinite place or a prime not dividing N , then we have that $\chi_v^{(i)}(u_v) = 1$ because $u_v = 1$ at such places. Next, at some prime $p_j \neq p_i$ appearing in the prime factorization of N , $u_{p_j} = n$. Since we assumed that $(n, N) = 1$, we have that $p_j \nmid n$, so $n \in \mathbb{Z}_{p_j}^{\times}$. Therefore,

$$\chi_{p_j}^{(i)}(n) = \chi(p_j)^0 = 1.$$

Finally, we calculate $\chi_{p_i}^{(i)}(n)$. Since $(n, N) = 1$, we have that $n \in n + p_i^{e_i} \mathbb{Z}_p$, so $\chi_{p_i}^{(i)}(n) = \chi(n)^{-1}$. Ultimately, we find that

$$\chi_{\mathbb{A}}^{(i)}(u) = \chi^{(i)}(n)^{-1}.$$

Therefore, it follows by definition that

$$\chi_{\mathbb{A}}(u) = \prod_{i=1}^r \chi^{(i)}(n)^{-1} = \chi(n)^{-1}.$$

Therefore, the idelic lift recovers the original Dirichlet character, up to the inverse convention used in Definition 6.31.

$\chi_{\mathbb{A}}$ also provides a way to write the original character χ in terms of the local components χ_v , as defined in Definition 6.31.

Lemma 6.34. *Let N be an integer with prime decomposition $N = \prod_{i=1}^r p_i^{e_i}$. If $\gcd(d, N) = 1$, then $\chi(d) = \prod_{i=1}^r \chi_{p_i}(d)^{-1}$.*

Proof. It suffices to prove this when N is a prime power, for then multiplicativity will finish the proof. Therefore, the statement we wish to prove is that if $N = p^e$ and $(d, p^e) = 1$, then $\chi(d) = \chi_p(d)^{-1}$. Write d as $\pm \prod_{j=1}^s \ell_j^{f_j}$. Note that since $(d, p^e) = 1$, $\ell_j \neq p$, for all j . By Definition 6.31, it follows that $\chi_{\ell_j}(\ell_j) = \chi(\ell_j)$. If $j \neq k$, then ℓ_j is a ℓ_k -adic unit, so $\chi_{\ell_k}(\ell_j) = 1$. By multiplicativity, it follows that $\chi_{\ell_i}(d) = \chi(\ell_i)^{f_i}$. Next, we note that $\chi_q(d) = 1$ if $q \neq \ell_j$, for all j , because d is a q -adic unit then. Finally, $\chi(\pm 1) = \chi_{\infty}(d)$ because χ_{∞} measures the sign of d . Therefore, we have the following:

$$\chi(d) = \chi\left(\pm \prod_{j=1}^s \ell_j^{f_j}\right) = \chi(\pm 1) \cdot \prod_{j=1}^s \chi(\ell_j)^{f_j} = \chi_{\infty}(d) \prod_{i=1}^s \chi_{\ell_i}(d) = \prod_{\text{places } v \neq p} \chi_v(d).$$

To conclude, we use the fact that $\chi_{\mathbb{A}}$ is a unitary Hecke character, so $\chi_{\mathbb{A}}$ is trivial on $\mathbb{Q}^{\times}$. Therefore, $\chi_{\mathbb{A}}(d) = 1$, and Definition 6.31 implies

$$\chi(d)\chi_p(d) = \chi_{\mathbb{A}}(d) = 1.$$

Therefore, $\chi(d) = \chi_p(d)^{-1}$. □

Definition 6.35 (Definition 4.12.12 in [GH11]). Let N be an integer with prime decomposition $N = \prod_{i=1}^r p_i^{e_i}$ and let χ be as in Lemma 6.34. Assume that $k_f \in U_0(N)$ takes the form

$$k_f = \left\{ \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}, \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}, \begin{pmatrix} a_5 & b_5 \\ c_5 & d_5 \end{pmatrix}, \dots \right\}.$$

We then define

$$\omega_{\chi}(k_f) = \prod_{i=1}^r \chi_{p_i}(d_{p_i}).$$

To provide some motivation for Definition 6.35, we recall Definition 2.27. In the classical case, Dirichlet characters acted on the bottom right entry of a matrix in $\Gamma_0(N)$. Since $U_0(N)$ plays the role of $\Gamma_0(N)$, as described in Remark 6.30, we repeat the construction. Moreover, ω_{χ} was defined in this way because if $q \neq p_i$, for all i , then $\chi_q(d_q)$ is trivial.

6.3.3. The Adelic Lift is an Automorphic Form. We shall finally show that a classical modular form lifts to an adelic automorphic form. In doing this, we will first formalize our idea at the beginning of this section, which shifted from viewing a classical modular form as a function on $\mathbb{H}$ to a function on $\mathrm{GL}(2, \mathbb{R})$. We will then define the adelic lift of a classical automorphic form and then check that this adelic lift satisfies each axiom of Definition 6.25 in a series of lemmas.

Definition 6.36 (Definition 4.12.1 in [GH11]). Let f be a classical automorphic form of weight k on $\mathbb{H}$. We define $\tilde{f}: \mathrm{GL}(2, \mathbb{R})^+ \rightarrow \mathbb{C}$ as

$$\tilde{f}(g) := (f|_k g)(i).$$

Remark 6.37. In this subsection, if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{R})^+$, we take the normalization

$$(f|_k g)(\tau) = \det(g)^{\frac{k}{2}} j(g, \tau)^{-k} f(g \cdot \tau), \quad \tau \in \mathbb{H}, \quad (40)$$

where $j(g, \tau)$ is the factor of automorphy as in Definition 2.6. We note that this normalization differs from the normalization taken in Section 4.1.

Remark 6.38. The Iwasawa decomposition (Proposition 4.4.1 in [GH11]) tells us that $g \in \mathrm{GL}(2, \mathbb{R})^+$ can be uniquely written in the form

$$g = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r & 0 \\ 0 & r \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix},$$

where $x, y, r, \theta \in \mathbb{R}$ with $y, r > 0$, and $0 \leq \theta < 2\pi$. Therefore, $\tilde{f}$ is dependent on the parameters x, y, r, θ . In particular, we find that

$$\tilde{f}(g) = y^{\frac{k}{2}} e^{ik\theta} f(x + iy).$$

Lemma 6.39 (Equation 4.12.2 in [GH11]). *Let f be an automorphic form of weight- k on $\mathbb{H}$. Let $r_{\theta'} \in \mathrm{SO}(2, \mathbb{R})$. Then, $\tilde{f}(gr_{\theta'}) = e^{ik\theta'} \tilde{f}(g)$. If $r_{\theta'} \in \mathrm{O}(2, \mathbb{R}) \setminus \mathrm{SO}(2, \mathbb{R})$, then $\tilde{f}(gr_{\theta'}) = (-i)^k e^{ik\theta'} \tilde{f}(g)$.*

Proof. This is clear from the definition of the slash- k operator. $\square$

Definition 6.40 (Definition 4.12.13 in [GH11]). Let N be a positive integer with prime factorization $N = \prod_{i=1}^r p_i^{e_i}$. Let χ be a Dirichlet character modulo N , and let k be an integer. Assume that $f \in \mathcal{A}_k(N, \chi)$ and $\tilde{f}$ be its lift to $\mathrm{GL}(2, \mathbb{R})^+$. Let ω_χ be as in Definition 6.35. We define the *adelic lift* of f , denoted $f_{\mathbb{A}}: \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}}) \rightarrow \mathbb{C}$ as

$$f_{\mathbb{A}}(\iota_{\mathrm{diag}}(\gamma) \cdot \iota_{\infty}(g_{\infty}) \cdot k_f) := \tilde{f}(g_{\infty}) \cdot \omega_{\chi}(k_f),$$

where $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g_{\infty} \in \mathrm{GL}(2, \mathbb{R})^+$, and $k_f \in U_0(N)$ come from the strong approximation (Theorem 6.28) for an element $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$.

Moving forward, let us fix the notation N, k, χ from Definition 6.40 and any other notation from Definition 6.25. We will also fix a classical modular form $f \in \mathcal{M}_k(N, \chi)$.

Lemma 6.41 (Smoothness). *$f_{\mathbb{A}}$ is smooth.*

Proof. Given $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, write g as $\iota_{\mathrm{diag}}(\gamma) \cdot \iota_{\infty}(g_{\infty}) \cdot k_f$ where $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g_{\infty} \in \mathrm{GL}(2, \mathbb{R})^+$ and $k_f \in U_0(N)$. Then, since ω_{χ} is locally constant, we can choose an open neighborhood $U_f \subseteq U_0(N)$ of k_f such that ω_{χ} is constant on U_f . Then, we may choose any open neighborhood $U_{\infty} \subseteq \mathrm{GL}(2, \mathbb{R})^+$ containing g_{∞} . To conclude, $f_{\mathbb{A}}$ will be smooth on $U_{\infty} \times U_f$, since $\tilde{f}$ is smooth on U_{∞} . $\square$

Lemma 6.42 ($\mathrm{GL}(2, \mathbb{Q})$ -Invariance). *$f_{\mathbb{A}}$ is $\mathrm{GL}(2, \mathbb{Q})$ -invariant.*

Proof. Let $\gamma' \in \mathrm{GL}(2, \mathbb{Q})$ and $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. Then, write $g = \iota_{\mathrm{diag}}(\gamma) \cdot \iota_{\infty}(g_{\infty}) \cdot k_f$ by strong approximation, where $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g_{\infty} \in \mathrm{GL}(2, \mathbb{R})^+$, and $k_f \in U_0(N)$. We have as desired:

$$f_{\mathbb{A}}(\gamma'g) = f_{\mathbb{A}}(\iota_{\mathrm{diag}}(\gamma'\gamma) \cdot \iota_{\infty}(g_{\infty}) \cdot k_f) = \tilde{f}(g_{\infty}) \cdot \tilde{\chi}_{\mathrm{adelic}}(k_f) = f_{\mathbb{A}}(g).$$

$\square$

Lemma 6.43 (Central Character). *$f_{\mathbb{A}}$ has central character $\chi_{\mathbb{A}}$.*

Proof. Let $r \in \mathbb{A}_{\mathbb{Q}}^{\times}$ and $g \in \mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. We will first expand the product rg for clarity. By the strong approximation theorem for ideles (Proposition 1.4.5 in [GH11]), we can write $r = \iota_{\mathrm{diag}}(\alpha) i_{\infty}(r_{\infty}) r_f$, with $\alpha \in \mathbb{Q}^{\times}$, $r_{\infty} \in \mathbb{R}_{>0}$, and $r_f = (1, r_2, r_3, \dots)$ with $r_p \in \mathbb{Z}_p^{\times}$ for all p . Here, ι_{diag} and ι_{∞} are the $\mathbb{A}_{\mathbb{Q}}^{\times}$ analogues of the maps defined for $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ as in Equations (34) and (35), respectively. The strong approximation theorem for $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$ (Theorem 6.28) tells us that we may express g as $g = \iota_{\mathrm{diag}}(\gamma) i_{\infty}(g_{\infty}) k_f$, where $\gamma \in \mathrm{GL}(2, \mathbb{Q})$, $g_{\infty} \in \mathrm{GL}(2, \mathbb{R})^+$, and $k_f \in U_0(N)$. Then, we can write rg as

$$\iota_{\mathrm{diag}} \left(\begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \gamma \right) i_{\infty} \left(\begin{pmatrix} r_{\infty} & 0 \\ 0 & r_{\infty} \end{pmatrix} g_{\infty} \right) \begin{pmatrix} r_f & 0 \\ 0 & r_f \end{pmatrix} k_f.$$

Therefore, we have that

$$f_{\mathbb{A}}(rg) = \left(f|_k \begin{pmatrix} r_{\infty} & 0 \\ 0 & r_{\infty} \end{pmatrix} g_{\infty} \right) (i) \cdot \omega_{\chi} \left(\begin{pmatrix} r_f & 0 \\ 0 & r_f \end{pmatrix} k_f \right).$$

We wish to show this equals $\chi_{\mathbb{A}}(r)f_{\mathbb{A}}(g)$. It is clear that

$$f|_k \begin{pmatrix} r_{\infty} & 0 \\ 0 & r_{\infty} \end{pmatrix} = f,$$

by the normalization defined in Equation (40), so we just need to show that the remaining term equals $\chi_{\mathbb{A}}(r)$. To show this, we recall Definition 6.31; before proceeding, note that $\chi_{\mathbb{A}}(\iota_{\text{diag}}(\alpha)) = 1$ because $\chi_{\mathbb{A}}$ is defined on $\mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}}^{\times}$. At the infinite place, $\chi_{\infty}(r_{\infty}) = 1$ because $r_{\infty} > 0$. Further, for all primes not dividing N , $\chi_p(r_p) = 1$ because r_p is a p -adic unit then. Therefore,

$$\chi_{\mathbb{A}}(r) = \prod_{p|N} \chi_p(r_p) = \omega_{\chi} \left(\begin{pmatrix} r_f & 0 \\ 0 & r_f \end{pmatrix} \right)$$

Note that the last equality is definitional, as per Definition 6.35; we are done. $\square$

Lemma 6.44 (Right- $U_0(N)$ -Finiteness). *$f_{\mathbb{A}}$ satisfies right- $U_0(N)$ -finiteness.*

Proof. Let $x = (x_{\infty}, x_f) \in K := O(2, \mathbb{R}) \prod_p GL(2, \mathbb{Z}_p)$ and $g \in GL(2, \mathbb{A}_{\mathbb{Q}})$. At the infinite place, $x_{\infty} \in O(2, \mathbb{R})$, so Lemma 6.39 implies that

$$f_{\mathbb{A}}(gx_{\infty}) = ce^{ik\theta'} f_{\mathbb{A}}(g),$$

where $c \in \{\pm 1, \pm i\}$ and θ' is some angle associated to the rotation matrix x_{∞} . Thus right-translation by x_{∞} only multiplies $f_{\mathbb{A}}$ by a scalar.

Now consider the finite component. Since $U_0(N) \subseteq \prod_p GL(2, \mathbb{Z}_p)$ is a compact open subgroup, it has finite index in $\prod_p GL(2, \mathbb{Z}_p)$. Therefore, we may choose representatives $\kappa_1, \dots, \kappa_m$ for the finite quotient $\prod_p GL(2, \mathbb{Z}_p)/U_0(N)$. Then every $x_f \in \prod_p GL(2, \mathbb{Z}_p)$ can be written in the form $x_f = \kappa_j u$, for some j and some $u \in U_0(N)$. Now, we can calculate that

$$f_{\mathbb{A}}(g\kappa_j u) = f_{\mathbb{A}}(g\kappa_j) \omega_{\chi}(u).$$

It follows that every right translate of $f_{\mathbb{A}}(g)$ by k_f is a scalar multiple of one of the following finitely many functions:

$$f_{\mathbb{A}}(g\kappa_1), \dots, f_{\mathbb{A}}(g\kappa_m)$$

Combining this with the infinite-place calculation, every right translate by an element of K lies in the finite-dimensional $\mathbb{C}$ -vector space $\text{Span}\{f_{\mathbb{A}}(g\kappa_j) : 1 \leq j \leq m\}$. Therefore, $f_{\mathbb{A}}$ is right- $U_0(N)$ -finite. $\square$

Remark 6.45. Note that the group K in the proof above is a maximal compact subgroup of $GL(2, \mathbb{A}_{\mathbb{Q}})$. Compactness follows from Tychonoff's theorem and a similar proof technique to Proposition 5.26 to unify the subspace topology and product topology on K . As for maximality, one can show this using the projections onto each place.

Lemma 6.46 ($Z(U(\mathfrak{gl}(2, \mathbb{C})))$ -Finiteness). *$f_{\mathbb{A}}$ satisfies $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ -finiteness.*

Proof. We will use the coordinates coming from the strong approximation theorem (Theorem 6.28) again. Recall from Section 6.1.3 that $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ is a polynomial algebra generated by D_I and D_{Ω} . To calculate $(D_I f_{\mathbb{A}})(g)$, we want to differentiate the expression $f_{\mathbb{A}}((1+t)g)$ by t and then evaluate the expression at $t = 0$. We have that

$$f_{\mathbb{A}}((1+t)g) = \tilde{f}((1+t)g_{\infty}) \cdot \omega_{\chi}(k_f).$$

Then, we can calculate that

$$\tilde{f}((1+t)g_{\infty}) = (f|_k(1+t)g_{\infty})(i) = \det(g_{\infty})^{\frac{k}{2}} j(g_{\infty}, i)^{-k} f(g_{\infty} \cdot i) = \tilde{f}(g_{\infty}).$$

Thus,

$$\frac{\partial}{\partial t} f_{\mathbb{A}}((1+t)g_{\infty}) \Big|_{t=0} = \frac{\partial}{\partial t} \left(\tilde{f}((1+t)g_{\infty}) \cdot \omega_{\chi}(k_f) \right) \Big|_{t=0} = \frac{\partial}{\partial t} \left(\tilde{f}(g_{\infty}) \cdot \omega_{\chi}(k_f) \right) \Big|_{t=0} = 0.$$

Therefore, D_I acts on $f_{\mathbb{A}}$ by 0.

Next, we calculate the effect of the Laplace-Beltrami operator. For this calculation, we use Iwasawa decomposition on g_∞ and Equation (39). Using Iwasawa decomposition, we find that we can write

$$f_{\mathbb{A}}(g) = f(|_k g_\infty)(i) \cdot \omega_\chi(k_f) = e^{ik\theta} \cdot f(x + iy) \cdot \omega_\chi(k_f),$$

where $x, y, \theta \in \mathbb{R}$ with $y > 0$ and $0 \leq \theta < 2\pi$ as in Remark 6.38. Then,

$$\begin{aligned} D_\Omega(f_{\mathbb{A}}(g)) &= \left(-y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + y \frac{\partial^2}{\partial x \partial \theta} \right) (y^{\frac{k}{2}} e^{ik\theta} f(x + iy) \cdot \omega_\chi(k_f)) \\ &= \frac{k}{2} \left(1 - \frac{k}{2} \right) y^{\frac{k}{2}} e^{ik\theta} f(x + iy) \cdot \omega_\chi(k_f) \\ &= \frac{k}{2} \left(1 - \frac{k}{2} \right) f_{\mathbb{A}}(g). \end{aligned}$$

Therefore, the Laplace-Beltrami operator acts by a scalar. Since D_I and D_Ω both act by scalars, we have $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ -finiteness. $\square$

Lemma 6.47 (Moderate Growth). *$f_{\mathbb{A}}$ has moderate growth.*

Proof. We want to find some positive real constants P, Q such that $|f_{\mathbb{A}}| \leq P|g|^Q$. We use the coordinates from the Iwasawa decomposition again, so that we can write $\tilde{f}(g) = y^{\frac{k}{2}} e^{ik\theta} f(x + iy)$. Here, we note that $x + iy = g_\infty \cdot i$. We will show that we can bound $f_{\mathbb{A}}$ depending on some expression in y , which we will then bound by $|g|$.

Since $\chi_{\mathbb{A}}$ is a unitary Hecke character, we have the following equality:

$$|f_{\mathbb{A}}(g)|_{\mathbb{C}} = |\tilde{f}(g)|_{\mathbb{C}} = |y^{\frac{k}{2}} f(x + iy)|_{\mathbb{C}} \quad (41)$$

Since $y^{\frac{k}{2}}$ is of polynomial growth, we just need to show that $|f(x + iy)|_{\mathbb{C}}$ is of polynomial growth. Write $\tau := x + iy$ and consider an arbitrary element $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$. By basic transformation laws, we can deduce that

$$\mathrm{Im}(\gamma\tau)^{k/2} |f(\gamma\tau)| = \frac{y^{k/2}}{|c\tau + d|^k} |c\tau + d|^k |f(\tau)| = y^{k/2} |f(\tau)|.$$

It follows that $y^{k/2} |f(\tau)|$ is invariant under an $\mathrm{SL}(2, \mathbb{Z})$ -action. Thus, let $\omega = \gamma\tau$, where ω lies in the fundamental domain for $\mathrm{SL}(2, \mathbb{Z})$, and write $\omega = s + it$. Now, note that f is holomorphic at the cusps, so f is bounded on the fundamental domain, so we have $|f(\omega)|_{\mathbb{C}} \leq C_0$. Therefore,

$$y^{k/2} |f(\tau)|_{\mathbb{C}} = t^{k/2} |f(\omega)|_{\mathbb{C}} \leq C_0 t^{k/2}. \quad (42)$$

Now, we note that $t \leq y + y^{-1}$. This follows from the fact that $t = \frac{y}{|c\tau + d|^2}$ and case-working. If $c = 0$, then $\gamma \in \mathrm{SL}(2, \mathbb{Z})$ implies that $d = \pm 1$, so

$$t = \frac{y}{|c\tau + d|^2} = y \leq y + y^{-1}.$$

If $c \neq 0$, then we have the following chain of inequalities

$$y^2 \leq c^2 y^2 \leq c^2 y^2 + (cx + d)^2 = |c\tau + d|^2.$$

This implies the following:

$$t = \frac{y}{|c\tau + d|^2} \leq \frac{y}{y^2} = y^{-1} \leq y + y^{-1}.$$

Therefore, $t \leq y + y^{-1}$ in all cases. Using this inequality along with Equations (41) and (42), we attain

$$|f_{\mathbb{A}}(g)|_{\mathbb{C}} \leq C_0 (y + y^{-1})^{\frac{k}{2}}. \quad (43)$$

To conclude, it suffices to show that $y + y^{-1} < P|g|^Q$, for some positive real P, Q . If $g_\infty = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, we shall denote by $M := \max\{|A|_\infty, |B|_\infty, |C|_\infty, |D|_\infty, |AD - BC|_\infty^{-1}\}$. First, we note that y is the imaginary part of $g_\infty \cdot i$. Thus,

$$y = \operatorname{Im}(g_\infty \cdot i) = \frac{\det(g_\infty)}{C^2 + D^2}. \quad (44)$$

Therefore,

$$y^{-1} = \frac{C^2 + D^2}{\det(g_\infty)} \leq 2M^2 |\det(g_\infty)|^{-1} \leq 2M^3. \quad (45)$$

To bound y , we must first establish the following inequality:

$$|\det(g_\infty)| = |AD - BC| \leq \sqrt{A^2 + B^2} \sqrt{C^2 + D^2} \leq M \sqrt{2(C^2 + D^2)}.$$

Note that the first inequality above is due to the Cauchy-Schwartz inequality. In turn, we find that

$$\frac{|\det(g_\infty)|^2}{C^2 + D^2} \leq 2M^2.$$

Using Equation (44) again, we have as follows:

$$y = \frac{\det(g_\infty)}{C^2 + D^2} \leq 2M^2 |\det(g_\infty)|^{-1} \leq 2M^3. \quad (46)$$

Combining Equations (45) and (46), we have that

$$y + y^{-1} \leq 4M^3 \leq 4|g|^3.$$

By Equation (43), we have

$$|f_{\mathbb{A}}(g)|_{\mathbb{C}} \leq C_0(4|g|^3)^{\frac{k}{2}} = P|g|^Q,$$

for suitable P, Q , so we are done. $\square$

Theorem 6.48 (Proposition 4.12.14 in [GH11]). *$f_{\mathbb{A}}$ is an adelic automorphic form with central character $\chi_{\mathbb{A}}$.*

Proof. This follows by the previous lemmas. $\square$

Remark 6.49. In fact, by restricting enough conditions on the adelic automorphic side, we can *descend* to a classical modular form. That is, by restricting to a subspace of adelic automorphic forms, we can construct an inverse to the adelic lift. The sufficient conditions are expressed in Proposition 5.5 in [Eme].

7. QUATERNIONIC MODULAR FORMS AND THE JACQUET-LANGLANDS CORRESPONDENCE

To conclude this thesis, we shall discuss quaternionic modular forms and the Jacquet-Langlands correspondence. We will certainly not prove the Jacquet-Langlands correspondence because it is far out of scope, but given the statement, we will run through an example which exemplifies its statement.

7.1. Quaternion Algebras. Here, we will briefly introduce quaternion algebras. We will try to include as many proofs as possible, but some results rely on the theory of central simple algebras, which would divert us too much from the main subject. For more details and justifications, we refer to [Voi21]. We will also note that any field appearing in this section are assumed to have characteristic different from 2. This is a technicality which we shall not consider, as we ultimately restrict our base field to $\mathbb{Q}$ from Section 7.3.

7.1.1. *Basic Definitions and Results.* Throughout this section, let F be a field.

Definition 7.1 (Definition 2.2.1 in [Voi21]). An algebra B over F is a *quaternion algebra* if there exist elements $i, j \in B$ such that $1, i, j, ij$ form a F -basis for B subject to the relations

$$i^2 = a, \quad j^2 = b, \quad ij = -ji,$$

for some $a, b \in F^\times$. We denote this by $(a, b)_F$. We note that we will also refer to the basis element ij as k .

Definition 7.2. An algebra B over F is a *central simple algebra* if the center of B is precisely F and B admits no nontrivial two-sided ideals.

Remark 7.3. Proposition 7.6.1 in [Voi21] provides an equivalent characterization of a quaternion algebra, which is that a quaternion algebra over F is a central simple F -algebra of dimension 4.

Remark 7.4. Note that a quaternion algebra need not be commutative at all. In the notation in Definition 7.1, we can calculate that $ij = k = -ji$.

Example 7.5. The ring of *Hamiltonian quaternions* is the ring $\mathbb{R} \oplus i\mathbb{R} \oplus j\mathbb{R} \oplus k\mathbb{R}$ subject to the relations

$$i^2 = j^2 = k^2 = ijk = -1.$$

Note that this is just $(-1, -1)_\mathbb{R}$ in the notation above. The Hamiltonian quaternions form a division ring, and are a quaternion algebra over $\mathbb{R}$. Given a nonzero element $x = a + bi + cj + dk \in \mathbb{R} \oplus i\mathbb{R} \oplus j\mathbb{R} \oplus k\mathbb{R}$, the inverse of x is given by

$$x^{-1} := \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}.$$

The main proposition of this subsection will be the one below, which states that the ring of matrices is another example of a quaternion algebra.

Proposition 7.6 (Proposition 2.3.1 in [Voi21]). *Let $B = (a, b)_F$. Further, let $F(\sqrt{a})$ be the splitting field for the polynomial $x^2 - a$. Then, there is an injection of F -algebras as below.*

$$\begin{aligned} \varphi: B &\rightarrow M(2, F(\sqrt{a})) \\ 1 &\mapsto I_2 \\ i &\mapsto \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix} \\ j &\mapsto \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix} \end{aligned}$$

Here, φ is extended multiplicatively to be a map over B . Moreover, if $a = 1$, we have an isomorphism $B \cong M(2, F)$.

Proof. To verify that this is a valid morphism of F -algebras, we need to show that it's a morphism of rings and respects the multiplication on both sides. The former is true by definition of φ . As for the latter, this follows because we extend φ multiplicatively. For $i^2 = a$, we have that $\varphi(a) = aI$, so we check that $\varphi(i)^2 = aI$.

$$\varphi(i^2) = \varphi(i)\varphi(i) = \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix} \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$$

Note that the check for $\varphi(j)^2 = bI$ is analogous to the calculation above. Next, for $ij = -ji$.

$$\begin{aligned} \varphi(ij) &= \varphi(i)\varphi(j) = \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix} \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & b\sqrt{a} \\ -\sqrt{a} & 0 \end{pmatrix} \\ \varphi(-ji) &= -\varphi(j)\varphi(i) = -\begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix} = -\begin{pmatrix} 0 & -b\sqrt{a} \\ \sqrt{a} & 0 \end{pmatrix} \end{aligned}$$

Next, we shall prove that this is an injection. Suppose that an element $t + xi + yj + zij \in B$ maps to the matrix

$$\begin{pmatrix} t + x\sqrt{a} & b(y + z\sqrt{a}) \\ y - z\sqrt{a} & t - x\sqrt{a} \end{pmatrix} \in M(2, F(\sqrt{a}))$$

through φ . If this matrix is equal to zero, then it is clear that $t = x = y = z = 0$. To illustrate this, we have that $t - x\sqrt{a} = t + x\sqrt{a} = 0$ from the top left and bottom right entries, so $2t = 0$ by summing the two relations. Thus, $t = 0$ and so $x = 0$. The argument for y and z is similar. Therefore, φ is injective.

In the case that $a = 1$, we have that $F(\sqrt{1}) = F$, so $\dim_F(B) = 4 = \dim_F(M(2, F))$. We therefore have a linear injection between spaces of the same dimension; hence, φ is a surjection as well. $\square$

Remark 7.7. Let B, D be quaternion algebras over F and write $B = (a, b)_F$. Let $\varphi: B \rightarrow D$ be a map sending a basis of B to a basis of D . If we want to show that φ is a valid map of quaternion algebras over F , we must verify that φ is a valid map of F -algebras that respects the multiplicative relations $i^2 = a$, $j^2 = b$, $ij = -ji$. While F -linearity is usually immediate, we must manually check the relations $\varphi(i)^2 = a \cdot \varphi(1)$, $\varphi(j)^2 = b \cdot \varphi(1)$, and $\varphi(i)\varphi(j) = -\varphi(j)\varphi(i)$ hold. Moving forward, we shall omit these checks, as they are similar to the calculations performed in the proof of Proposition 7.6.

Remark 7.8. We now have two examples of quaternion algebras over $\mathbb{R}$: the ring of Hamiltonian quaternions, and $M(2, \mathbb{R})$. However, we shall note that these are non-isomorphic quaternion algebras. The easiest way to see this is that the ring of Hamiltonian quaternions is a division algebra and $M(2, \mathbb{R})$ is clearly not, since non-invertible matrices have no inverse.

Lemma 7.9. *Let F be a field, and $a, b \in F$. Then, $(a, b)_F \cong (b, a)_F$ as F -algebras.*

Proof. Let $1, i, j, ij$ be a basis for $(a, b)_F$ and let $1, x, y, xy$ be a basis for $(b, a)_F$. Consider the map $\varphi: (a, b)_F \rightarrow (b, a)_F$ defined on the basis elements as $1 \mapsto 1$, $i \mapsto y$, $j \mapsto x$ extended F -linearly and multiplicatively. We can also define the map $\psi: (b, a)_F \rightarrow (a, b)_F$, by mapping $1 \mapsto 1$, $x \mapsto j$, $y \mapsto i$, then extending F -linearly and multiplicatively. By the method in Remark 7.7, one can show that φ and ψ are valid maps of quaternion algebras. Furthermore, it is clear from the definition of φ and ψ on the basis elements that they are left and right inverses to each other. Therefore, we have a linear bijection between spaces of the same dimension, so we have an isomorphism of quaternion algebras. $\square$

Corollary 7.10.

- (i) *There is a unique quaternion algebra over any algebraically closed field (up to isomorphism).*
- (ii) *There are exactly two quaternion algebras over $\mathbb{R}$ (up to isomorphism):*

$$(a, b)_{\mathbb{R}} \cong \begin{cases} (-1, -1)_{\mathbb{R}} & \text{if } a < 0, b < 0, \\ M(2, \mathbb{R}) & \text{otherwise.} \end{cases}$$

Proof. Let F be an algebraically closed field, with $a, b \in F$. Then, $F = F(\sqrt{a})$ because F is algebraically closed. By algebraic closure again, $\sqrt{a} \in F$, and so we may rescale the basis element i to $\frac{i}{\sqrt{a}}$. This gives the isomorphism $(a, b)_F \cong (1, b)_F$. Then, by using Proposition 7.6 for when $a = 1$, we attain that $(1, b)_F \cong M(2, F)$. Therefore, $(a, b)_F \cong (1, b)_F$. Since our choice of a, b was arbitrary and $(a, b)_F \cong (b, a)_F$ by Lemma 7.9, it follows that we have a unique quaternion algebra up to isomorphism.

We now proceed to the second statement. First, suppose $a < 0, b < 0$. Recall that $(-1, -1)_{\mathbb{R}}$ were the Hamiltonian quaternions. Let $1, x, y, xy$ be a basis for $B := (a, b)_{\mathbb{R}}$ (so $x^2 = a, y^2 = b, xy = -yx$) and $1, i, j, k$ be a basis for the Hamiltonian quaternions. Then, the map $\varphi: B \rightarrow \mathbb{R} \oplus i\mathbb{R} \oplus j\mathbb{R} \oplus ij\mathbb{R}$ defined on the basis elements

$$1 \mapsto 1, \quad \frac{x}{\sqrt{-a}} \mapsto i, \quad \frac{y}{\sqrt{-b}} \mapsto j, \quad \frac{xy}{\sqrt{ab}} \mapsto k.$$

is a valid map of quaternion algebras, via the check stated in Remark 7.7. We may define an inverse to φ by sending $1 \mapsto 1$, $i \mapsto \frac{x}{\sqrt{-a}}$, $j \mapsto \frac{y}{\sqrt{-b}}$, and extending F -linearly and multiplicatively. This is also a valid map of quaternion algebras, via Remark 7.7, so we have an isomorphism.

Next, suppose, without loss of generality, that $a > 0$. Note that generality is not lost due to Lemma 7.9. Since a is positive, $\sqrt{a} \in \mathbb{R}$, so we can rescale the basis element i for $(a, b)_{\mathbb{R}}$ by $\frac{i}{\sqrt{a}}$ to attain the isomorphism $B \cong (1, b)_{\mathbb{R}}$. Then, by specifying Proposition 7.6 to $a = 1$, we have the desired isomorphism $B \cong M(2, \mathbb{R})$. $\square$

We now restrict to when the base field is $\mathbb{Q}$, as this is the case we are ultimately interested in our study of the Jacquet-Langlands correspondence.

Lemma 7.11. *Let B be a quaternion algebra over $\mathbb{Q}$ and $v \in M_{\mathbb{Q}}$. Then, $B \otimes_{\mathbb{Q}} \mathbb{Q}_v$ is a quaternion algebra over $\mathbb{Q}_v$.*

Proof. Let $B = (a, b)_{\mathbb{Q}}$, for $a, b \in \mathbb{Q}$. We will show that $B \otimes_{\mathbb{Q}} \mathbb{Q}_v \cong (a, b)_{\mathbb{Q}_v}$. Consider the map $\varphi: (a, b)_{\mathbb{Q}_v} \rightarrow B \otimes_{\mathbb{Q}} \mathbb{Q}_v$ defined as follows on the basis elements:

$$1 \mapsto 1 \otimes 1, \quad i \mapsto i \otimes 1, \quad 1 \mapsto j \otimes 1,$$

and extended $\mathbb{Q}_v$ -linearly and multiplicatively. By the check in Remark 7.7, this is a valid map of quaternion algebras over $\mathbb{Q}_v$. Furthermore, since $1 \otimes 1$, $i \otimes 1$, $j \otimes 1$, and $k \otimes 1$ acts as a basis for $B \otimes_{\mathbb{Q}} \mathbb{Q}_v$, this is a linear bijection between spaces of the same dimension. Therefore, we have the desired isomorphism. $\square$

7.1.2. Reduced Norm and Trace. In this subsection, we fix B to be a central simple algebra over a field F . Recall the notion of the separable closure of F , which we shall denote by F^{sep} . It is the set of separable elements inside the algebraic closure of F .

Lemma 7.12. *There exists a unique quaternion algebra over a separably closed field (up to isomorphism).*

Proof. Let F^{sep} be our separably closed field. Let $a, b \in F^{\text{sep}}$ and consider the quaternion algebra $(a, b)_{F^{\text{sep}}}$. Note that a must be nonzero. We claim that the polynomial $x^2 - a$ is separable. As we have assumed at the very beginning of the section that $\text{char} F \neq 2$, the (formal) derivative of this polynomial is $2x$. Since $\gcd(x^2 - a, 2x) = 1$, $x^2 - a$ is indeed a separable polynomial. Therefore, a admits a square root inside F^{sep} . Now, if we repeat the argument for the first part in Corollary 7.10, we attain the result. $\square$

Corollary 7.13. *We have a F -algebra isomorphism $\phi: B \otimes_F F^{\text{sep}} \cong M_n(F^{\text{sep}})$.*

Proof. This follows from Lemma 7.12, as both $B \otimes_F F^{\text{sep}}$ and $M_n(F^{\text{sep}})$ are quaternion algebras over F^{sep} . $\square$

Definition 7.14 (7.8.1 in [Voi21]). Let ϕ be the isomorphism from Corollary 7.13. The *reduced norm* $\text{nrd}: B \rightarrow F$ is defined by $\text{nrd}(b) = \det(\phi(b \otimes 1))$. The *reduced trace* $\text{trd}: B \rightarrow F$ is defined by $\text{trd}(b) = \text{tr}(\phi(b \otimes 1))$.

Remark 7.15. We note that the reduced norm actually lies in F . For this, we appeal to the action of the absolute Galois group $G = \text{Gal}(F^{\text{sep}}/F)$ on $B \otimes_F F^{\text{sep}} \cong M_n(F^{\text{sep}})$ defined by

$$\sigma(b \otimes x) := b \otimes \sigma(x),$$

where $\sigma \in G$, $b \in B$, $x \in F^{\text{sep}}$. Therefore, $\sigma(b \otimes 1) = b \otimes 1$. Noting that $\sigma\phi\sigma^{-1}$ is another F -algebra isomorphism $B \otimes_F F^{\text{sep}} \cong M_n(F^{\text{sep}})$, the Skolem-Noether theorem (Theorem 7.1.3 in [Voi21]) tells us that $\sigma(\text{nrd}(b)) = \text{nrd}(b)$. Therefore, $\text{nrd}(b)$ is invariant under the Galois action, so it is in F . Since the trace of a matrix is invariant under conjugation, we attain that $\text{trd}(b) \in F$ for the same reason. In the case when B is a quaternion algebra B over F , we can provide a more explicit description of the reduced norm and reduced trace, as is in the following proposition.

Proposition 7.16. *Let B be a quaternion algebra over F and $x = x_1 + ix_2 + jx_3 + kx_4 \in B$. Define $\bar{x} := x_1 - ix_2 - jx_3 - ikx_4$. Then,*

$$\begin{aligned}\mathrm{trd}(x) &= x + \bar{x} = 2x_1 \\ \mathrm{nrd}(x) &= x\bar{x} = x_1^2 - ax_2^2 - bx_3^2 + abx_4^2\end{aligned}$$

Proof. Suppose that $B = (a, b)_F$ and let ϕ be the isomorphism from Corollary 7.13. If $x \in B$, $\phi(x \otimes 1) \in M(2, F^{\mathrm{sep}})$, so $\phi(x \otimes 1)$ satisfies the characteristic polynomial

$$X^2 - \mathrm{tr}(\phi(x \otimes 1))X + \det(\phi(x \otimes 1)).$$

However, $\mathrm{tr}(\phi(x \otimes 1)) = \mathrm{trd}(x)$ and $\det(\phi(x \otimes 1)) = \mathrm{nrd}(x)$, and so x satisfies what is referred to as the reduced characteristic polynomial

$$X^2 - \mathrm{trd}(x)X + \mathrm{nrd}(x),$$

since ϕ is a F -algebra isomorphism. Further, by Remark 7.15, this is a quadratic polynomial over F . Switching gears, we also observe that x satisfies the identity

$$x^2 + (x + \bar{x})x - x\bar{x} = 0,$$

so x satisfies the polynomial

$$X^2 + (x + \bar{x})X - x\bar{x}.$$

This is also a quadratic polynomial in F . Now, the minimal polynomial of x over F is not of degree 1 because $x \notin F$, and the characteristic polynomial for $\phi(x \otimes 1)$ implies that $1, x, x^2$ are linearly independent; therefore, these three polynomials must coincide, proving the result. $\square$

7.1.3. Ramification. We will now restrict to a quaternion algebra B over $\mathbb{Q}$. We will also fix this notation for this subsubsection.

Notation 7.17. Let $v \in M_{\mathbb{Q}}$. We denote $B_v := B \otimes_{\mathbb{Q}} \mathbb{Q}_v$. In particular, if v is the infinite place, we denote $B_{\infty} := B \otimes_{\mathbb{Q}} \mathbb{R}$.

Now that we know that B_v is a quaternion algebra, we define the following notion.

Definition 7.18. Let $v \in M_{\mathbb{Q}}$. B is *ramified* at v if B_v is a division ring. We say that B is *split* or *unramified* at v if $B_v \cong M(2, \mathbb{Q}_v)$.

Definition 7.19. The *discriminant* of B , denoted $\mathrm{disc}(B)$, is the product of primes that ramify in B .

Remark 7.20. We note that $\mathrm{disc}(B)$ is well-defined, in the sense that it is actually a finite product of primes. This is expressed in Corollary 14.2.3 of [Voi21], which states the stronger statement that every quaternion algebra over $\mathbb{Q}$ has an even number of ramified places.

Definition 7.21 (Definition 14.5.7 in [Voi21]). B is said to be *totally definite* if the infinite place is ramified in B . Otherwise, we say that B is *indefinite*.

Remark 7.22. More generally, $\mathbb{Q}$ fits into a class of number fields called totally real fields. In this broader setting, Corollary 7.10 states that if we have a totally definite quaternion algebra over a number field, then the number field is actually totally real. While we will not study quaternionic modular forms in this setting, it is the correct generalization to consider if one wishes to generalize many definitions in Section 7.3.

Using this notion, we will be interested in totally definite quaternion algebras over $\mathbb{Q}$ later. This will be of great significance later because this allows us to apply Fujisaki's lemma, which is the subject of Section 7.2. Before this though, we must introduce a few more preliminaries from quaternion algebras.

7.1.4. *Orders.* We will now introduce the notion of an order and a maximal order.

Definition 7.23 (Definitions 10.2.1 and 10.4.1 in [Voi21]). Let R be a PID with $F := \text{Frac}R$, and let B be a quaternion algebra over F .

- An R -lattice in B is a finitely generated R -submodule of B of the form $Rx_1 \oplus Rx_2 \oplus Rx_3 \oplus Rx_4$, where x_1, x_2, x_3, x_4 form a basis for B as a F -vector space.
- An R -order $O \subseteq B$ is a R -lattice in B which is also a subring of B . If O is not properly contained in another R -order, then we say that O is *maximal*.

Notation 7.24. Given an order $O \subseteq B$, we shall denote $O_p := O \otimes_{\mathbb{Z}} \mathbb{Z}_p$.

Lemma 7.25. Let O be an order of B . Then, $O \otimes_{\mathbb{Z}} \mathbb{Z}_p$ is an order of B_p , where p is a prime.

Proof. The first claim follows immediately from O being an order of B and the fact that tensor products and direct sums commute. Explicitly, O_p is a subring of B_p and is a $\mathbb{Z}_p$ -module of rank 4. $\square$

Lemma 7.26 (Lemma 10.4.3 in [Voi21]). An order $O \subseteq B$ is maximal if and only if $O_p \subseteq B_p$ is maximal for all p .

Proof. We refer to Lemma 10.4.3 in [Voi21]. $\square$

Example 7.27. $M(2, \mathbb{Z}_p)$ is a $\mathbb{Z}_p$ -order in the quaternion algebra $M(2, \mathbb{Q}_p)$ over $\mathbb{Q}_p$. Firstly, it is clearly a subring of $M(2, \mathbb{Q}_p)$. Next, we can write

$$M(2, \mathbb{Z}_p) = \mathbb{Z}_p \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \oplus \mathbb{Z}_p \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \oplus \mathbb{Z}_p \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \oplus \mathbb{Z}_p \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

By Lemma 7.26, we also attain that $M(2, \mathbb{Z}_p)$ is a maximal order in $M(2, \mathbb{Q}_p)$.

Proposition 7.28. Let p be a prime not dividing $\text{disc}(B)$ and O be a maximal order of B . Then, $O_p \cong M(2, \mathbb{Z}_p)$ and $O_p^\times \cong \text{GL}(2, \mathbb{Z}_p)$.

Proof. $p \nmid \text{disc}(B)$ implies that $B_p \cong M(2, \mathbb{Q}_p)$ by definition. By Lemma 7.26, it follows that O_p is a maximal $\mathbb{Z}_p$ -order in B_p . To conclude, we must use Corollary 10.5.5 in [Voi21], which states that any two maximal $\mathbb{Z}_p$ -orders in B are conjugate to each other. Then, by Example 7.27, we have that $M(2, \mathbb{Z}_p)$ is a maximal order of $M(2, \mathbb{Q}_p)$, so O_p and $M(2, \mathbb{Z}_p)$ are conjugate to each other. In particular, conjugation provides an isomorphism of $\mathbb{Z}_p$ -orders, we have the desired result. Furthermore, the isomorphism $O_p^\times \cong \text{GL}(2, \mathbb{Z}_p)$ follows by taking the group of units. $\square$

7.2. Fujisaki's Lemma. Let B be a finite dimensional simple $\mathbb{Q}$ -algebra. We will also reuse the notation in Section 5.3. That is, if v is the infinite place of $\mathbb{Q}$, $\mathbb{Q}_v := \mathbb{R}$ and $\mathcal{O}_v := \mathbb{R}$. If v is a finite place of $\mathbb{Q}$, then $\mathbb{Q}_v$ denotes the usual v -adic numbers and $\mathcal{O}_v := \mathbb{Z}_v$. Consider the ring $B_{\mathbb{A}} := B \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$. Element-wise, this makes sense, but we also wish to endow $B \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$ with a topology.

Remark 7.29. $B_{\mathbb{A}}$ is not exactly a quaternion algebra, but is an $\mathbb{A}_{\mathbb{Q}}$ -module of rank 4.

Lemma 7.30. If $\dim_{\mathbb{Q}}(B) = n$, then we have an algebraic isomorphism

$$B_{\mathbb{A}} \cong (\mathbb{A}_{\mathbb{Q}})^n.$$

Proof. Since $\dim_{\mathbb{Q}}(B) = n$, we have that $B \cong \bigoplus_{i=1}^n \mathbb{Q}$. Then, we have that $B_{\mathbb{A}} \cong \bigoplus_{i=1}^n (\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}})$ because direct sums commute with tensor products. Then, using the canonical isomorphism $(\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}) \cong \mathbb{A}_{\mathbb{Q}}$, we have that $B_{\mathbb{A}} \cong \bigoplus_{i=1}^n \mathbb{A}_{\mathbb{Q}}$. The proof finishes by the fact that finite direct sums and products are isomorphic. $\square$

Definition 7.31. $B_{\mathbb{A}}$ is a topological ring, with the topology on $B_{\mathbb{A}}$ defined so that the isomorphism becomes a homeomorphism as well. Explicitly, the topology on $B_{\mathbb{A}}$ is induced by the product topology on $(\mathbb{A}_{\mathbb{Q}})^n$.

Proposition 7.32. $B_{\mathbb{A}}$ is locally compact.

Proof. This follows by the fact that $(\mathbb{A}_{\mathbb{Q}})^n$ is a finite product of locally compact spaces. $\square$

We shall also consider the group of units $B_{\mathbb{A}}^{\times}$.

Definition 7.33. $B_{\mathbb{A}}^{\times}$ is a topological group, with the topology being the unit topology as described in Section 5.4.

Remark 7.34. We shall note that the topology defined on $B_{\mathbb{A}}$ and $B_{\mathbb{A}}^{\times}$ are analogous to how we defined the topology on $M(2, \mathbb{A}_{\mathbb{Q}})$ and $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$. This similarity motivates if we can interpret $B_{\mathbb{A}}$ and $B_{\mathbb{A}}^{\times}$ using the restricted product topology, and indeed we can. We have the following isomorphisms of topological rings and groups, respectively:

$$B_{\mathbb{A}} \cong \prod_{v \in M_{\mathbb{Q}}} (B_v, O_v)$$

$$B_{\mathbb{A}}^{\times} \cong \prod_{v \in M_F} (B_v^{\times}, O_v^{\times})$$

Here, the $O_v := O \otimes_{\mathbb{Z}} \mathcal{O}_v$, where O is a maximal order of B . We note that the proofs of these statements are analogous to the proofs of Propositions 5.33 and 5.35, so we will omit them.

Definition 7.35. Let us define the following morphism of topological groups:

$$\nu: B_{\mathbb{A}}^{\times} \rightarrow \mathbb{R}_{>0}$$

$$(\alpha_v) \mapsto \prod_v |\mathrm{nrd}_v(\alpha_v)|_v$$

In the above the maps nrd_v are the reduced norm maps at the place v ; that is, $\mathrm{nrd}_v: B_v \rightarrow \mathbb{Q}_v$. Further more, the absolute values $|\cdot|_v$ are taken to be normalized.

Proposition 7.36. *The map ν as above is well-defined.*

Proof. Let $\alpha = (\alpha_v) \in B_{\mathbb{A}}^{\times}$. From the restricted product in Remark 7.34, $\alpha_v \in O_v^{\times}$ for all but finitely many v . Fix some v such that $\alpha_v \in O_v^{\times}$. Then, since nrd_v is multiplicative, we have

$$\mathrm{nrd}_v(\alpha_v) \mathrm{nrd}_v(\alpha_v^{-1}) = \mathrm{nrd}_v(1) = 1.$$

Then, because $\alpha_v \in O_v$, it follows that $\mathrm{nrd}_v(\alpha_v) \in \mathbb{Z}_v$. Furthermore, since $\mathrm{nrd}_v(\alpha_v^{-1}) = \mathrm{nrd}_v(\alpha_v)^{-1}$, we find that $\mathrm{nrd}_v(\alpha_v) \in \mathbb{Z}_v^{\times}$. Therefore, $|\mathrm{nrd}_v(\alpha_v)|_v = 1$. This occurs for all but finitely many v , so we have shown that the product appearing in $\nu((\alpha_v))$ is actually a finite product. $\square$

Definition 7.37. We define $B_{\mathbb{A}}^{(1)}$ to be the kernel under the map ν .

Lemma 7.38. *The map ν from Definition 7.35 is continuous.*

Proof. We first note that $\mathrm{nrd}_v: B_v \rightarrow \mathbb{Q}_v$ is continuous because it is given by a polynomial in B_v , as can be seen through the discussion in Section 7.1.2. Further, $|\cdot|_v$ is continuous. Therefore, for a fixed place v , $|\mathrm{nrd}_v(\cdot)|_v$ is continuous. Now, let $U \subseteq \mathbb{R}_{>0}$ be open and $\alpha = (\alpha_v) \in \nu^{-1}(U)$. Then, there exists some finite set S such that $\alpha_v \in \mathcal{O}_v^{\times}$ for $v \notin S$. Hence, $|\mathrm{nrd}(\alpha_v)|_v = 1$ for $v \notin S$. Now, since each map $|\mathrm{nrd}_v(\cdot)|_v$ is continuous, the finite product $\prod_{v \in S} |\mathrm{nrd}_v(\cdot)|_v$ is continuous. Since

$$\prod_{v \in S} |\mathrm{nrd}_v(\alpha_v)|_v = \nu(\alpha) \in U,$$

there exist open neighborhoods $W_v \subseteq B_v^{\times}$ containing α_v for $v \in S$, such that $\prod_{v \in S} |\mathrm{nrd}_v(\beta_v)|_v \in U$ whenever $\beta_v \in W_v$ for all $v \in S$. Now set

$$V := \prod_{v \in S} W_v \times \prod_{v \notin S} \mathcal{O}_v^{\times},$$

so that V is an open neighborhood of α in $B_{\mathbb{A}}^{\times}$. We just check that $\nu(V) \subseteq U$. For any $\beta = (\beta_v)_v \in V$, we have $\beta_v \in \mathcal{O}_v^{\times}$ for all $v \notin S$. It follows that

$$\nu(\beta) = \prod_{v \in S} |\mathrm{nrd}_v(\beta_v)|_v \in U.$$

Therefore, we are done. $\square$

Theorem 7.39 (Fujisaki's Lemma; Theorem 27.6.14 in [Voi21]). *If B is a finite-dimensional division algebra over $\mathbb{Q}$, then $B^\times \backslash B_{\mathbb{A}}^{(1)}$ is compact.*

Proof. We refer to Theorem 27.6.14 in [Voi21]. $\square$

Notation 7.40. We denote by $B^f := B \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}^f$, where $\mathbb{A}_{\mathbb{Q}}^f$ is the ring of finite adeles from Definition 5.22.

Lemma 7.41. $B_{\mathbb{A}} \cong B^f \times B_{\infty}$ as topological rings.

Proof. This follows from Lemma 5.23 and the fact that finite products commute with tensor products. $\square$

Lemma 7.42. $B_{\mathbb{A}}^\times \cong (B^f)^\times \times B_{\infty}^\times$ as topological groups.

Proof. This follows from restricting the isomorphism in Lemma 7.41 to its group of units. Then, since the topologies on these groups are induced from the corresponding product topologies, the restriction remains a homeomorphism. $\square$

Corollary 7.43. $B^\times \backslash (B^f)^\times$ is compact.

Proof. We follow the argument in [Imp] and alter it to suit our setting. If we can construct a surjective continuous map $B^\times \backslash B_{\mathbb{A}}^{(1)} \rightarrow B^\times \backslash (B^f)^\times$, it then follows by Fujisaki's lemma since the continuous image of a compact space is compact. Let $x \in (B^f)^\times$. By Lemma 7.42, we want to extend this to some element $(x, y) \in (B^f)^\times \times B_{\infty}^\times$ lying in the kernel of ν . Consider the map

$$\begin{aligned} \nu^f : (B^f)^\times &\rightarrow \mathbb{R}_{>0} \\ (\alpha_v)_v &\mapsto \prod_v |\mathrm{nrd}_v(\alpha_v)|_v \end{aligned} \tag{47}$$

This is just the map ν from Definition 7.35 but restricted to the finite adeles, so is continuous by Lemma 7.38. Then, $\nu^f(x) = r > 0$. We claim that (x, y) with $y = (y_1, \dots, y_n)$ and

$$\mathrm{nrd}(y_i) = r^{-\frac{1}{n}}, \quad \text{for all } i,$$

is a valid preimage of x under the induced map, where $n = \dim_F(B)$. Indeed, we can check that

$$\nu((x, y)) = r \cdot |r^{-\frac{1}{n}}|^n = 1,$$

so $(x, y) \in B_{\mathbb{A}}^{(1)}$. $\square$

Corollary 7.44. *Given an open subgroup $U \subseteq (B^f)^\times$, the double coset space $B^\times \backslash (B^f)^\times / U$ is finite.*

Proof. The double coset provides a disjoint open cover of $(B^f)^\times$. Then, the compactness of $B^\times \backslash (B^f)^\times$ implies that this open cover is actually finite. $\square$

7.3. Quaternionic Modular Forms. Now, let B be a totally definite quaternion algebra over $\mathbb{Q}$ until Section 7.5.1. We will also denote by $Z((B^f)^\times)$ as the center of $(B^f)^\times$.

Definition 7.45. Let F be a field. The space of *quaternionic modular forms* over F for $B^\times$ is the set of functions $f : (B^f)^\times \rightarrow F$ that satisfy the following conditions.

- (1) $f(bg) = f(g)$, for all $b \in B^\times$ and $g \in (B^f)^\times$.
- (2) $f(gz) = f(g)$, for all $z \in Z((B^f)^\times)$ and $g \in (B^f)^\times$.
- (3) There exists a compact open subgroup $U \subseteq (B^f)^\times$ such that $f(gu) = f(g)$, for all $g \in (B^f)^\times$ and $u \in U$.

We denote this set as $L_2^B(F)$.

Remark 7.46. The '2' in the notation for $L_2^B(F)$ reflects the phenomenon described in Theorem 7.65.

Remark 7.47. Aside from the fact that a quaternionic modular form and an automorphic form (as in Definition 6.25) are defined on different spaces, there is another distinction that is noteworthy; quaternionic modular forms are merely functions, while automorphic forms satisfy some analytic conditions. We loosely discuss why these conditions are neglected. For example, the notions of smoothness and $Z(U(\mathfrak{gl}(2, \mathbb{C})))$ do not make sense because, as in Definition 6.17, these are conditions involving the infinite place; however, the infinite place is not present in $(B^f)^\times$. Further, Fujisaki's lemma tells us that the relevant double coset space is compact, so bounded. In turn, the right- K -finiteness and moderate growth conditions are redundant.

Since F is naturally a F -module, we shall note that we can make $L_2^B(F)$ into an additive abelian group by pointwise addition, and we can also make it into an F -module by defining

$$(r \cdot f)(x) = r \cdot f(x), \quad r \in F, x \in (B^f)^\times.$$

Furthermore, the group $(B^f)^\times$ acts on $L_2^B(F)$ by

$$(g \cdot f)(x) = f(xg), \quad g, x \in (B^f)^\times.$$

Definition 7.48. We refer to a compact open subgroup $U \subseteq (B^f)^\times$ as a *level* of $(B^f)^\times$.

Example 7.49. We provide the main example of the level which we shall consider. Let $d = \text{disc}(B)$ and N be an integer coprime to d . Further, let p be a prime dividing N , and e be the order of p dividing N . In particular, $p \nmid d$, so B splits at p and we have an isomorphism $\varphi_p: O_p^\times \xrightarrow{\sim} \text{GL}(2, \mathbb{Z}_p)$ from Proposition 7.28. Define the sets

$$U_{0,p} := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbb{Z}_p) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{p^e} \right\},$$

$$U_{1,p} := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbb{Z}_p) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix} \pmod{p^e} \right\}.$$

Then, define $O_{0,p} := \varphi_p^{-1}(U_{0,p})$ and $O_{1,p} := \varphi_p^{-1}(U_{1,p})$. Then, we claim that the following sets are compact open subgroups of $(B^f)^\times$, so are levels:

$$U_0(N) := \prod_{p \nmid N} O_p^\times \times \prod_{p \mid N} O_{0,p}$$

$$U_1(N) := \prod_{p \nmid N} O_p^\times \times \prod_{p \mid N} O_{1,p}$$

We will prove this here for $U_0(N)$. Firstly, this is clearly open in $(B^f)^\times$ because for all but finitely many primes, the local factor is $O_p^\times$. For compactness, this relies on the fact that $O_p^\times$ and $O_{0,p}$ are compact; thus, Tychonoff's theorem says that the product is compact. We then conclude by the fact that the product topology and the subspace topology induced from $(B^f)^\times$ on $U_0(N)$ coincide, so $U_0(N)$ is compact as a subspace of $(B^f)^\times$. Note that this fact can be proven using similar techniques to Proposition 5.26, so we are done.

Definition 7.50. Let U be a level of $(B^f)^\times$. The space of U -invariants for the $(B^f)^\times$ -action on $L_2^B(F)$ is denoted by $L_2^B(U; F)$.

Example 7.51. We note that constant functions are valid examples of quaternionic modular forms. If $f: (B^f)^\times \rightarrow F$ is a constant function, then it is immediate that all three axioms in Definition 7.45 are met. Note that this contrasts how constant functions are generally not automorphic forms over $\text{GL}(2, \mathbb{A}_\mathbb{Q})$. If we want a constant function to be a valid automorphic form over $\text{GL}(2, \mathbb{A}_\mathbb{Q})$, the central character must be trivial.

Definition 7.52. Let U be a level of $(B^f)^\times$. We define $\mathcal{S}_2^B(U; F)$ as the quotient of $L_2^B(U; F)$ by the space of constant functions.

Remark 7.53. Note that the space of constant functions is 1-dimensional, since they can be identified with F .

Now, since quaternionic modular forms are invariant on the left by $B^\times$ by definition, we may view U -invariant quaternionic modular forms as functions from the double coset space $B^\times \backslash (B^f)^\times / U$ to F . Given this viewpoint, the following theorem tells us that the space of U -invariant quaternionic modular forms is finite dimensional.

Theorem 7.54. *Let F be a field, and U be a level of $(B^f)^\times$. The space $L_2^B(U; F)$ is a finite-dimensional F -vector space.*

Proof. We follow the argument provided in [Imp]. The crux of this is Corollary 7.44. The corollary tells us that we attain a finite disjoint open cover of $(B^f)^\times$ due to the $B^\times$ action and U action. Thus, we write

$$(B^f)^\times = \bigsqcup_{i=1}^n B^\times g_i U.$$

We now define the map

$$\begin{aligned} L_2^B(U; F) &\rightarrow F^n \\ f &\mapsto (f(g_1), \dots, f(g_n)). \end{aligned}$$

We claim that this is an injective F -linear map. F -linearity is immediate, so we shall prove injectivity. Suppose that we have $f_1, f_2 \in L_2^B(U; F)$ such that $f_1(g_i) = f_2(g_i)$, for all $i = 1, \dots, n$. Since we want to show that $f_1 = f_2$, take some arbitrary $g \in (B^f)^\times$. By the disjoint cover, we can write $g = dg_i u$, for some $d \in B^\times$ and $u \in U$. Then, it follows that

$$f_1(dg_i u) = f_1(g_i) = f_2(g_i) = f_2(g),$$

by $B^\times$ and U invariance. Thus, we achieve injectivity. It follows that the dimension of $L_2^B(U; F)$ is bounded by n , so we have that this is a finite-dimensional F -vector space. $\square$

7.4. The Hecke Operators. We shall continue with our set up that B is a totally definite quaternion algebra over $\mathbb{Q}$. Further, let us fix a maximal order O of B , $d = \text{disc}(B)$, and an integer N coprime to d . The objective of this subsection is to define the appropriate Hecke operators on the space $L_2^B(U; \mathbb{C})$, where U is either $U_0(N)$ or $U_1(N)$ from Example 7.49. The level will play the role of the congruence subgroups in the case for classical modular forms; more specifically, $U_0(N)$ will play the role of $\Gamma_0(N)$ and $U_1(N)$ will play the role of $\Gamma_1(N)$. Note that $L_2^B(U; \mathbb{C})$ is finite dimensional by Theorem 7.54.

The way we define the Hecke operator on $L_2^B(U; \mathbb{C})$ will be very similar to that for classical modular forms in Section 4.1. However, we note that we will only do this for *good primes*, that is, primes which do not divide d or N . In the classical case, we did this by defining Hecke operators as double coset operators on some congruence subgroup of $\text{SL}(2, \mathbb{Z})$. One of the first results introduced in Section 4.1 was that the relevant double coset space was a finite disjoint union, so we do the same here. We note that moving forward, we will use Proposition 7.28 fluently. That is, if we have a good prime p , then we will identify $O_p^\times$ with $\text{GL}(2, \mathbb{Z}_p)$.

Notation 7.55. Let p be a good prime. We denote by η_p to be the element $(B^f)^\times$ which is 1 at all places not p , and corresponding to the matrix $\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$ at the place p .

Lemma 7.56. *Let p be a good prime. We can write $U\eta_p U$ as a finite disjoint union*

$$\left(\bigsqcup_{j=0}^{p-1} \alpha_{p,j} U \right) \sqcup \alpha_{p,\infty} U,$$

where $\alpha_{p,j}$ (for $j = 0, \dots, p-1$) and $\alpha_{p,\infty}$ are elements of $(B^f)^\times$ which are 1 at every place except p . At the place p , $\alpha_{p,j}$ corresponds to $\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix}$ (for each $j = 0, \dots, p-1$), and $\alpha_{p,\infty}$ corresponds to $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$.

Proof. We only need to focus on the place p , since η_p is just 1 at all other places. To prove this, we will use the p -adic Iwasawa decomposition theorem, as in Proposition 4.2.1 of [GH11]. First, since $p \nmid N$, the factor corresponding to the place p is $\mathrm{GL}(2, \mathbb{Z}_p)$ for both $U_0(N)$ and $U_1(N)$. Therefore, the p -adic absolute value of the determinant of every element of $\mathrm{GL}(2, \mathbb{Z}_p) \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$ is p^{-1} . Then, the Iwasawa decomposition theorem states that every element of $\mathrm{GL}(2, \mathbb{Z}_p) \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$ can be uniquely written as

$$\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} p^a & 0 \\ 0 & p^b \end{pmatrix} \cdot k,$$

where $u \in \mathbb{Z}$, $a + b = 1$ (for nonnegative integers a, b), and $k \in \mathrm{GL}(2, \mathbb{Z}_p)$. If $a = 1$, $b = 0$, then the two left factors give

$$\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & u \\ 0 & 1 \end{pmatrix}.$$

Then, we have that $\begin{pmatrix} p & u \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p) = \begin{pmatrix} p & u+pk \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$ for any integer k , by noting that we have the equality $\begin{pmatrix} p & u \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & u+pk \\ 0 & 1 \end{pmatrix}$, and that $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}(2, \mathbb{Z}_p)$. Therefore, we can just take u to be its equivalence value modulo p . Consequently, this provides the cosets $\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$, for $j = 0, \dots, p-1$. Repeating the argument for $a = 0$ and $b = 1$, we attain the remaining coset $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$.

To conclude, we show that these cosets do not coincide. We will show this claim is true for $\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$ (for a fixed j) and $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \mathrm{GL}(2, \mathbb{Z}_p)$ and leave the remaining cases, as they can be proven analogously. For this, it suffices to show that $\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \notin \mathrm{GL}(2, \mathbb{Z}_p)$. We can calculate that $\begin{pmatrix} p & j \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} p^{-1} & jp^{-1} \\ 0 & p \end{pmatrix}$, so we find that

$$\begin{pmatrix} p^{-1} & jp^{-1} \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} p^{-1} & -j \\ 0 & p \end{pmatrix},$$

so this is not in $\mathrm{GL}(2, \mathbb{Z}_p)$. We are done. $\square$

Definition 7.57. Let $p \nmid d$ and $f \in L_2^B(U; \mathbb{C})$. We define the Hecke operator T_p as

$$(T_p f)(\gamma) = \sum_{j=0}^{p-1} f(\gamma \alpha_{p,j}) + f(\gamma \alpha_{p,\infty}),$$

where $\gamma \in B^\times \backslash (B^f)^\times / U$.

As suggested by the notation, this is analogous to the Hecke operators T_p acting on classical modular forms. Remember that we also defined the diamond operator in the classical setting. This operator acted nontrivially on modular forms over $\Gamma_1(N)$, so we shall define the analogous operator for quaternionic modular forms over $U_1(N)$.

Notation 7.58. Let p be a good prime. We denote by ν_p be the element of $(B^f)^\times$ which is 1 at all places not p and corresponding to the matrix $\begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}$ at the place p .

Lemma 7.59. $U_1(N) \nu_p U_1(N) = \nu_p U_1(N)$.

Proof. This follows from the fact that ν_p is a central element in $(B^f)^\times$. $\square$

Definition 7.60. Let $U = U_1(N)$ and $f \in L_2^B(U; \mathbb{C})$. Then, $\langle p \rangle f$ is defined by

$$(\langle p \rangle f)(\gamma) = f(\gamma \nu_p),$$

where $\gamma \in B^\times \backslash (B^f)^\times / U_1(N)$.

Remark 7.61. We note that this operator depends only on $p \pmod{N}$. If q is a prime such that $p \equiv q \pmod{N}$, then we can rewrite $\gamma \nu_p$ as $\frac{p}{q} \cdot \gamma \nu_q u$, where $u \in U_1(N)$ and $\frac{p}{q} \in B^\times$.

In practice, determining the Hecke action on $L_2^B(U; \mathbb{C})$ can be done in an algorithmic way. As we will see below, the Hecke action is combinatorial, in the sense that it boils down to a counting argument. First, the double coset space $B^\times \backslash (B^f)^\times / U$ is finite, so suppose it is of size h , for some positive integer h . Then, we can find h representatives of the space, so we can write

$$B^\times \backslash (B^f)^\times / U = \bigsqcup_{i=1}^h B^\times g_i U. \quad (48)$$

Then, we can find a basis $f_1, \dots, f_h$ on $L_2^B(U; \mathbb{C})$ such that

$$f_i(g_j) = \delta_{ij}, \quad (49)$$

where δ_{ij} denotes the Kronecker-delta function. If we want to calculate the Hecke action T_p on f_i , it suffices to calculate $(T_p f_i)(g_j)$, for $j = 1, \dots, h$, due to the disjoint union in Equation (48). Fixing j , we have, by definition, that

$$(T_p f_i)(g_j) = \sum_{k=0}^{p-1} f_i(g_j \alpha_{p,k}) + f_i(g_j \alpha_{p,\infty}).$$

By our definition of f_i as in Equation (49), f_i equals 1 if $g_j \alpha_{p,k}$ or $g_j \alpha_{p,\infty}$ lies in the same class as g_i , and 0 otherwise. Therefore,

$$(T_p f_i)(g_j) = |\{r \in \{0, 1, \dots, p-1, \infty\} : g_j \alpha_{p,r} \in B^\times g_i U\}|. \quad (50)$$

The analogous check can be done for the diamond operator.

Proposition 7.62. *Fix $j \in \{1, \dots, h\}$. Then,*

$$\sum_{i=1}^h (T_p f_i)(g_j) = p + 1.$$

Proof. For every $r \in \{0, 1, \dots, p-1, \infty\}$, $g_j \alpha_{p,r}$ lies in exactly one coset $B^\times g_i U$. Since there are $p+1$ choices for r , counting every single one provides the desired formula, by Equation (50). $\square$

Definition 7.63. We refer to the matrix with entries $((T_p f_i)(g_j))_{ij}$ as the *Hecke matrix* associated to T_p .

Corollary 7.64. *$p+1$ is always an eigenvalue of the Hecke matrix, which corresponds to the space of constant functions.*

Proof. The proposition above implies that the rows of the Hecke matrices sum to $p+1$, so $(1, 1, \dots, 1)^T$ is an eigenvector always present, which has eigenvalue $p+1$. $\square$

7.5. The Jacquet-Langlands Correspondence. Here, we bring our knowledge together and conclude by providing an example to illustrate a version of the Jacquet-Langlands correspondence. I will note that the version of the Jacquet-Langlands correspondence written here is a great simplification of what the correspondence fully entails. The version I have focused on is found in Theorem 3.9 in [DV11] and on Page 8 in [Buz].

Theorem 7.65. *We have an injection of Hecke modules*

$$\mathcal{S}_2^B(U_0(N); \mathbb{C}) \hookrightarrow \mathcal{S}_2(\Gamma_0(Nd))$$

and

$$\mathcal{S}_2^B(U_1(N); \mathbb{C}) \hookrightarrow \mathcal{S}_2(\Gamma_1(N) \cap \Gamma_0(d)),$$

whose images are newforms at all primes dividing d .

In Theorem 7.65, by an injection of Hecke modules, we mean that the above maps are compatible with the Hecke operators at the good primes. More precisely in the case for the level $U_0(N)$, the diagram

$$\begin{array}{ccc} \mathcal{S}_2^B(U_0(N); \mathbb{C}) & \hookrightarrow & \mathcal{S}_2(\Gamma_0(Nd)) \\ T_p \downarrow & & \downarrow T_p \\ \mathcal{S}_2^B(U_0(N); \mathbb{C}) & \hookrightarrow & \mathcal{S}_2(\Gamma_0(Nd)) \end{array}$$

commutes when p is a good prime. The same commutative diagram holds for the level $U_1(N)$, with $\Gamma_0(Nd)$ replaced by $\Gamma_1(N) \cap \Gamma_0(d)$. Therefore, if $f \in S_2^B(U; \mathbb{C})$ is a Hecke eigenform with

$$T_p f = a_p(f) f$$

for all good primes p , then its image g is also a Hecke eigenform and satisfies

$$T_p g = a_p(f) g$$

for all such p . Therefore, the system of Hecke eigenvalues is preserved exactly. However, eigenforms are only determined up to multiplication by a nonzero scalar. In other words, if $\{f_i\}$ is a basis of eigenforms on the quaternionic side and $\{g_i\}$ is a chosen basis of newforms on the classical side, then even after permuting the indices, the image of f_i need not equal g_i ; instead the image may be a nonzero scalar multiple of g_i . This scalar does not affect the Hecke eigenvalues. After normalizing the classical eigenforms, then the corresponding system of eigenvalues can be read off uniquely from the Fourier expansion of the normalized newform.

7.5.1. An Example: $d = 19$, $U = U_0(1)$. We will now carry out a basic example which illustrates Theorem 7.65. In particular, by considering normalized eigenforms, we will show that the Hecke eigenvalues for T_p line up as expected. For this, we follow the algorithm discussed at the end of Section 7.4.

We consider the case where B is a quaternion algebra of discriminant 19 and level $U_0(1)$. Let us denote $U := U_0(1)$. By Example 15.5.7 in [Voi21], $B = (-1, -19)_{\mathbb{Q}}$ is a quaternion algebra over $\mathbb{Q}$ of discriminant 19. Let $1, i, j, ij$ be a basis for B . Then, Example 15.5.7 in [Voi21] tells us that $O = \mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}\beta \oplus \mathbb{Z}i\beta$, where $\beta := \frac{1+j}{2}$ is a maximal order for B . We will fix this notation for the rest of this subsection. By Theorem 7.65, we have the injection of Hecke modules $S_2(U; \mathbb{C}) \hookrightarrow S_2(\Gamma_0(19))$. Since the level of $\Gamma_0(19)$ is prime, there are no old forms in the space $S_2(\Gamma_0(19))$. Thus, the image of this injection is $S_2(\Gamma_0(19))$. Furthermore, Example 3.41 tells us that $\dim S_2(\Gamma_0(19)) = 1$, and the normalized cusp form in $S_2(\Gamma_0(19))$ is given as follows.

$$q - 2q^3 - 2q^4 + 3q^5 - q^7 + q^9 + 3q^{11} + 4q^{12} - 4q^{13} - 6q^{15} + 4q^{16} - 3q^{17} + q^{19} + O(q^{20}) \quad (51)$$

This cusp form is taken from [LMF26]. Reading off the coefficients of the prime exponent terms except for 19, these are what we expect to get for our Hecke eigenvalues on the quaternionic side.

While I would like to dive straight into the algorithm in Section 7.4, carrying out this algorithm requires us to have already found a set of representatives in the disjoint union in Equation (48). Therefore, let us first search for a set of representatives. In doing so, the isomorphism in Proposition 7.28 will become very useful. First, we shall make this isomorphism concrete. For this, we first require the following lemma.

Lemma 7.66. *Let p be a good prime. Then, there exists nontrivial $x, y \in \mathbb{Z}_p$ such that $x^2 + y^2 = -19$. Explicitly, when $p = 2$, we may take $y = 2$ and $x \equiv 1 \pmod{4}$.*

Proof. Since p is a good prime, $p \nmid 19$. We claim there exists nontrivial elements $x_0, y_0 \in \mathbb{F}_p$ such that $x_0^2 + y_0^2 = -19 \pmod{p}$. We first assume that $p \neq 2$. There are $\frac{p+1}{2}$ values x^2 and $-19 - y^2$ can take modulo p . Then, $2 \cdot \frac{p+1}{2} = p + 1$, which is strictly bigger than p , so there must be some x_0 and y_0 which agree modulo p . Moreover, they are clearly nontrivial. Then, we can apply Hensel's lemma to the polynomial $f(x) := x^2 + y_0^2 + 19 \in \mathbb{Z}_p[X]$, since $|f(x_0)|_p < 1$ and $|f'(x_0)|_p = 1$. Thus, (x_0, y_0) lifts to a solution (x, y_0) in $\mathbb{Z}_p$.

Next, assume $p = 2$. For this, we will be more concrete and show that $y = 2$ works. Setting $y = 2$, the quadratic equation becomes $x^2 = -23$. Using the fact that a 2-adic unit u is a square in $\mathbb{Z}_2$ if and only if $u^2 \equiv 1 \pmod{8}$, we find that -23 is a square in $\mathbb{Z}_2$. Thus, a solution $x \in \mathbb{Z}_2$ exists. To specify $x \equiv 1 \pmod{4}$, all we need to do is just choose the solution satisfying $x \equiv 1 \pmod{8}$, which reduces to $x \equiv 1 \pmod{4}$. $\square$

Proposition 7.67. *Let p be a good prime. Then, the map $\varphi_p: O_p \rightarrow M(2, \mathbb{Z}_p)$ defined on the basis elements as*

$$1 \mapsto I_2, \quad i \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} x & y \\ y & -x \end{pmatrix},$$

where $x, y \in \mathbb{Z}_p$ such that $x^2 + y^2 = -19$, is a valid isomorphism of $\mathbb{Z}_p$ -orders. Further, this restricts to an isomorphism of groups $O_p^\times \rightarrow \text{GL}(2, \mathbb{Z}_p)$.

Proof. Lemma 7.66 tells us that such x, y exist in the first place. To prove φ_p is an isomorphism between $O_p^\times$ and $\text{GL}(2, \mathbb{Z}_p)$, we will first show that this mapping of bases provides a $\mathbb{Q}_p$ -algebra isomorphism between B_p and $M(2, \mathbb{Q}_p)$. Then, we will show this descends to a $\mathbb{Z}_p$ -algebra isomorphism which further descends to its unit groups.

First, this is a valid map of $\mathbb{Q}_p$ -algebras, as can be verified using the check expressed in Remark 7.7. To show that this provides an isomorphism, we just check that the images of the basis elements are linearly independent, since both B_p and $M(2, \mathbb{Q}_p)$ are $\mathbb{Q}_p$ -algebras of dimension 4. Let $a, b, c, d \in \mathbb{Q}_p$ such that

$$a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + c \begin{pmatrix} x & y \\ y & -x \end{pmatrix} + d \begin{pmatrix} y & -x \\ -x & -y \end{pmatrix} = \begin{pmatrix} a + cx + dy & b + cy - dx \\ -b + cy - x & a - cx - dy \end{pmatrix} = 0.$$

Summing the top left and bottom right conditions, we find that $a = 0$. If we perform similar comparisons of the matrix above, we find that $a = b = c = d = 0$, so we have a valid basis. Therefore, this provides an isomorphism between B_p and $M(2, \mathbb{Q}_p)$.

To show that this restricts to an isomorphism of $\mathbb{Z}_p$ -orders, we show this is bijective. First, note that

$$\varphi_p(\beta) = \frac{1}{2} \begin{pmatrix} 1+x & y \\ y & 1-x \end{pmatrix}.$$

For $p \neq 2$, it is clear this is contained in $M(2, \mathbb{Z}_p)$. When $p = 2$, recall from the proof of Lemma 7.66 that we could've chosen $y = 2$ and $x \equiv 1 \pmod{2}$. Therefore, $\varphi_2(\beta) \in M(2, \mathbb{Z}_2)$ as well. Therefore, we see that $\varphi_p(O_p) \subseteq M(2, \mathbb{Z}_p)$. Further, $\varphi_p(O_p)$ must be maximal in $M(2, \mathbb{Q}_p)$; otherwise, this would contradict maximality of O_p , as in Lemma 7.26. Since $M(2, \mathbb{Z}_p)$ and $\varphi_p(O_p)$ are both maximal orders, they must therefore be equal. Consequently, $\varphi_p(O_p) = M(2, \mathbb{Z}_p)$, so φ_p is bijective. This shows that we have an isomorphism of $\mathbb{Z}_p$ -orders.

To conclude, this isomorphism of $\mathbb{Z}_p$ -orders restricts to an isomorphism between the unit groups, proving the claim. $\square$

We will note that due to the isomorphism above, the check of whether an element lies in the same coset or not can be done using congruence relations. For this reason, it will be useful to provide congruence relations which provide a condition for when an element $a + bi + cj + dk \in B$ is actually in the maximal order O .

Proposition 7.68. *$a + bi + cj + dk \in B$ is in the maximal order O if and only if $2a, 2b, 2c, 2d \in \mathbb{Z}$ and satisfies $2a \equiv 2c \pmod{2}$, $2b \equiv 2d \pmod{2}$.*

Proof. A generic element in O is of the form $a' + b'i + c'\frac{1+i}{2} + d'\frac{i+ij}{2}$, for $a', b', c', d' \in \mathbb{Z}$. Collecting terms, we have

$$\underbrace{\left(a' + \frac{1}{2}c'\right)}_{=:a} + \underbrace{\left(b' + \frac{d'}{2}\right)}_{=:b}i + \underbrace{\frac{c'}{2}}_{=:c}j + \underbrace{\frac{d'}{2}}_{=:d}ij.$$

Relabeling the coefficients as done above, we see that $2a, 2b, 2c, 2d$ are integers. Moreover, the claimed congruence relations hold. The reverse implication is immediate from the presentation above. $\square$

Let us finally find our disjoint union as in Equation (48).

Proposition 7.69. *We can write $B^\times \backslash (B^f)^\times / U = B^\times U \sqcup B^\times \gamma U$, where γ is the element 1 everywhere and corresponding to the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ at the place 2.*

Proof. As stated above, $\dim \mathcal{S}_2(\Gamma_0(19)) = 1$, so Theorem 7.65 tells us that the size of $B^\times \backslash (B^f)^\times / U$ is 2, because $\mathcal{S}_2^B(U; \mathbb{C})$ is the quotient of $L_2^B(U; \mathbb{C})$ by constant functions. Therefore, it suffices to show that γ and 1 don't lie in the same coset.

We proceed by contradiction; suppose that γ lies in the trivial coset. Then, there exist elements $\delta = (\delta, \delta, \dots) \in B^\times$ and $u \in U$ such that $\gamma = \delta u$. Now, we shall use subscripts for δ, γ, u to denote the component at that place (for instance, δ_2 is the component of δ at the place 2). Since γ is the identity at all places not 2, $\delta_p \in O_p$ for all p , so $\delta \in O$. Therefore, writing $\delta = a + bi + cj + dk$, the coefficients must satisfy the conditions in Proposition 7.68. Focus now at the place 2; since 2 is a good prime, we can use the map $\varphi_2: O_2^\times \xrightarrow{\sim} \text{GL}(2, \mathbb{Z}_2)$ from Proposition 7.67. We also have that $\delta_2 = \gamma_2 u_2$, where $\gamma_2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

Since $\text{nrd}_2(\delta_2)$ corresponds to the determinant of the matrix we have that $\text{nrd}_2(\delta_2) = 2$ because the determinant of γ_2 is 2. Further, u maps to some matrix in $\text{GL}(2, \mathbb{Z}_2)$, so has determinant a unit. By Proposition 7.16, we can write this explicitly as

$$a^2 + b^2 + 19c^2 + 19d^2 = 2.$$

Note that a, b, c, d are not necessarily integers, so we multiply this by 4 to solve the equivalent equation

$$4a^2 + 4b^2 + 4 \cdot 19c^2 + 4 \cdot 19d^2 = 8,$$

but subject to the parity conditions in Proposition 7.68. It is clear that the only possible solution sets (a, b, c, d) are $(\pm 1, \pm 1, 0, 0)$ and $(\pm 1, \mp 1, 0, 0)$.

We can also study the explicit matrix δ maps to in $\text{GL}(2, \mathbb{Z}_2)$, which is what provides the contradiction. We have $\delta_2 = \gamma_2 u_2^{-1}$. Using φ_2 , we get an equality of matrices. If we write $\varphi_2(u_2^{-1})$ as $\begin{pmatrix} r & s \\ t & u \end{pmatrix} \in \text{GL}(2, \mathbb{Z}_2)$, we have the following. We can explicitly find that $\varphi_2(\delta_2)$ has top row entries vanishing modulo 2:

$$\varphi_2(\delta_2) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix} = \begin{pmatrix} 2r & 2s \\ t & u \end{pmatrix} \equiv \begin{pmatrix} 0 & 0 \\ * & * \end{pmatrix} \pmod{2}$$

We now use the explicit construction of φ_2 given as in Proposition 7.67. Note that Lemma 7.66 tells us that we can set $y = 2$ and $x \equiv 1 \pmod{4}$ for φ_2 . Therefore,

$$\varphi_2(\delta_2) = \begin{pmatrix} a + xc + 2d & b + 2c - dx \\ -b + 2c - xd & a - xc - 2d \end{pmatrix} \equiv \begin{pmatrix} a + xc & b + 2c \\ * & * \end{pmatrix} \pmod{2} \quad (52)$$

Plugging in our solutions $(a, b, c, d) = (\pm 1, \pm 1, 0, 0), (\pm 1, \mp 1, 0, 0)$ into this matrix, none of these allow for the top row to vanish. Therefore, we reach a contradiction, so γ represents a nontrivial representative class. $\square$

Remark 7.70. As for some motivation as to why I tested this element denoted γ in Proposition 7.69, it has a nontrivial entry at the smallest possible good prime. At the place 2, it also corresponds to one of the more simple matrices $\alpha_{p,r}$, for $r \in \{0, 1, \dots, p-1, \infty\}$ from Lemma 7.56. More generally, we note that one of the elements $\alpha_{2,r}$ will work, due to the theory of Brandt matrices, as described in Chapter 41 in [Voi21].

Corollary 7.71. *Let $\delta = \frac{A}{2} + \frac{B}{2}i + \frac{C}{2}j + \frac{D}{2}ij \in O$, where $A, B, C, D \in \mathbb{Z}$ are subject to the congruences in Lemma 7.68. Then, δ lies in the coset $B^\times \gamma U$ if and only if*

$$A + C + 2D \equiv B + 2C - D \equiv 0 \pmod{4}.$$

Proof. This comes from Equation (52). Multiplying everything by 2, we attain that

$$A + xC + 2D \equiv B + 2C - Dx \equiv 0 \pmod{4}.$$

Then, we use the fact that $x \equiv 1 \pmod{4}$ from Lemma 7.66. $\square$

We now have all the data we need to compute the Hecke action for any good prime. We shall specify a basis f, g for $L_B^2(U; \mathbb{C})$ such that

$$\begin{aligned} f(1) &= 1, & f(\gamma) &= 0, \\ g(1) &= 0, & g(\gamma) &= 1. \end{aligned}$$

In the example below, we carry out the algorithm described at the end of Section 7.4.

Example 7.72. We carry this out explicitly for $p = 3$. We first calculate $(T_p f)(\gamma)$. Using Proposition 7.67, we fix an explicit isomorphism φ_3 . In particular, we may take $y = 1$ and $x \equiv 1 \pmod{3}$. To check if $\gamma\alpha_{3,r}$ is contained in $B^\times U$, we can check if $\delta \in \gamma\alpha_{3,r}U$. Writing $\gamma\alpha_{3,r}u = \delta$, we find that $\delta_p \in O_p$ for all p , so $\delta \in O$. Therefore, we can consider $\delta = \frac{A}{2} + \frac{B}{2}i + \frac{C}{2}j + \frac{D}{2}ij \in O$, where $A, B, C, D \in \mathbb{Z}$ are subject to the congruences in Lemma 7.68. Now, note that $\text{nr}_3(\gamma\alpha_{3,r}u) = 6$ because the determinant of the matrices corresponding to $\gamma, \alpha_{3,r}, u$ are 2, 3, 1, respectively. Therefore, the reduced norm on δ provides the quadratic

$$A^2 + B^2 + 19C^2 + 19D^2 = 24, \quad (53)$$

where we have already multiplied by 4 to get an integral condition. Thus, the valid solution sets are $(\pm 1, \pm 2, \pm 1, 0)$ and $(\pm 2, \pm 1, 0, \pm 1)$. Up to units, we have the solutions $(1, \pm 2, \pm 1, 0)$. Now, amongst these solutions, the ones lying in $B^\times U$ cannot satisfy the congruences in Corollary 7.71. Checking one by one, we see that $(1, 2, 1, 0)$ and $(1, -2, 1, 0)$ are trivial.

Now, we check which $\alpha_{3,r}$ lie in the trivial coset. We note that

$$\varphi_3(\delta_3) = \begin{pmatrix} \frac{A+C+D}{2} & \frac{B+C-D}{2} \\ -\frac{B+C-D}{2} & \frac{A-C-D}{2} \end{pmatrix} =: M,$$

since $y = 1$ and $x \equiv 1 \pmod{3}$. Checking if $\delta \in \gamma\alpha_{3,r}U$ is now equivalent to seeing if there exists some matrix $\begin{pmatrix} s & t \\ u & v \end{pmatrix} \in \text{GL}(2, \mathbb{Z})$ such that

$$M = \begin{cases} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} s & t \\ u & v \end{pmatrix} = \begin{pmatrix} 6s + 2ru & 6t + 2rv \\ u & v \end{pmatrix} & \text{if } r = 0, 1, 2, \\ \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} s & t \\ u & v \end{pmatrix} = \begin{pmatrix} 2s & 2t \\ 3u & 3v \end{pmatrix} & \text{if } r = \infty. \end{cases}$$

If we consider the matrices above modulo 3, we get a defining relation for each element $\alpha_{3,r}$. If $r = 0, 1, 2$, then $\delta \in \gamma\alpha_{3,r}U$ if $M \pmod{3}$ has top row a multiple of $2r$ of the bottom row. If $r = \infty$, then $\delta \in \gamma\alpha_{3,r}U$ if $M \pmod{3}$ has bottom row vanishing. Using this condition, we find that $(1, 2, 1, 0)$ corresponds with $r = 2$ and $(1, -2, 1, 0)$ corresponds with $r = \infty$. Therefore, $(T_3 f)(\gamma) = 2$ and $(T_3 g)(\gamma) = 2$ by Proposition 7.62. We can go through the similar calculation for $(T_3 f)(1)$, but this simplifies drastically. The difference in calculation will be that Equation (53) will now have a 6 instead of a 24, but then it is clear that no solutions for A, B, C, D exist. Thus, we find that $(T_3 f)(1) = 0$ and so $(T_3 g)(1) = 4$. Ultimately, our Hecke matrix is

$$\begin{pmatrix} 0 & 4 \\ 2 & 2 \end{pmatrix},$$

which gives the eigenvalues 4 and -2 . Corollary 7.64 says that we always expected 4 to be an eigenvalue associated to the constant eigenform, so -2 is the eigenvalue which corresponds with the classical newform. Indeed, from Equation (51), we see the eigenvalues aligning.

By repeating Example 7.72 for all primes other than 19, we can compute the Hecke actions. With the help of artificial intelligence tools, I have also written a program using Python 3.11.5 ([Pyt26]), `itertools`, NumPy ([HMvdW⁺20]), pandas ([McK10]), and SageMath ([S⁺26]). While I initially wrote my program for $B = (-1, -19)_{\mathbb{Q}}$, it works for the example $B = (-1, -11)_{\mathbb{Q}}$ and level $U_0(1)$ as well. The reason for this is that almost all results in Section 7.5.1 hold for $B = (-1, -11)_{\mathbb{Q}}$ and level $U_0(1)$ as well. In Table 1 below, I have included the eigenvalues for primes up to 100. This table includes what my program outputs for the bad primes enough;

interestingly, it calculates the correct eigenvalue. My code is available on GitHub¹, and in Appendix A, I have included documentation. Below is the table of eigenvalues calculated from my program.

TABLE 1. The columns “EV” record the remaining eigenvalue of T_p , which is expected to agree with the entries of the next column “ a_p ”. “ a_p ” contains the p -th coefficient of the cusp form. The level for both columns is $U_0(1)$. The shaded entries are the program’s outputs at the bad primes.

p	$d = 19$		$d = 11$	
	EV	a_p	EV	a_p
2	0	0	−2	−2
3	−2	−2	−1	−1
5	3	3	1	1
7	−1	−1	−2	−2
11	3	3	1	1
13	−4	−4	4	4
17	−3	−3	−2	−2
19	1	1	0	0
23	0	0	−1	−1
29	6	6	0	0
31	−4	−4	7	7
37	2	2	3	3
41	−6	−6	−8	−8
43	−1	−1	−6	−6
47	−3	−3	8	8
53	12	12	−6	−6
59	−6	−6	5	5
61	−1	−1	12	12
67	−4	−4	−7	−7
71	6	6	−3	−3
73	−7	−7	4	4
79	8	8	−10	−10
83	12	12	−6	−6
89	12	12	15	15
97	8	8	−7	−7

8. CONCLUSION AND FUTURE DIRECTIONS

To conclude, this thesis has explored several related settings in the theory of modular and automorphic forms: classical modular forms, automorphic forms on $\mathrm{GL}(2, \mathbb{A}_{\mathbb{Q}})$, and quaternionic modular forms. After establishing the necessary background in the classical setting, we introduced the adelic framework and explained how classical modular forms may be viewed as automorphic forms. We then considered quaternionic modular forms and studied the Jacquet–Langlands correspondence to describe their relationship with classical modular forms. Finally, this correspondence was illustrated through an explicit computational example. In this way, the thesis has shown how these apparently different formulations are connected within a common broader framework.

There are many natural directions in which the work of this dissertation may be extended. First, there remains significant scope to deepen the study of classical modular forms themselves.

¹<https://github.com/joeorii/M4R.git>

In particular, one may dive deeper into the major problem of modularity, which we have barely touched upon in this thesis. This is because we require knowledge about abelian varieties for this. Chapters 6 of [DS05] naturally continues where we finished in Section 4 in this dissertation, and equips the reader with the necessary background to understand what the modularity theorem is stating.

A second natural direction is the study of automorphic representations. While this dissertation discusses modular forms and automorphic forms as functions, the modern language in which these objects become useful is through automorphic representations. Chapter 5 of [GH11] picks up from where we finished in Section 6. This is crucial because the conjectures by Langlands himself are crafted using this language. In its full power, the Jacquet-Langlands correspondence, whose computational aspects we studied in Section 7, is also phrased using this terminology.

Finally, we could continue pursuing the computational aspects of the Jacquet-Langlands correspondence done in Section 7. The most accessible direction would be to investigate why my program correctly calculates the Hecke eigenvalues for bad primes as well, which would require us to study and define appropriate Hecke operators at bad primes. A more ambitious task would be to generalize the program so that it returns the Hecke matrices after inputting only the quaternion algebra, the level, and the prime of the Hecke action. The difficulty in this is that if the size of the relevant double coset space increases, we need to find more nontrivial representatives. This is easier said than done because considerable care is required in deriving the congruence relations which defined these nontrivial representatives. Alternatively, we could switch directions and aim to write a program which calculates the p -Brandt matrices, as described in Chapter 41 of [Voi21]. While we have not considered this option in writing the program, it may be easier to implement as the size of the double coset space expands.

We hope this thesis has provided a glimpse into the depth and versatility of modular forms. We would like to reiterate how multifaceted this subject can be—possessing analytic, arithmetic, and geometric features, which we have seen even at such an early stage. To conclude, we will end with a quote from Langlands himself in [Lan94], speaking about the area which he envisioned.

“It is a deeper subject than I appreciated and, I begin to suspect, deeper than anyone yet appreciates. To see it whole is certainly a daunting, for the moment even impossible, task.”

APPENDIX A. REFERENCE DOCUMENTATION FOR CODE

- `algebra_info(a, b)` takes in the integers a, b which define the quaternion algebra and returns the quaternion algebra, along with the elements i, j, k which generate the quaternion algebra.
- `maximal_info(a, b)` returns a maximal order of the quaternion algebra $(a, b)_{\mathbb{Q}}$.
- `get_units(a, b)` returns the units of the quaternion algebra $(a, b)_{\mathbb{Q}}$.
- `number_of_units(a, b)` returns the number of units of the quaternion algebra $(a, b)_{\mathbb{Q}}$.
- `sols_of_norm(a, b, m)` returns elements of the quaternion algebra having norm m , where m is the inputted integer.
- `max_order_solutions_of_norm(a, b, m, maximal_prop)` returns elements in the maximal order with norm m . `maximal_prop` is a set of congruences which is the defining condition for an element in the quaternion algebra to be in the maximal order.
- `nontrivial_filter_norm(a, b, m, maximal_prop, nontrivial_prop)` returns elements in the maximal order and in the nontrivial representative of the double coset space. `nontrivial_prop` is the defining condition for an element in the quaternion algebra to be in the nontrivial representative.
- `orbits_mod_maximal_order(a, b, m, maximal_prop)` returns the number of orbits in the maximal order which have norm m .
- `nontrivial_orbits_mod_maximal_order(m, a, b, maximal_prop, nontrivial_prop)` returns the number of orbits in the nontrivial representative which have norm m .
- `hecke_operator_matrix(a, b, maximal_prop, nontrivial_prop, p)` returns the Hecke action matrix for the Hecke action T_p . We note that this p must be a good prime for this function to work properly.
- `maximal_order_congruences(a, b)` returns the congruences which define the maximal order.
- `bad_primes(a, b, N)` returns the set of bad primes. Here, N is the level.
- `hecke_eigenvalues_by_prime(a, b, maximal_prop, nontrivial_prop, N, prime_bound)` returns the eigenvalues of the Hecke action. Here, `prime_bound` is the upper bound on the primes p which we wish to verify Jacquet-Langlands for. For instance, in Table 1, we set `prime_bound = 100`.

REFERENCES

- [Ant26] Anthropic, *Claude*, <https://claude.ai/>, 2026, Large language model, accessed 7 June 2026.
- [Bum97] Daniel Bump, *Automorphic forms and representations*, Cambridge Studies in Advanced Mathematics, Cambridge University Press, 1997.
- [Buz] Kevin Buzzard, *Old notes about computing modular forms on definite quaternion algebras*, https://www.ma.imperial.ac.uk/~buzzard/maths/research/notes/old_notes_about_computing_modular_forms_on_def_quat_algs.pdf.
- [BvdGHZ08] Jan Hendrik Bruinier, Gerard van der Geer, Günter Harder, and Don Zagier, *The 1-2-3 of modular forms: Lectures at a summer school in nordfjordeid, norway*, Universitext, Springer-Verlag Berlin Heidelberg, 2008.
- [Car16] Ana Caraiani, *Modularity: from fermat to the langlands programme*, EMS Newsletter (2016), no. 102, 10–14.
- [CFSU10] J. W. S. Cassels, A. Frohlich, London Mathematical Society., and International Mathematical Union., *Algebraic number theory*, London Mathematical Society, 2010 (eng).
- [Con] Keith Conrad, *Maximal compact subgroups of $GL_n(\mathbb{Q}_p)$* , <https://kconrad.math.uconn.edu/blurbs/gradnumthy/maxcompact.pdf>, Online notes, accessed 2026.
- [DS05] Fred Diamond and Jerry Shurman, *Modular forms, elliptic curves, and modular curves*, Springer New York, New York, NY, 2005.
- [DV11] Lassina Dembele and John Voight, *Explicit methods for hilbert modular forms*, 2011.
- [Eme] Matthew Emerton, *Classical modular forms as automorphic forms*, <https://www.math.uchicago.edu/~emerton/gl2.pdf>, Lecture notes.
- [Gel75] Stephen Gelbart, *Automorphic forms on adele groups*, Annals of Mathematics Studies, vol. 83, Princeton University Press, Princeton, NJ, 1975.
- [GH11] Dorian Goldfeld and Joseph Hundley, *Automorphic representations and L -functions for the general linear group*, Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2011.
- [Gol06] Dorian Goldfeld, *Automorphic forms and L -functions for the group $gl(n, r)$* , Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2006.
- [Gre15] Ben P. Green, *Jacquet–langlands for GL_2* , [https://pub.math.leidenuniv.nl/~vonkjb/other/automorphic/GL\(2\).pdf](https://pub.math.leidenuniv.nl/~vonkjb/other/automorphic/GL(2).pdf), 2015, Lecture notes, University of Oxford.
- [HMvdW⁺20] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant, *Array programming with NumPy*, Nature **585** (2020), no. 7825, 357–362.
- [Hum72] James E. Humphreys, *Introduction to lie algebras and representation theory*, Graduate Texts in Mathematics, Springer-Verlag, New York, 1972 (eng).
- [Imp] Imperial College London, *FLT: Fermat’s Last Theorem Project Repository*, <https://github.com/ImperialCollegeLondon/FLT>, GitHub repository, accessed March, 2026.
- [JL70] Hervé Jacquet and Robert P Langlands, *Automorphic forms on $gl(2)$* , Lecture Notes in Mathematics, vol. 114, Springer-Verlag, Berlin, New York, 1970.
- [Kna01] Anthony W. Knap, *Representation theory of semisimple groups: An overview based on examples*, Princeton Mathematical Series, vol. 36, Princeton University Press, Princeton, NJ, 2001.
- [Kob93] Neal Koblitz, *Introduction to elliptic curves and modular forms*, 2 ed., Graduate Texts in Mathematics, vol. 97, Springer-Verlag, New York, 1993.
- [Lan67] Robert P. Langlands, *Letter to andré weil (1967)*, 1967, <https://publications.ias.edu/rpl/paper/43>.
- [Lan70] ———, *Problems in the theory of automorphic forms*, Lectures in Modern Analysis and Applications III (C. T. Taam, ed.), Lecture Notes in Mathematics, vol. 170, Springer, Berlin, Heidelberg, 1970, pp. 18–61.
- [Lan94] Robert P. Langlands, *Review of Modular forms by Haruzo Hida*, Bulletin of the American Mathematical Society **30** (1994), no. 1, 122–127.
- [LMF26] The LMFDB Collaboration, *The L -functions and modular forms database*, <https://www.lmfdb.org>, 2026, [Online; accessed 2 June 2026].

- [McK10] Wes McKinney, *Data structures for statistical computing in python*, Proceedings of the 9th Python in Science Conference, 2010, pp. 56–61.
- [Mil09] James S. Milne, *Modular functions and modular forms*, <https://www.jmilne.org/math/CourseNotes/MF.pdf>, 2009, Course notes.
- [Mil20] James S. Milne, *Algebraic number theory (v3.08)*, 2020, Available at www.jmilne.org/math/, p. 166.
- [Miy06] Toshitsune Miyake, *Modular forms*, Springer Monographs in Mathematics, Springer, Berlin, Heidelberg, 2006.
- [Mun75] J.R. Munkres, *Topology: a first course*, Prentice-Hall, New Jersey, 1975.
- [Ope26] OpenAI, *Chatgpt*, <https://chatgpt.com/>, 2026, Large language model, accessed 7 June 2026.
- [Pyt26] Python Software Foundation, *Python 3 documentation*, Python Software Foundation, 2026, Accessed 7 June 2026.
- [RV98] Dinakar Ramakrishnan and Robert J. Valenza, *Fourier analysis on number fields*, 1998.
- [S⁺26] W. A. Stein et al., *Sage Mathematics Software (Version 10.8)*, The Sage Development Team, 2026, <http://www.sagemath.org>.
- [Ser73] Jean-Pierre Serre, *A course in arithmetic*, Graduate Texts in Mathematics, vol. 7, Springer-Verlag, New York, 1973.
- [Shi71] Goro Shimura, *Introduction to the arithmetic theory of automorphic functions*, Princeton University Press, Princeton, NJ, 1971.
- [Sut21] Andrew Sutherland, *18.785 Number Theory I*, Massachusetts Institute of Technology: MIT OpenCourseWare, 2021, Fall 2021.
- [Voi21] John Voight, *Quaternion algebras*, Graduate Texts in Mathematics, Springer, Cham, 2021.
- [Wil95] Andrew Wiles, *Modular elliptic curves and fermat’s last theorem*, Annals of Mathematics **141** (1995), no. 3, 443–551.
- [Zag08] Don Zagier, *Elliptic modular forms and their applications*, The 1-2-3 of Modular Forms (Jan Hendrik Bruinier, Gerard van der Geer, Günter Harder, and Don Zagier, eds.), Springer, Berlin, Heidelberg, 2008, pp. 1–103.